

# Epistemic Debt

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## *The Accounting Layer of Computational Semantics*

*Unpaid Warrant in Bounded Observer  
Architectures: From Scientific Representation to  
Governed Semantic Use*

Gianluca Conte

*Conceptual – methodological research monograph.  
Position and framework paper*

AUTHOR MANUSCRIPT  
2026

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Epistemic Debt: The Accounting Layer of Computational Semantics. Unpaid Warrant in Bounded Observer Architectures: From Scientific Representation to Governed Semantic Use

**ISBN: 9798187686445**

Independent author manuscript / pre-publication research monograph.

This work is conceptual, formal-methodological and programmatic. It is not legal, tax, financial, medical or engineering certification advice. It is not undergone formal journal peer review. It is circulated for scholarly review, criticism and further development.

AI assistance in preparing this edition is disclosed in full in the Methodological Note at the end of the book.





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## Scientific Status and Non-Claims

This ledger fixes evidence class before the argument begins. Typography, theorem labels, internal audits, implementation witnesses, and release polish cannot upgrade an item beyond the evidence class shown here.

| Evidence class                                         | Current status / boundary                                                                                                                                               |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mathematical statements — LOCALLY ARGUED / MIXED CLASS | Credited standard results, local derivations, definition-level consequences, non-entailments, and conditional conformance invariants remain separated at point of use.  |
| Conformance properties — CONDITIONAL                   | Hold only under the stated mediation, receipt, authority, typing, relevance, dependency-coverage, and propagation-policy premises.                                      |
| Empirical validation — OPEN                            | No independent outcome study is reported; workflow benefit, threshold adequacy, reviewer capacity, dependency recall, and cross-domain payoff remain empirical targets. |

| Evidence class                                                      | Current status / boundary                                                                                                                                                            |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Instrument reliability<br>— OPEN                                    | The nine-component debt vector remains a candidate coding instrument pending independent componentwise reliability and incremental utility.                                          |
| Independent peer review — ABSENT                                    | Internal hostile review and AI-assisted self-critique are pre-publication controls, not external scientific authority.                                                               |
| Implementation and reproducibility — WITNESS / SUPPLEMENT-DEPENDENT | Constructibility is witnessed; the argument is standalone, while numerical audits and counterexample-search receipts require named supplements. Independent validation remains open. |

*Interpretive rule. A status changes only when evidence of the corresponding class enters the record. Editing may reduce ambiguity and overclaim; it cannot pay empirical-validation or independent-authority debt.*

## Abstract

Artificial intelligence can produce polished semantic artefacts before their sources, transformations, limits, and authorized uses are reconstructable. This monograph treats that problem as a sequence rather than a pile of theories. Reticular Local Abstraction (RLA) is used as a general grammar for a scientific commonplace: the same domain is represented at multiple levels, and every transmission preserves some distinctions while collapsing others relative to declared questions. Compact Reticular Computability (CRC) then asks when such a multi-level description has actually been compiled into an effective, bounded computational object. A literature-compiled bryophyte model supplies the stress case: when equations, parameters, taxonomic scope, or calibration are incomplete, the act of compilation exposes approximations and transfers that must either remain declared or become hidden epistemic liabilities. From that hinge, epistemic convolution is defined as a CNN-inspired movement over semantic fields - local operators, aggregation, col-

lapse, candidate stabilization, and typed non-answer - without claiming a new neural primitive. Reticular Observer Architectures (ROA) formalize the resulting bounded observer: horizon, reticulum, local transducers, memory, monitoring, receipts, validation, and human authority. Epistemic Debt is the accounting consequence: unpaid warrant attached to semantic objects before use, while a separate named-use gate decides whether residual debt is tolerable, review-requiring, waived, or blocking. An optional bounded-simulation bridge to cellular automata remains outside the causal spine. The formal core mixes credited standard results, local derivations, non-entailments, and conditional conformance invariants; the debt vector remains a candidate instrument; empirical superiority, independent peer review, proof-assistant verification, calibration, dependency completeness, and institutional legitimacy remain open. The programme should be simplified or rejected if matched, simpler baselines deliver equal reconstruction and protection against unsupported authoritative use at lower burden.



## **Significance and Scientific Contribution**

The core problem is not merely that AI can emit false sentences. It is that any representation can become actionable before the warrant for its use is reconstructable. The proposed architecture joins abstraction, computability, typed refusal, provenance, assurance, and use-sensitive governance without collapsing them into one score. Its strongest claim is conditional and testable: making horizon, lineage, debt, and gate outcomes first-class should reduce unsupported authoritative use and reconstruction cost relative to simpler baselines, while preserving human accountability. The maturity ledger above is binding: candidate compositional contribution, constructibility, and internal consistency are not substitutes for instrument reliability, empirical validation, or independent authority.



## The Wager in One Page

A machine can now produce a polished answer before an institution can reconstruct how that answer was made. The old problem of knowledge is that every map leaves something out; the new operational problem is that machines can draw and circulate maps faster than people can recover what was omitted, borrowed, approximated, collapsed, or never checked. The failure is therefore larger than false sentences. It is semantic work becoming actionable before its warrant is reconstructable.

This book names the unpaid remainder Epistemic Debt. A claim, label, summary, model, equation, or risk object carries debt when some warrant required to understand or rely on it remains unresolved: source, trace, transformation, collapse, label, reification, validation, authority, or transfer. Debt is not a synonym for falsehood. A statement may be true and still carry debt because its path cannot be reconstructed; an approximation may be useful and honest when declared, yet hazardous when promoted as established fact without provenance or validation.

The companion operation is a bounded semantics of claim admissibility. Under a declared horizon of sources, scope, transformations, receipts, support rules, validations, and resource limits, a system can compute what status a claim has earned and can return answer, unknown, contradiction, review, timeout, failure, or out-of-horizon rather than manufacturing completion. That computation does not decide truth. It makes the observer and its limits explicit.

The wager is therefore modest and testable: if representations carry their own horizon, lineage, unresolved debt, and use boundary, machine-assisted work should become easier to reconstruct and harder to launder into unsupported authority. If simpler provenance, assurance, selective-prediction, or workflow systems achieve the same protection with less burden, this architecture should contract to those simpler forms.

*Genealogical spine. Established multi-level scientific practice -> RLA as a common grammar -> CRC as an executability criterion -> bryophyte compilation as a stress case for approximation and transfer -> epistemic convolution as a semantic compilation method -> ROA as the bounded observer architecture -> Epistemic Debt as the accounting remainder -> named-use governance. The PCE/cellular-automata bridge is optional and enters only when a cross-formalism simulation claim is made.*



## A Scene Before the Theory

A compliance team asks an AI system to summarize supplier-payment controls. The output states that payments above a threshold require approval by the chief financial officer. The sentence is plausible and immediately reusable, but an auditor asks: which interview or policy supports it, what population it covers, whether exceptions were tested, what transformation produced the wording, and who reviewed the conclusion. If those questions cannot be answered, the defect is not merely possible hallucination; it is unreconstructable semantic work. The surface is ready before the warrant is.

Now move the same problem into science. A literature compiler needs a physiological relation for a bryophyte model. The relevant literature may supply a species-specific equation, a range, a qualitative dependency, a relation measured in a nearby taxon, or no directly usable function at all. The honest description is not one exclusive label but a small profile: `source_basis` records whether support is DIRECT, TRANSFERRED,

SYNTHESIZED, or UNRESOLVED; compiler\_ops records every declared operation applied by the compiler, including approximation when one is introduced; calibration\_state records what has actually been calibrated or tested. A transferred relation can therefore also be approximated, and a direct relation can still be unvalidated for the intended conditions. Failure begins when one of those dimensions is silently omitted or promoted into a stronger evidential state.

This is the bridge from CRC to Epistemic Debt. Compiling a scientific domain does not manufacture missing science; it reveals where a computable reticulum needs more warrant than the literature presently supplies. An AI system can make that gap more dangerous because it can generate a plausible bridge. The architecture does not claim to eliminate such bridges. It requires them to carry their papers.



## The Seven Questions

- 1. What does a representation preserve, and for which questions? RLA supplies a common grammar for stating the answer.*
- 2. When does a collection of models become an executable object? CRC requires effective operators, scheduling, resource bounds, terminals, and explicit failure.*
- 3. What happens when the science does not supply every required equation or parameter? The bryophyte case separates source basis, compiler transformations, calibration, and unresolved relations so approximation and transfer can remain inspectable obligations rather than mutually exclusive labels.*
- 4. How can AI assemble and navigate semantic structure without pretending to be a new neural theory? Epistemic convolution reuses the local operator / aggregation / collapse gesture while keeping candidate and certified objects distinct.*

5. *What is the observer? ROA is the bounded contract that joins horizon, reticulum, local operators, receipts, memory, monitoring, validation, and human authority.*
6. *What remains unpaid after a semantic transformation? Epistemic Debt records the unresolved warrant component by component; it does not estimate truth.*
7. *Who finally decides whether the object may be used? A named-use gate evaluates debt and evidence under a governance context; waiver permits without paying, while discharge changes the evidence state.*

## How to Read This Book

The book can be read linearly without prior knowledge of the ROA corpus. The main thread is a genealogy of different object types, not six theories competing at the same level: a scientific grammar, an executability criterion, a stress case, a semantic compilation method, an observer architecture, and an accounting discipline. Formal readers can descend into the local theorem chapters; other occasional readers can follow the chapter questions, running examples, and boundary boxes. Every load-bearing claim is tagged by what it is: definition, credited standard result, local derivation, conditional conformance invariant, empirical target, implementation witness, or open problem.

*Two sentences to carry. A representation is useful because it leaves something out; it becomes dangerous when its omissions are spent as warrant. ROA constructs the bounded observer, and Epistemic Debt records what its semantic artefacts still owe before use.*

## Genealogy and Type Contract

Mandatory spine.  $RLA \rightarrow CRC \rightarrow$  bryophyte compilation  $\rightarrow$  epistemic convolution / ECU / ECNN  $\rightarrow$  ROA  $\rightarrow$  Epistemic Debt  $\rightarrow$  named-use gate. Later objects may consume earlier ones, but no layer borrows scientific maturity from another.

Type contract. RLA is a common grammar of levels, queries, transmissions and collapse; CRC is an executability criterion; the bryophyte is a stress case for approximation and transfer; epistemic convolution is a CNN-inspired compilation/navigation method, not a new neural primitive; ROA is the bounded observer architecture; Epistemic Debt is the accounting consequence; the gate alone decides one named use.

*Optional bridge. Cellular-automata / PCE language enters only when a cross-formalism simulation claim is actually made and must then carry source model, target model, encoding, observables, distortion, overhead, horizon, and any reverse obligation required by an equivalence claim.*



## Reading Paths for Different Audiences

| Reader                           | Shortest complete route                                                                                                                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theoretical computer scientist   | Track domains, codomains, effectivity, termination, reductions, and exact hypotheses. Route: Bounded Observer → RLA → CRC → Formal Core.                                                                            |
| Information scientist / engineer | Track source objects, transformations, receipts, status transitions, deterministic replay, and proposal/control separation. Route: Bounded Observer → Epistemic Convolution → Epistemic Debt → Reference Semantics. |

| Reader                 | Shortest complete route                                                                                                                                                                                                                                        |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Philosopher of science | Track observer relativity, purpose dependence, severe tests, theory-ladenness, and the separation of truth, support, validation, warrant, authority, and use. Route: Problem $\rightarrow$ RLA $\rightarrow$ Propagation/Transfer $\rightarrow$ Falsification. |







**PART I**

# **The Problem: Representation, Evidence, and Warrant**

---



## CHAPTER 1

# Every Representation Preserves Some Distinctions and Loses Others

*What is lost whenever anything is represented?*

On the wall of almost every classroom hangs a lie. The world map we grew up with shows Greenland roughly the size of Africa. In truth Africa is about fourteen times larger. The map is not mistaken — it is a Mercator projection, built in 1569 to do one thing supremely well: let a sailor draw a straight line and hold a true compass bearing along it. To buy that gift, the map gives something up. It stretches the far north and south, swelling Greenland and shrinking the tropics. The distortion is not an error. It is a choice — a particular human point of view, made for a particular purpose.

A representation is useful because it leaves something out. A map cannot keep every property of the territory. A scientific model cannot keep

every feature of the system. A legal memo cannot keep every possible reading. A dashboard cannot keep all the evidence behind a score. The loss is not the failure; the failure begins when the loss is forgotten.

So the correct question about a representation is not whether it is false, but what it was made for and what it cannot support. A navigation projection may be excellent for bearings and poor for comparing the area of continents. Used for the second purpose, it is being pushed beyond the warrant it earned. Epistemic Debt names exactly that excess of use over warrant.

The debt is not always a lie. It may be a missing caveat, a lost scale, a summary without source density, a label that grew too broad, or a model rule whose empirical status is unclear. Error suggests a binary defect; debt names a use-independent obligation attached to an object under a horizon. The same debt vector can be tolerable for orientation and blocking for authoritative use because the gate, not the vector, is use-sensitive. The ledger does not moralize the object;

it records what remains unpaid and governs the use separately.

The same rule will govern every later example. A map, a model, a dashboard tile and a generated answer are not first judged by asking whether they are globally true. They are first judged by asking what use they are entitled to support. The moment a representation is used outside that entitlement, its omissions stop being harmless compression and become debt.







## CHAPTER 2

# Why Artificial Intelligence Makes an Old Epistemic Problem Operationally Urgent

*Why does a very old epistemic problem become urgent now?*

Humans have always summarized, classified and inferred. What changed is speed, volume and institutional dependence. An AI system now produces policies, risk objects, scientific syntheses, legal issue lists, code reviews and board memos faster than review structures can reconstruct their evidence paths. The most visible failure is hallucination, but it is not the deepest one. The deeper failure is unreconstructable semantic work: a polished artefact exists, but the organization cannot say which source supports each claim, what was inferred, which caveats were lost, who reviewed the conclusion, or what residual risk was accepted. The output survives; the warrant trail evaporates.

This moves the problem from private virtue to public infrastructure. It is no longer enough for a careful person to know a claim is uncertain. The uncertainty must travel with the artefact. A system must be able to represent not only an answer but also unknown, contradiction, review-required, out-of-scope and debt-bearing states — so that machine-assisted work accumulates instead of evaporating.

The central acceleration is not only that AI writes faster. It is that AI can produce socially usable surfaces - summaries, labels, tables, decisions, risk categories - before anyone has reconstructed their source chain. That inversion changes the governance problem: review arrives after use unless the proof object travels with the claim — sources, receipts, transformations, debt vector, and the identifier of the horizon they were computed under, so status can be recomputed rather than assumed.

Reader payoff: this chapter is not only about hallucination. It reframes hallucination as one case of a larger institutional problem: semantic surfaces become actionable before their warrant be-

comes reconstructable. Once that is clear, the rest of the monograph can stop chasing every possible model failure and focus on the status layer that should travel with every output.

A particularly revealing special case appears when AI is used to compile science. If a required relation is unavailable in usable form, a model may offer a plausible proxy, transfer a relation from a nearby organism or scale, or synthesize a functional form from several sources. None of those moves is automatically a hallucination. The epistemic failure is the undeclared promotion: an approximation, transfer, or synthesis presented downstream as if it were a directly sourced, locally validated relation. The bryophyte case later makes this distinction concrete.



## CHAPTER 3

# Epistemic Status Without Ontological Overreach

*Does the system decide what exists, or only what may be used?*

This monograph lives on the epistemological side. It is about the status of claims and artefacts — what is supported, unsupported, contradiction, or out-of-scope, and what it still owes — not a theory of what ultimately exists. When a system defines an object called invoice, species, risk, control, hormone or legal issue, it has not discovered the furniture of reality; it has declared a working vocabulary for a horizon.

The computational semantics of claim admissibility therefore computes admissibility. It answers a restricted question: given these sources, this scope, these transformations, these receipts, this support matrix and these validation predicates, what state may this claim carry now? The discipline protects both sides. It guards against naive automation, where a computed answer is treated

as true, and against sterile scepticism, where every imperfect representation is dismissed as useless.

## **The Unintegrated Problem Across Adjacent Research Fields**

The programme stands downstream of mature traditions. Information theory explains limits on transmission and compression; sufficient statistics and Blackwell comparison formalize task-relative informativeness; abstract interpretation provides sound abstraction of program semantics; computability theory bounds effective procedures; simulation and bisimulation specify cross-system correspondence; selective prediction and conformal methods formalize abstention and risk control; paraconsistent logic and argumentation treat contradiction; provenance standards record derivation; assurance cases organize claims, arguments, and evidence; proof-carrying code attaches certificates to artefacts; model cards, datasheets, FactSheets, and algorithmic audits document intended use and risk; philosophy of science supplies model-relative representation, severe testing, and social criticism.

No novelty is claimed for those components. The differential claim is architectural. Existing disciplines usually answer different questions: *what information was lost?*; *what program property is soundly approximated?*; *which conclusion survives attack?*; *where did the data come from?*; *may the classifier abstain?*; *does the safety argument support the claim?* ROA/ED attempts to keep these questions typed but connected across one lifecycle: a representation is constructed under a horizon, transmitted across levels, executed by local units, challenged by terminal states, persisted with provenance, assigned unpaid warrant, and finally compared with a named use at a gate.

The selected neighbours constrain the contribution claim. RLA must not be presented as replacing abstract interpretation, information bottleneck, sufficient statistics, causal abstraction, or coarse-graining. CRC must not be presented as a new computability thesis. ECNN must not be reduced to neural uncertainty or allowed to borrow the maturity of selective prediction. The optional bounded-simulation bridge must use ordinary simulation obligations when invoked.



ED must not be presented as the invention of provenance, assurance, audit, abstention, or technical debt. The research contribution is the explicit composition and the hypothesis that this composition improves reconstructability and prevents unsupported authoritative use at acceptable cost.

The state of the art also supplies the reporting standard. High-quality theoretical work separates definitions, assumptions, credited standard results, derived results, conjectures, implementation witnesses, empirical targets, and open problems. Reproducibility guidance distinguishes computational reproducibility from empirical replicability. NIST AI RMF and its Generative AI Profile, ISO/IEC 23894:2023, and ISO/IEC 42001:2023 provide organizational risk-management and management-system obligations; W3C PROV distinguishes entities, activities, agents, derivations, and responsibility. These frameworks constrain ROA/ED, but they do not by themselves define the manuscript's claim-local EvidenceStatus  $\times$  DebtVector  $\times$  named-use GateOutcome

state. That difference is a scope distinction to be tested, not evidence of superiority.

Recent causal-abstraction work makes the novelty boundary sharper. Intervention-sensitive mappings, soft interventions, distributed alignment, graded faithfulness, combinations of partial high-level models, inference under lossy representations, distributionally robust causal abstractions, compositional abstraction, and learned causal-abstraction networks already provide mathematically serious accounts of cross-level fidelity (Massidda et al., 2023; Geiger et al., 2024, 2025; Pîslar, Magliacane, and Geiger, 2025; Xia and Bareinboim, 2025; Felekis, Damoulas, and Giampouras, 2025; D’Acunto, Di Lorenzo, and Barbarossa, 2026; Lorenz and Tull, 2026). RLA therefore does not claim to invent faithful abstraction or the idea of a network of abstractions. Its narrower proposed delta is to join query-relative factorization to typed fibers and side information, receipts, explicit collapse debt, typed terminals, and named-use governance across one lifecycle. It is defeated as a distinct contribution when a matched causal-abstraction, compositional, or

provenance formalism supplies those same interfaces with equal diagnostic power and lower burden.

Work published in 2026 sharpens five additional boundaries. Context compression can preserve reconstruction quality while weakening refusal behaviour, so semantic recoverability cannot be treated as behavioural or governance preservation (Nguyen and Lin, 2026). Local diagnostic regions can reveal where a causal abstraction succeeds rather than forcing one global yes/no verdict, reinforcing the need for horizon- and query-indexed comparison (Li et al., 2026). Sequential prediction with abstentions exposes a formal trade-off between errors and erroneous abstentions, while clinical selective-prediction studies show that subgroup-dependent miscalibration can make abstention policies appear safer globally than they are locally (Yu and Blanchard, 2026; Lechuga Lopez, Shamout, and Rudner, 2026). Search augmentation can improve answerable-query accuracy while degrading abstention on unanswerable queries, especially under noisy retrieval, so adding evidence channels is not automatically epistemically

monotone (Xie et al., 2026). NIST’s post-deployment monitoring report likewise treats monitoring methods as incomplete and scattered across the lifecycle (Rao et al., 2026). These results constrain the ROA/ED test design; they do not validate the architecture.

New work is not promoted into a new ROA layer merely because it is newer, more capable, or more mathematically expressive. It is first typed by the interface it changes: abstraction and fidelity belong at RLA; executability, tools, memory, scheduling, and resource control at CRC/ECU; model-side uncertainty at the epistemic head; assurance, monitoring, provenance, and threat arguments at ROA; residual warrant at Epistemic Debt; permission at the named-use gate. A comparator changes the mandatory spine only when it reconstructs one of those interfaces with equal or greater diagnostic power at lower burden; otherwise it is a stronger local formalism, an input signal, an implementation pattern, or an empirical baseline. This rolling rule lets the literature grow without turning literature growth into ontology growth.



## CHAPTER 4

# Scientific Lineage: What Is Inherited and What Is Actually Claimed

*What is inherited, what is relabeled, and what is actually claimed as new?*

The framework is strongest when it refuses novelty by vocabulary. Levels, abstractions, partitions, sufficient statistics, coarse-graining, simulation relations, partial functions, provenance graphs, abstention, assurance cases, technical debt, and human authorization all have mature scientific homes. The contribution claimed here is therefore neither a new law of nature nor the discovery of these ingredients. It is a typed composition that makes their interfaces explicit and attaches use-sensitive unpaid warrant to the semantic objects that cross them. The layer notes below state selected neighbours, the relabeling risk, the claimed delta, and the condition under which the delta should be rejected.

## Layer Contribution Matrix

| Layer | Inherited neighbours                                                                                                                                                                                                                                                                    | Candidate delta                                                                                                      | Defeat condition                                                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| RLA   | Model hierarchies, coarse-graining, sufficient statistics, information bottleneck, abstract interpretation, causal abstraction, observational equivalence (Shannon 1948; Cousot & Cousot 1977; Tishby et al. 1999; Rubenstein et al. 2017; Beckers & Halpern 2019; Geiger et al. 2023). | Query-indexed contract: state types, fibers/side information, factorization or recovery maps, and recorded collapse. | Relabeling if an existing abstraction or causal-model formalism supplies the same contract with equal diagnostic power and lower burden. |

| Layer | Inherited neighbours                                                                                                                                                                                                            | Candidate delta                                                                                                                                      | Defeat condition                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| CRC   | Classical computability, type soundness, operational semantics, model checking, scheduling, resource-bounded evaluation (Turing 1936/37; Rice 1953; Hoare 1969; Cousot & Cousot 1977; Clarke et al. 1999; Baier & Katoen 2008). | One acceptance object coupling effective operators, scheduler, budgets, progress/acyclicity, typed terminals, receipts, and cross-layer obligations. | Defeated if the package reduces without loss to an ordinary workflow or transition-system specification. |



| Layer      | Inherited neighbours                                                                                                                                   | Candidate delta                                                                                                                                     | Defeat condition                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| PCE bridge | Reductions, simulation, bi-simulation, compiler correctness, behavioural equivalence (Park 1981; Milner 1989; Nacula 1997; Cook 2004; Sangiorgi 2011). | Only a bounded, observable-specific simulation certificate with distortion, overhead, horizon, and reverse obligations when equivalence is claimed. | Adds nothing if no non-trivial observable-preserving simulation exists or “equivalence” is resemblance by construction. |

| Layer          | Inherited neighbours                                                                                                                                                                                                                          | Candidate delta                                                                                                                                            | Defeat condition                                                                                                                                               |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ECU / ECN<br>N | Pipelines, neural-symbolic systems, selective prediction, calibration, conformal prediction, paraconsistent reasoning (Dung 1995; d’Avila Garcez et al. 2012; Guo et al. 2017; Vovk et al. 2005; Angelopoulos & Bates 2023; Wen et al. 2025). | Typed partial epistemic transducers with non-answer terminals, receipts, debt updates, and candidate/certified separation; neural implementation optional. | Empirically unnecessary if matched selective-generation, workflow, or assurance baselines deliver equal diagnosis, traceability, and safety with lower burden. |

| Layer | Inherited neighbours                                                                                                                                                                                                   | Candidate delta                                                                                                                                                                | Defeat condition                                                                                                                                          |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROA   | Observer-relative modelling, provenance, knowledge graphs, runtime monitoring, assurance arguments, sociotechnical governance (Giere 1988; W3C 2013; Kelly & Weaver 2004; Raji et al. 2020; Bloomfield & Rushby 2024). | Compile a versioned horizon, reticulum, local units, memory, monitor, threat model, governance context, and human authority into one reconstructable observer server contract. | Weakened if conventional retrieval + provenance + assurance yields the same outcomes, or if added types do not improve reconstruction/error localization. |

| Layer                  | Inherited neighbours                                                                                                                                                                                                                     | Candidate delta                                                                                                                                          | Defeat condition                                                                                                                                                 |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Epis-<br>temic<br>Debt | Debt metaphors, evidence sufficiency, provenance gaps, assurance deficits, documentation/model cards, risk acceptance (Cunningham 1992; Kruchten et al. 2012; Mitchell et al. 2019; Sculley et al. 2015; W3C 2013; Longpre et al. 2024). | Use-independent componentwise unpaid-warrant vector + separate use-sensitive gate + typed discharge + debt-preserving waiver + non-laundering invariant. | Fails as an instrument if coders cannot apply it reliably, hidden dependencies create false cleanliness, or simpler workflows match outcomes with less friction. |

**Close Comparators and Residual Delta**

Truth-maintenance systems, justification logic, nanopublications/micropublications, provenance semirings, information-flow/taint analysis, and digital-twin credibility practice all overlap material parts of the architecture. Their detailed compari-

son is retained in Appendix G rather than repeated here.

*Residual claim.* After those comparators are admitted, the defensible hypothesis is a typed composition that separates evidence status from named-use authorization, attaches residual warrant to semantic transformations, preserves debt across waiver, parameterizes propagation by relevance and dependency coverage, and requires a governed transition before authoritative use. This is useful composition only if comparative tests discriminate it.

*Differential-contribution rule.* A concept counts as novel here only when the manuscript states (i) inherited prior art, (ii) the additional relation or constraint, (iii) a case the addition discriminates, and (iv) a falsifier. New terminology without all four is relabeling, not novelty.



**PART II**

# **The Bounded Observer: Horizon, Status, and Refusal**

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## CHAPTER 5

# From Generated Answers to Explicit Bounded Observers

*What is missing when an answer arrives without its observer?*

The bottleneck in AI-assisted work is no longer fluent generation; it is reconstruction, provenance, proof and governance. A prompt-first workflow generates a convincing report, analysis or compliance artefact, but afterwards an organisation often cannot say which evidence supported each claim, what was inferred rather than supported, what was missing, or whether the result was approved, merely reviewed, or only drafted. The dominant pattern prompt → answer hides the observer behind the answer. The observer-compiler picture is different. Many of the most useful outputs of AI work are not final answers but intermediate, typed artefacts — definitions, taxonomies, blueprints, issue chains, graph fragments, challenge suites, review checklists — which

a human curator corrects, compresses, validates and reuses as input for later steps. Cumulative intelligence is not located inside a single model call; it is distributed across calls, human decisions, files, versions, traces and reviews. A prompt, on this reading, is the design of a natural-language operator over a field, not a request for a sentence. This monograph proposes a shape that makes that structure explicit: bounded domain material  $\rightarrow$  explicit observer structure  $\rightarrow$  typed epistemic artefacts  $\rightarrow$  validation state  $\rightarrow$  proof / witness / review / governance. A reticular observer architecture is a finite, horizon-relative structure specifying what can be represented, how distinctions are transmitted or collapsed, which artefacts can be produced, and under which validation and governance conditions they may be used.

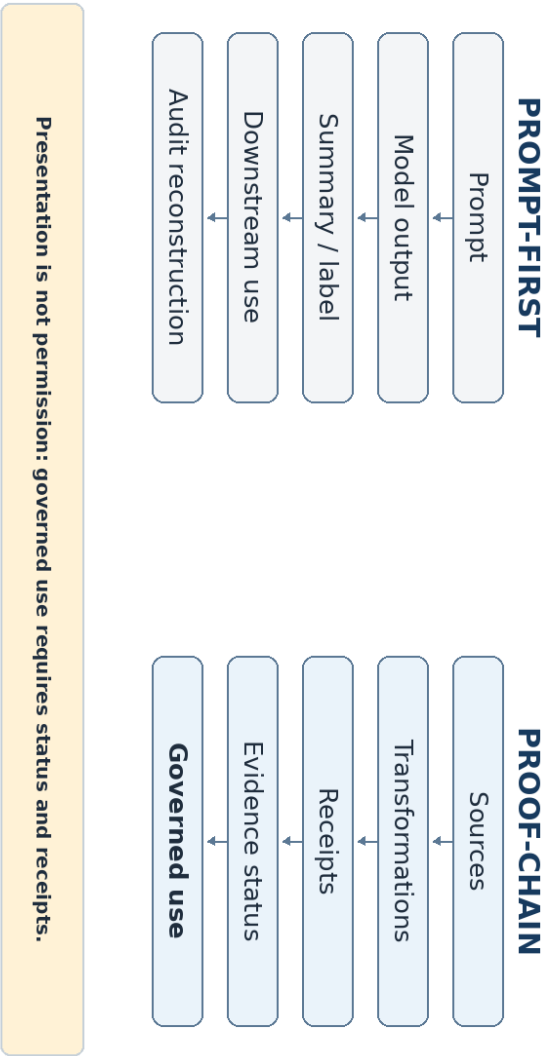


Figure 1. Prompt-first output versus proof-chain flow. The proof-chain route keeps sources, transformations, receipts, evidence status, and governed use explicit; presentation alone does not establish warrant.

A single example runs through the monograph, drawn from compliance review because that domain stresses evidence, its absence, accountability and human review. An organisation reviews whether supplier payments are adequately controlled; an analyst conducts interviews, and the transcripts

are the bounded domain material. The horizon admits questions about which controls exist, where evidence is missing and what follow-up is required, and excludes a final legal opinion on liability. The levels are raw answers, atomic facts, links between them, and a compact synthesis; the interviews yield, among others, the atom “a chief-financial-officer (CFO) approval threshold exists” and the evidence gap “no formal exception-handling procedure was found.” The monograph returns to this example at each layer — collapse, abstention, reification — so that the abstract machinery always has a concrete referent.

## CHAPTER 6

# Computational Semantics of Claim Admissibility Within a Declared Epis- temic Horizon

*What, exactly, is claimed to be computable?*

The framework uses “computational semantics” in a deliberately restricted sense. Classical operational semantics maps programs to transition behaviour; denotational semantics maps expressions to mathematical objects; axiomatic semantics relates programs to assertions (Scott and Strachey, 1971; Plotkin, 1981/2004; Hoare, 1969; Winskel, 1993). Here the object is different: a versioned claim or semantic artefact under a declared horizon. The local computation maps that object to an evidence status, debt vector, typed terminal, and receipt. Formally, for a horizon  $H$  and claim  $c$ ,  $CS\_H(c) = \langle \text{status\_H}(c), \text{debt\_H}(c), \text{terminal\_H}(c), \text{receipt\_H}(c) \rangle$  when the admitted

source enumeration, transformations, support rules, validation predicates, scheduler, and timeout policy are effective. This is a semantics of admissibility and use-readiness, not a universal semantics of truth. Answer, unknown, contradiction, out-of-horizon, review-required, timeout, and failure are all successful typed outputs when produced by the declared rules. The claim is conditional. A semantic operation is computational only after the state representation, admitted operators, stopping conditions, failure constructors, and resource envelope are declared. The architecture therefore rejects a common equivocation: a finite vocabulary does not make an evaluation computable, and a computed status does not make the represented claim true. Computability belongs to the evaluator; warrant belongs to evidence and use; truth remains external to the framework.

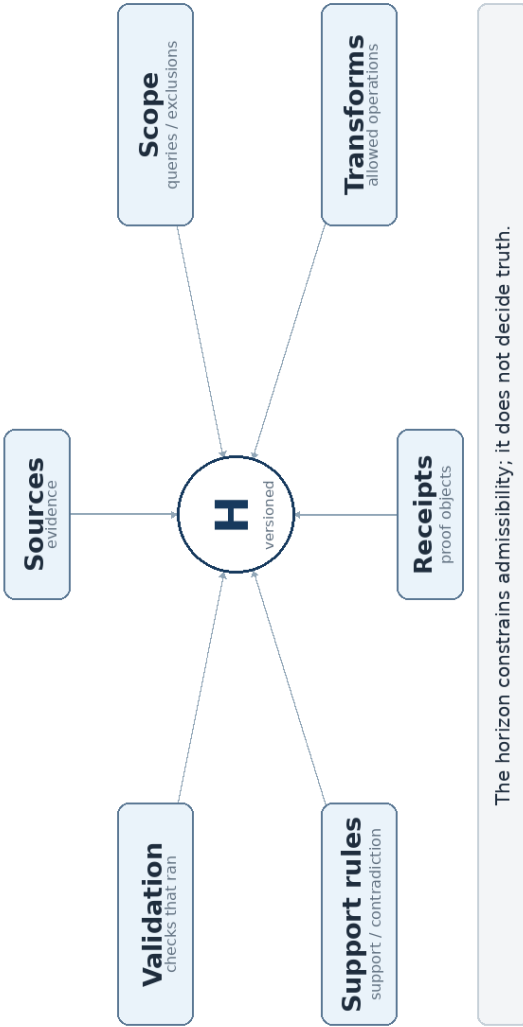
What must be declared before a system is allowed to answer?

A horizon is not a poetic boundary. It is an operational contract. It declares which sources are admitted, which scope is in force, which trans-

formations are allowed, which receipts count, which support-and-contradiction matrix is used, and which validation predicate must hold before a claim can move to a stronger status.

Without a horizon, semantic computation inflates: the system answers as if every possible source, law, theory and reviewer were available. With a horizon, it knows what it is allowed to know and what it must mark out-of-scope. The horizon also makes disagreement intelligible: two reviewers may assign different statuses to the same claim because they work under different admitted sources or scope. That is not a bug when the horizon difference is declared; it is the condition that makes status computable.

The horizon is deliberately contractual because a system cannot compute the status of a claim until it knows which sources count, which transformations are allowed, which receipts are required, which standards apply and which uses are excluded. “Answer the question” is not yet a computable instruction; “answer under this horizon and return a status” is.



*Figure 2. Declared epistemic horizon. Sources, scope, transformations, receipts, support rules, and validation define the perimeter within which claim admissibility is computed; the horizon does not decide truth.*



Standalone test: after reading this chapter, a reader should be able to reject any answer that does not declare its admissible source perimeter, intended use and review boundary. A system without horizon may still produce text, but it cannot yet compute permission.

Horizon adequacy — who audits the contract

One question remains, and the chapter cannot honestly close without raising it: the contract disciplines everything computed under it, and nothing about the drafting of the contract itself. Declare a narrow horizon — two friendly interviews admitted, sampled transactions excluded — and the CFO claim sits comfortably at supported. Every entry in the ledger is typed, every receipt genuine, every status correct; and the whole construction answers a question sized to guarantee its own comfort. “Supported under our declared horizon” is an honest sentence about the wrong contract.

The ledger cannot repair this from inside. Debt is computed under H, and a component computed under the horizon cannot price the choice of the horizon without circularity. The repair there-

fore lives at the gate, where use enters the computation: the intended use determines the minimum admissible horizon — the sources that must be admitted, the validations that must at least be possible, the reviews that must be reachable — and a gate asked to authorize a use under a horizon below that minimum returns neither allow nor block but horizon-inadequate: a review-required state whose subject is the contract, not the claim. The chapter on proof chains, use gates, discharge, and waiver wires the check into the gate; Part IX records it in the formal core; Appendix D carries its calibration — which uses require which minimum horizons — as an open obligation, because this monograph can name the check and cannot yet ship its thresholds. The check needs a shape before it can have settings, and the shape is what this monograph can responsibly supply. An adequacy contract declares twelve things, and Appendix C carries them as a blank form: who set the minimum horizon for a class of use, what that minimum requires, which version is in force from when, how a decision under it is contested, who may authorize an exception, and when it lapses.

Populated, that is a governance artifact belonging to the institution deploying the ledger; empty, it is a form. This monograph ships the type and a blank form — Appendix C carries the fields — marks both CANDIDATE, and does not pretend the values are calibrated — an uncalibrated protocol presented as ready would be the false green analyzed in the chapter on preventing unsupported success states committed at the level of governance. The principle compresses to one sentence, and it is the sentence a hostile reviewer should be handed first: the horizon protects the computation from inflation, and the gate protects the institution from the horizon.



## CHAPTER 7

# Information, Proof, and the Limits of Computation

*Why must a semantic architecture be able to stop without an answer?*

Gödel boundary. Let  $F$  be a consistent, effectively axiomatized formal theory capable of representing elementary arithmetic. Gödel's first incompleteness theorem implies that  $F$  is incomplete: some arithmetical sentence is not decided by  $F$ . Under the usual derivability conditions, the second theorem prevents  $F$  from proving its own consistency if it is consistent (Gödel, 1931). These results concern sufficiently expressive formal theories; they do not show that every practical model is incomplete, that empirical evidence is futile, or that a machine can never prove useful theorems. Their architectural consequence is narrower: internal derivability cannot be promoted into a complete certificate of truth.

Church–Turing boundary. Church and Turing established equivalent formal accounts of effective calculability and undecidable decision problems (Church, 1936; Turing, 1936/1937). In modern terms, no total computable procedure decides the halting behaviour of every program-input pair. A bounded observer therefore needs resource budgets, timeout, and non-answer terminals. Hiding nontermination behind a spinner or a guessed answer is not semantic completion.

Rice boundary. Rice’s theorem states that every non-trivial extensional property of the partial function computed by an arbitrary program is undecidable (Rice, 1953). The qualifier extensional matters: syntactic properties and many properties over restricted finite domains remain decidable. ROA therefore does not claim a universal checker of “meaning.” It checks declared predicates over typed, bounded objects; when a requested property escapes that contract, the correct result is out-of-horizon, timeout, review, or an explicit open proof obligation.

Shannon boundary. Shannon entropy quantifies uncertainty in a distribution and supports rig-

orous accounts of coding, channel capacity, and information loss (Shannon, 1948). It does not by itself encode reference, truth, justification, authority, or intended use. RLA can use entropy and mutual information to measure preservation under a distribution, but Epistemic Debt is not entropy: it records unpaid warrant, including source, validation, authority, and transfer obligations that may remain high even when a representation is information-rich.

Design inference. These are constraints, not decorations. Gödel blocks complete self-grounding of sufficiently expressive formal theories; Church–Turing blocks a universal effective decision procedure; Rice blocks total recognition of non-trivial program semantics; Shannon blocks the equation of information quantity with epistemic standing. The positive architecture is consequently local: declare the horizon, compute only decidable or budgeted predicates, return typed non-knowledge, preserve receipts, and route authorization outside the evaluator.

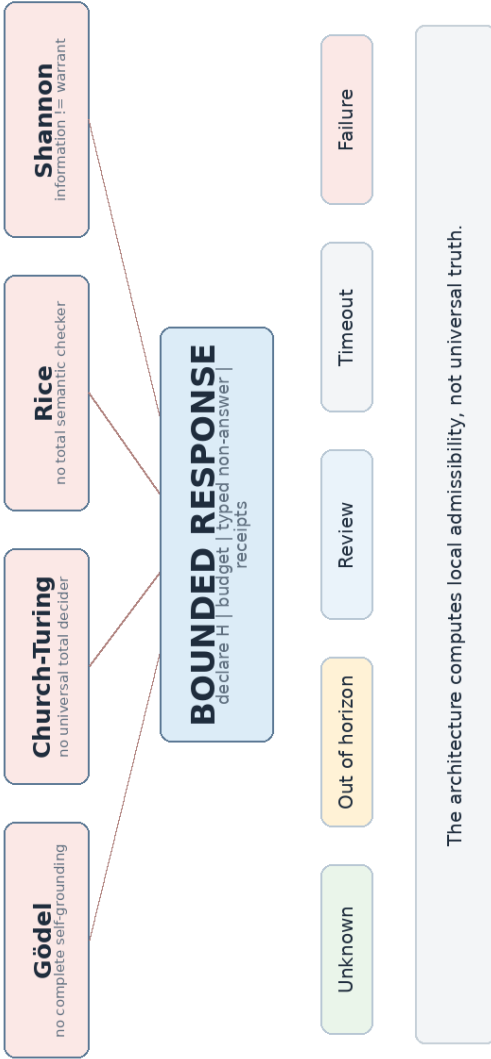


Figure 3. Formal limits and the bounded architectural response. Gödel, Church-Turing, Rice, and Shannon constrain what can be certified or decided; the response is local, budgeted computation with typed non-answer and preserved receipts.



## CHAPTER 8

# Operational Definition of the Computational Semantics of Claim Admissibility

*What is inherited, what is relabeling, and what is added?*

Inherited. Programming-language semantics already supplies rigorous relations among syntax, execution, mathematical meaning, and assertions; formal verification supplies proof obligations; abstract interpretation supplies sound approximation; information theory supplies measures of uncertainty and loss. Renaming any of these “computational semantics” would add nothing.

Local definition.

Let  $H = (S, \Omega, Q, T, R, M, V, \Pi)$

contain admitted sources  $S$ , scope  $\Omega$ , declared queries  $Q$ , effective transformations  $T$ , required receipts  $R$ , decidable support and contradiction

rules  $M$ , validation procedures  $V$ , and a priority or conflict policy  $\Pi$ . The resource envelope  $B$  is separate from  $H$  because the same horizon can be evaluated under different budgets. For claim  $c$ ,  $CS_{\{H,B\}}(c)$  returns  $\text{ClaimState} \times \text{Terminal} \times \text{Receipt}$ , where  $\text{ClaimState} = \text{EvidenceStatus} \times \text{DebtVector}$ . The computation is partial unless every invoked step is itself effective; it becomes terminalized and total only under the stronger budgeted-dispatcher assumptions stated in F06. This is the local computational semantics of claim admissibility and use-readiness, not a universal semantics of truth.

Claimed delta. The proposed composition couples a query-relative abstraction contract, an executable reticulum, typed epistemic terminals, provenance receipts, an observer context, a componentwise debt ledger, and a separate named-use gate. Its candidate contribution is architectural and relational. It is defeated as a distinct scientific contribution if an existing assurance, provenance, causal-abstraction, or workflow formalism supplies the same interfaces, failure localization, and empirical payoff with lower complexity.

**Falsifiability.** The formal part loses if a theorem fails under its declared hypotheses, a hidden totality or injectivity assumption is required, or an interface cannot be typed without circularity. The empirical part loses if matched simpler baselines achieve equal or lower unsupported authoritative use, reconstruction cost, false-green rate, and reviewer burden. The governance part loses if certification, waiver, and authority cannot be kept distinct in real workflows.

*Programme-level loss rule. The integrated architecture must be removed, collapsed to simpler components, or demoted to a domain-specific pattern if independent preregistered comparisons across heterogeneous tasks repeatedly show that simpler provenance or assurance workflows weakly dominate it on unsupported authoritative use and reconstruction quality while imposing lower review and operational burden. No internal audit can waive this loss condition.*

## **Reader checkpoint — the same boundary in four languages**

Computer science: not every semantic property is decidable. Information science: preserved bits do not imply preserved warrant. Philosophy of science: proof inside a model is not truth of the world. Engineering: every run needs a typed way to stop, fail, or ask for review.

## CHAPTER 9

# Observer Relativity Without Epistemic Relativism

*Does a declared perspective make every answer equally valid?*

Observer relativity means that representation and status are indexed by a declared observer tuple. Different horizons may admit different sources, languages, queries, tolerances, and authorities. The resulting statuses can differ without contradiction because the contracts differ. Relativism would follow only if no cross-observer comparison were possible. ROA instead requires explicit comparison of source manifests, query families, transmission contracts, validation rules, and governance contexts. A disagreement whose contract difference can be located is more auditable and more open to criticism than an apparently universal answer whose observer is hidden.

## **Interlude – The Ruliad as Limit-object for the bounded observer**

Wolfram's Ruliad is useful here not as a premise of ROA or as imported ontology, but as a maximal contrast. Wolfram defines it as the unique entangled limit of all possible computations; his observer theory then emphasizes computationally bounded observers that sample and equivalence only a restricted portion of that formal vastness, with experienced regularities depending on observer characteristics (Wolfram, 2021, 2023, 2026). ROA's H is a different kind of object: a versioned operational contract over admitted sources, queries, transmissions, validation, authority, and resources. The Ruliad is therefore treated here as a formal limit-object, not as a literal superset from which ROA is derived.

The author's earlier reply accepts that observer-relative framing and adds one guardrail: formal maximality does not become a scientific claim of ontological finality unless an identification proce-

dure survives the bounded-observer constraints already in force. It further proposes operational objectivity as replay, diff, and audit inside a fixed H - an RLA/ROA extension of the dialogue, not a claim made by Wolfram (Conte, 2026d). AI can enlarge practical reach by making translations, searches, and concept mappings feasible; it expands or revises H rather than abolishing the need for one.

## **BOUNDARY RULE - WHAT THE WOLFRAM BRIDGE MAY AND MAY NOT DO**

Use the bridge only to distinguish maximal formal possibility from an admitted horizon, observer-relative regularity from universal necessity, or translation reach from earned warrant. Do not use it to claim that RLA/ROA is derived from Wolfram, that the Principle of Computational Equivalence has been established here, that the Ruliad is empirically validated as ontology, or that replay constitutes metaphysical objectivity. If concept growth erases the constraints that make an observer identifiable, the observer contract has been lost rather than completed.



## CHAPTER 10

# Evidence States and Typed Epistemic Outputs

*What kind of object does the computation return?*

The core formal object is small. Given a claim and a horizon, the function returns one member of a finite set of admissible states. The vocabulary may be extended by domain, but the principle is fixed: the output is an admissibility state, not a truth value. This is what makes the term computational semantics defensible in a restricted sense — the system is not deciding all semantic properties of arbitrary programs or worlds; it is deciding whether a claim satisfies declared proof conditions in a bounded horizon. The most important result is not a state at all, but unpaid warrant recorded beside it: a transformation moved meaning in a way that touches the claim, but the required receipt or validation is missing. The claim may be promising, useful, even likely true; what it has earned is recorded, and whether a use is refused is the gate's

question. This is why a responsible refusal can be a successful computation. A confidence score is not a proof status: a model can be confident about an unsupported claim and uncertain about a supported but nuanced one.

The evidence status function is not a truth function, and it is not yet a permission either. It asks what a claim has earned: raw output, computed but unsupported, supported-not-validated, or validated by a check that actually ran. Permission is the next question and a different function: the named-use gate, which alone sees the intended use and returns a typed `GateOutcome` such as warranted, permitted-with-waiver, review-required, horizon-inadequate, or blocked.

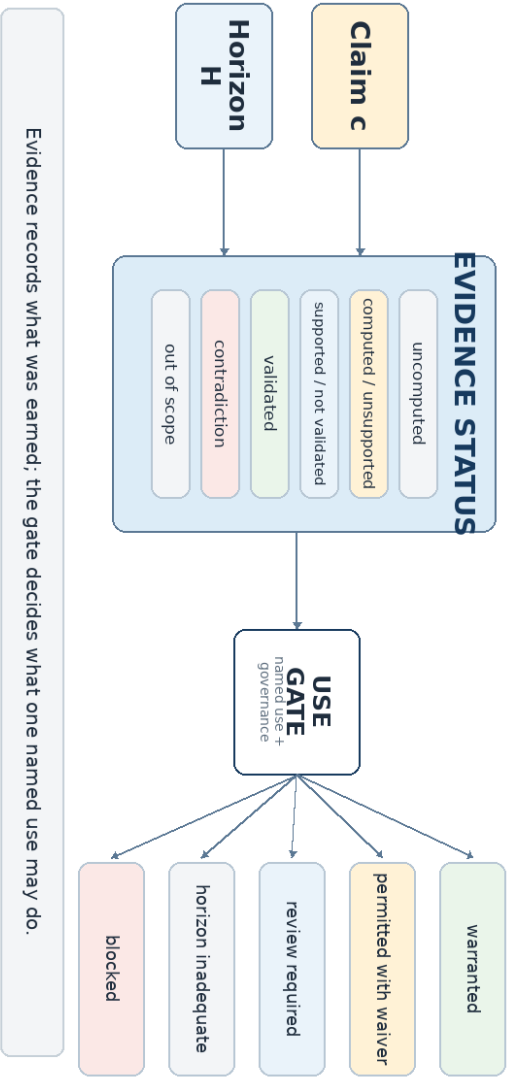


Figure 4. Evidence status and named-use gate. Evidence records what a claim has earned, while the gate separately decides what a named use may do under governance.

Status is therefore dynamic. A claim can move from unsupported to supported when a source is attached, and from supported to validated when a test is run. Warrant is not the next rung: a validated claim becomes warranted only at a gate, where an intended use is finally named, and the same claim returns to review-required at the gate when a stronger use is requested. The object did not change; the relation between object, horizon and use changed. Note the register: once a use is named the value is a gate outcome. The evidence status stays supported-not-validated in every row; the verdict is what moves.

The rest of the monograph spends these states and never again defines them, so they are worth one slow pass. Uncomputed is the honest starting fact: no one has asked the question under this horizon yet. Computed-unsupported is the state fluency hides best — the machine produced the sentence, and nothing yet stands behind it. Supported-not-validated marks the first real earning: a source is attached and admitted under the horizon. What that support is sufficient for is not yet

asked, because sufficiency is always sufficiency for something, and this function has not been told for what. Validated, the top of the evidential ladder, records that a declared check has actually run — a test executed, a sample drawn, a readback confirmed. The ladder stops there, and stopping is the point: no state of the evidence alone can say a claim is fit for a purpose, because fitness names a purpose and the evidence does not know one. Around the ladder stand the states that make refusal expressible: contradiction, a first-class result and not an embarrassment, because incompatible admitted sources are a finding; out-of-scope, which protects the contract by refusing questions the horizon never funded; and, beside all of them, the debt vector — which the entire accounting layer exists to be able to record. The vector does not say blocked: blocking is a verdict about a use, and neither status nor vector has been told one.

The set is finite by design and extensible by declaration. A domain may add states — a legal register may need privileged; a clinical one, escalate-to-clinician — but every extension is itself a declared act under the horizon, with criteria, because an

undeclared status is a label without papers, and the monograph has a word for that.

What the computation returns, then, is not the probability that a sentence is true but the earned answer to a narrower question: what this claim, under this contract, has paid for. A confidence value is a model-side score whose meaning depends on a specified calibration procedure and data distribution. Evidence status records what admitted evidence and declared checks support. Neither quantity is truth, and neither by itself authorizes a use.

Is “I do not know” a phrase, or a computed state?

A conventional system can answer, refuse, or say “I do not know.” In a proof environment that phrase is not enough: not-knowing must be typed. The epistemic head is the component that converts semantic input into an admissible state. It need not be a neural head in the machine-learning sense; it can be a rule layer, a review protocol, a database procedure, a typed workflow, a constrained model, or a human-machine checklist.

Four actions matter most. Answer when the status allows. Abstain when no support exists. Escalate when domain authority or review is required. Expose debt when the object is useful but carries unpaid warrant. Blocking is the strict case of exposing debt: the debt touches a use that must not proceed. This reframes humility: a good system does not merely sound cautious; it explains the operational reason for its caution and shows what evidence would move the claim to a stronger state.

The head's placement is the design decision that everything else assumes: it sits at the boundary where candidate meaning becomes institutional act, and the trust boundary sits with it — outside the model. The model may draft, summarize, classify and propose; the head consults objects that are governed differently from the model's output: sources, receipts, validation predicates, review states. That is why the head can be a rule layer or a checklist as easily as a neural component — its authority comes from what it reads, not from what it is.

The epistemic head is the part of the architecture that refuses to collapse every input into an

answer. Its action space must include ordinary answer, unknown, contradiction, out-of-horizon and review-required — not polite disclaimers but terminal states of the computation. Debt-open is not among them: it is an annotation that can accompany any of them, including answer.

### Epistemic head outputs

For technical readers, state the head's relation to the gate rather than inferring it: `EpistemicHeadOutput = Action x Annotations = render(evidence_status, debt_vector, gate_outcome)`.

It presents rather than decides — invoking the gate where a use is named, reporting the evidence status where none is. So debt-open means unpaid warrant exists, never that a use was refused. The head should also not be confused with a confidence threshold. Confidence is a model-side estimate. The epistemic head is a policy-and-evidence interface: it presents what the other components computed: the evidence status, the debt beside it, and where a use is named the gate's outcome. It does not form the judgement; it invokes and renders it.



Governed differently is a weaker claim than immutable, and the weaker claim is the true one. Receipts can be model-generated; sources can be stale, revoked or incomplete; validation predicates can be authored by the same pipeline they judge; review states can be selected to flatter. Six exposures follow — authorship, integrity, freshness, revocation, completeness and selection — with indicative controls: signing and role separation, effective dates, revocation lists, declared source manifests, and sampling under an authority the requester does not control. Indicative, because none is sufficient alone: a signature establishes who signed, not that the signer was competent to sign. This monograph names the exposures and the controls; it does not implement them, and Appendix D records the threat-model sketch as a TARGET. A trust boundary asserted in a diagram is a drawing. What makes it architectural is that the objects on the trusted side are the ones an adversary would have to attack in public.

Abstention deserves its rehabilitation in full. In a fluent system, unknown reads as failure; in a proof environment it is a successful computation

with a type — cheaper than a wrong answer, more informative than silence, and actionable, because a typed unknown says what would move it: which source to collect, which horizon to widen, which review to request. A system that cannot say unknown does not have more knowledge than one that can. It has fewer states.

In this architecture, then, “I do not know” is not the system running out of words. It is a terminal state with a type — the computation arriving, correctly, at the edge of what the horizon funds, and reporting the edge instead of crossing it.

**PART III**

# **Reticular Local Abstraction: A Common Grammar of Scientific Representation**

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## CHAPTER 11

# Scientific Levels as Typed Representations

*What is a level when ontology is not assumed?*

*RLA begins from a scientific commonplace rather than a novelty claim: researchers already use multiple descriptions, scales, variables, models, and coarse-grainings. The purpose of the reticular vocabulary is to make the contracts among those descriptions explicit and comparable across domains. The scientific burden is therefore not to prove that levels exist, but to state what each representation can answer, what each transmission preserves, and what each collapse makes unavailable.*

A level  $L$  is a pair  $\langle D(L), \text{Sigma}(L) \rangle$ .  $D(L)$  is a state domain: concentrations, texts, activation vectors, transactions, legal propositions, or any other declared representation.  $\text{Sigma}(L)$  is a family of admitted operations or dynamics over that do-

main. A level is an epistemic slice of practice, not a metaphysical layer of nature. It records the observables and rules under which a community or system performs work. This observer-relative treatment follows the model-purpose and scientific-representation tradition rather than asserting ontological strata (Giere, 1988; Weisberg, 2013; Frigg and Nguyen, 2020; Shalizi and Moore, 2003).

Levels are typed. A state in a molecular representation cannot be substituted for a population-level rate without a transmission contract. A label cannot be substituted for the evidence that generated it. This discipline is elementary and decisive: many semantic failures are type errors disguised by natural language. RLA makes the type boundary visible before asking whether the mapping is useful.

## CHAPTER 12

# What a Transmission Preserves: Query-Relative Fidelity

## Typed Reticula and Admissible Query Families

A **Reticular Local Abstraction** is a directed multigraph

$$R = (V, E, \{L_i\}_{i \in V}, \{\tau_e\}_{e \in E}),$$

whose level  $L_i = (X_i, \Sigma_i, Q_i)$  declares a state domain  $X_i$ , admitted operations or dynamics  $\Sigma_i$ , and a family of questions  $Q_i$ . Each edge  $e: i \rightarrow j$  carries a typed transmission  $\tau_e: X_i \rightarrow X_j$  together with a contract: domain, query family, side information, distortion, randomness, receipt, roll-back and open debt. A partial map may return a typed non-answer; it may not silently fabricate a value outside its domain.



For a declared query family  $Q$  on  $X$ , define query-equivalence by

$$x \equiv_Q x' \Leftrightarrow \forall q \in Q, q(x) = q(x').$$

A deterministic transmission  $\tau: X \rightarrow Y$  is  **$Q$ -faithful** when

$$\tau(x) = \tau(x') \Rightarrow x \equiv_Q x'.$$

Equivalently, every  $q \in Q$  factors through the image of  $\tau$ : there is a decoder  $\acute{q}: \tau(X) \rightarrow A_q$  such that  $q = \acute{q} \circ \tau$ . Injectivity is the special case in which  $Q$  separates all points of  $X$ . Thus losslessness is never asserted without naming what must be recoverable.

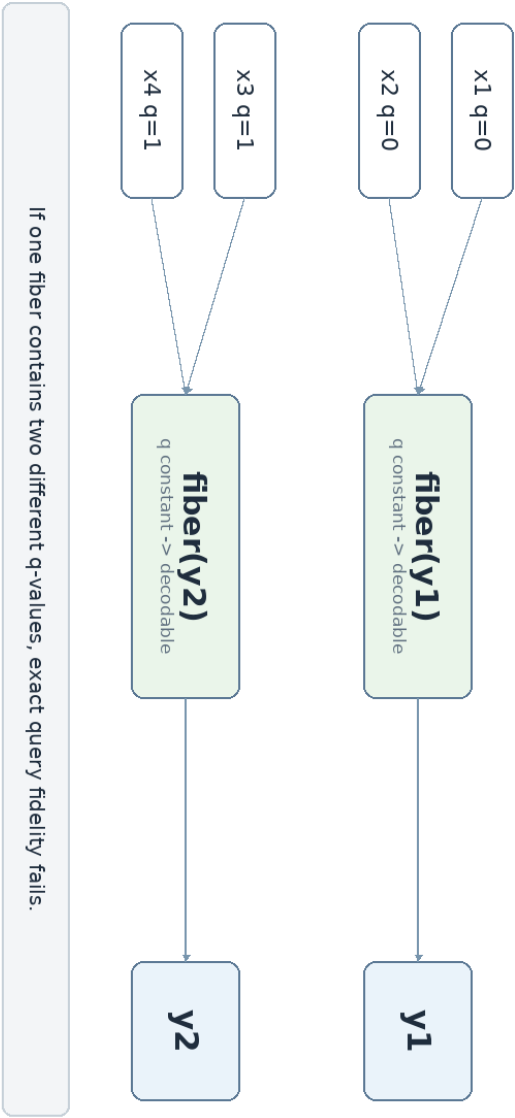


Figure 5. Query-relative fidelity in a transmission. A query is preserved when it is constant on each transmission fiber; full state reconstruction requires stronger assumptions.

## **Approximate and Stochastic Query Fidelity**

When the query codomain  $A_q$  carries a metric  $d_q$ , define the worst within-fiber query distortion by

$$\Delta_q(\tau) = \sup_{\{y \in \text{im}(\tau)\}} \sup_{\{x, x' \in \tau^{-1}(y)\}} d_q(q(x), q(x')).$$

The map is  $\varepsilon$ -faithful for  $q$  when  $\Delta_q(\tau) \leq \varepsilon$ , with the metric, tolerance, admissible domain, and composition rule declared. For a stochastic transmission represented by a Markov kernel  $K: X \rightsquigarrow Y$ , recoverability requires a decoder kernel  $D: Y \rightsquigarrow A_q$  and a declared loss or probability metric. Exact stochastic recovery means that the composite kernel  $D \circ K$  maps each admitted  $x$  to the point mass at  $q(x)$ ; approximate recovery bounds the chosen risk. A recorded random seed can make one realized run replayable, but it does not establish distributional validity, calibration, or robustness to a changed input law.

## **Scientific Scope and Falsification Criteria**

RLA does not claim that nature comes pre-partitioned into levels. It supplies a falsifiable bookkeeping grammar: a level or transmission is scientifically useful only if its domain, questions, preserved distinctions, collapsed distinctions and failure conditions can be inspected. An asserted cross-level relation fails when a declared query cannot be recovered within its tolerance, or when the mapping depends on undeclared side information. Query-indexed comparison is closest to sufficient statistics, Blackwell comparison, information bottleneck, abstract interpretation, and causal abstraction; RLA claims only the explicit coupling of these ideas to typed fibers, receipts, and downstream use (Blackwell, 1953; Cousot and Cousot, 1977; Tishby, Pereira and Bialek, 1999; Beckers and Halpern, 2019; Geiger et al., 2025).

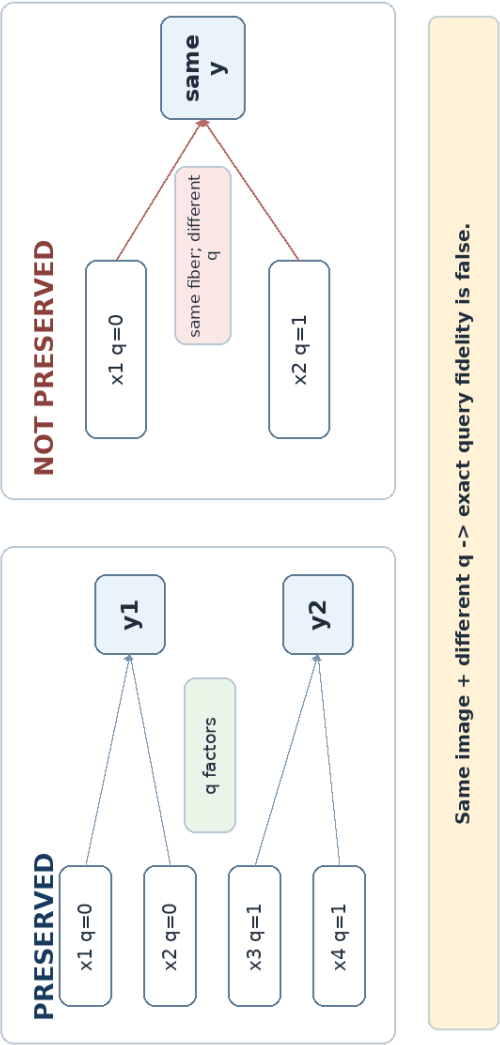


Figure 6. Fiber test for query fidelity. A single pair of source states with the same transmitted image but different query values defeats exact preservation for that query.

*Formal replacement. The mature term is query-faithful. “Injective” remains the special case that preserves all point distinctions; “quasi-injective” is retained only as historical vocabulary and must be unpacked into a query family, domain, side information, and tolerance.*

## **Query Preservation by Factorization**

Reminder — exact query preservation is an ordinary factorization condition: for each declared query  $q$ , exact preservation through  $\tau$  means there exists  $\bar{q}$  with  $q = \bar{q} \circ \tau$ , equivalently  $q$  is constant on every fiber of  $\tau$ . This is weaker than injectivity and stronger than saying “important information” survived. The canonical statements and proofs are F01–F04 in Part IX.

*Falsifier. One pair  $x, x'$  with  $\tau(x)=\tau(x')$  but  $q(x) \neq q(x')$  defeats exact preservation for that declared query.*



## CHAPTER 13

# Collapse, Compression, and the Limits of Emer- gence

*Does non-injectivity create emergence?*

Collapse is a many-to-one map. It can remove distinctions, reduce cost, stabilize a macro-description, and enable a new vocabulary. None of those consequences is equivalent to emergence. A defensible emergence claim must declare micro- and macro-languages, encodings, resource bounds, admissible algorithms, and invariance under reasonable recodings. Relative emergence then means that a macro-predicate is computable within the macro bound while no admitted micro procedure computes its pullback within the micro bound. The claim is relative to languages and resources, not absolute non-definability in every possible formalism. The boundary is informed by renormalization, coarse-graining, computational mechanics, and causal-emergence research, all of which require stronger structure than mere non-



injectivity (Anderson, 1972; Wilson, 1971; Crutchfield and Young, 1989; Shalizi and Crutchfield, 2001; Rosas et al., 2020; Mediano et al., 2022).

This correction matters because the early ROA corpus sometimes speaks as though non-injective collapse generically guarantees a macro-property not definable below. That is too strong without a language and resource contract. The revised architecture retains the insight - collapse can support macro-regularity - while replacing the generic theorem with a bounded, falsifiable definition.

Is compression wrong, or only undeclared compression?

Every summary is a collapse. So is every dashboard, every score, every model level, every executive one-pager, every traffic-light rating distilled from forty pages of findings. The opening chapter on representation made peace with this at the level of principle — representation is loss, and rightly; the chapter on transmissions and query-relative fidelity gave it structure — collapse is the non-injective transmission, the merging of states whose distinction carried information. This chapter

prices the everyday case: compression as the ordinary, indispensable, debt-generating act of institutional life.

Begin with what compression buys, because a chapter that only warned would be false to the argument. Compression makes work possible: no board reads every transcript and no engineer holds an entire state space. Collapse can make stable macro-descriptions and useful regularities available at a coarser level, but it does not by itself prove that those regularities are emergent, autonomous, or unavailable below. The trade is structural and cannot be refused, only declared: keep distinctions and pay in complexity; merge them and pay in lost queries, recovery cost, or downstream debt.

Collapse debt comes due at one specific moment: when distinctions that were compressed away are needed later but no longer visible. The heatmap that directed attention perfectly becomes dangerous the day it is cited as final proof; the coarse-grained model that made the simulation tractable becomes dangerous the day it is asked for micro-level explanation. Legitimacy has three conditions, and all three are declarations, not vir-

tues: the purpose of the collapse is declared; the lost distinctions are known and recorded; and the downstream use does not require them — or, where it might, the route back to detail is preserved. A collapse meeting all three is a paid abstraction. A collapse meeting none is a time bomb with a clean interface.

Collapse rarely travels alone, and its habitual companion is the fifth component of the vector. Label debt pairs with collapse debt so reliably that the coding manual treats them as dance partners: the compressed object receives a name more stable than the evidence supports. Watch the pairing work, because it is one mechanism, not two coincidences: pooling erases the internal variation — and the label then asserts the uniformity the pooling manufactured. Partial control, in two departments of five, per one interview collapses into a dashboard cell; the cell needs a short name; the name becomes “control exists.” Weak association, one cohort, unreplicated collapses into a slide; the slide says “cause.” Generated patch, sandbox-tested collapses into a ticket state; the state says “fixed.” In each case the collapse created the room

and the label moved in — and the reader downstream meets only the tenant, never the floor plan. This is why the remedy is never “avoid collapse” and never “avoid naming”: it is to record the pooling — what was merged, which caveats dropped, which uses require the drill-down — and to size the label to the record, not to the cell.

One paragraph now, owed since the chapter on undecidability across preserving and collapsing maps, that removes a comfort the previous pages might have left standing. It is tempting to read blocking as a purchase: collapse the channel, extinguish the undecidable question, enjoy the silence — a one-time payment for permanent relief. The corpus’s formal layer says otherwise, and the working translation is this: a limit blocked by one collapse can reappear after a later expansion. Silence bought by merging lasts exactly as long as the merging does. Re-introduce the distinctions — a finer log, a drill-down view, a de-aggregated feed, a richer model level grafted onto the coarse one — and the hard questions that the pooling silenced come back with them, undecidability included where the restored channel is critical. Insti-

tutions do this constantly and innocently: the executive dashboard is quietly wired back to transaction-level data “for the auditors,” and the organisation is once again, without noticing, hosting the questions the dashboard existed to retire. The ledger’s phrasing for the principle is short enough to keep: collapse is not amnesty. It is deferral with a visible price — and the price sheet must survive the collapse itself, which is exactly what the recorded-pooling discipline is for: the record of what was merged is also the early-warning register of what re-expansion will re-awaken.

This monograph, note, has already practised the discipline it is preaching, and the reader has seen it. The named-use gate performs a collapse — nine components must become one decision — and it performs it declaredly: the blocking-set rule is a merging whose purpose is stated (a decision must be reached), whose lost distinctions are known (the exact severities do not move the outcome once a threshold is crossed), and whose downstream need is preserved (every violating component’s identity survives, so every repair stays

addressable). One collapse, three conditions, all met in public. That is what a paid abstraction looks like from the inside.

So — is compression wrong, or only undeclared compression? Only undeclared: a declared collapse is how finite minds and finite institutions afford the world, and its debt is a fair price openly booked — while an undeclared one is a loan taken out in the reader's name, discovered at the worst moment, with interest, by whoever needed the distinction the interface no longer remembers losing.



## CHAPTER 14

# Diagnostics for Reticular Models: Exploratory, Not Validated

## Status and Limitations of the Original Reticular Metrics

The source corpus proposed a coefficient of collapse, an emergence index, and a reticular irreducibility index. These are retained as exploratory diagnostics, not validated measurement scales. A publishable metric must specify its scale type, invariances, estimator, error model, calibration data, and decision relevance. Until those obligations are met, the three indices may rank examples inside a frozen toy model but may not be compared across domains or cited as evidence of emergence. Their falsifier is straightforward: if independent coders or perturbation tests show that the ranking changes under representation choices that should be immaterial, the metric lacks construct validity.



***What should be measured before a reticulum is trusted?***

Useful diagnostics include fiber cardinality or entropy, distortion on declared query families, collision rates, contradiction rates, abstention coverage, dependency completeness, and the cost of reconstructing a source path. These are not one scalar. A transmission can have low geometric distortion and high authority debt; an epistemic head can be well calibrated and poorly sourced. ROA therefore treats metrics as a vector indexed by layer and intended use.

**Fiber distortion.** How much can a declared query vary inside one fiber? This measures query-relevant loss rather than generic compression.

**Collision profile.** Which distinctions are merged, at which transmission, and for which later uses do those collisions matter?

**Decoder success.** Can a declared query value—or, where required, a source state—be reconstructed from the target representation under the stated decoder contract?

**Transfer debt.** Which source assumptions fail, weaken, or remain untested at the destination horizon?

**Abstention recall.** Does the system return a typed non-answer on constructed hard cases that exceed support, authority, scope, time, or decidability?



**PART IV**

# **Compact Reticular Computability: Compil- ing Science into an Ef- fective Object**

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## CHAPTER 15

# When a Reticulum Becomes Executable

*CRC likewise does not propose a stronger Church-Turing thesis. It is a packaging discipline for a multi-level model: operators must be effective, the scheduler and resource policy must be declared, nontermination must have an honest terminal, and every claimed cross-level computation must be executable under the stated horizon. The broader research programme is constructive rather than foundational: ask whether enough of a scientific model can be compiled that its missing dependencies and approximations become inspectable.*

A **compact reticulum** is an effectively presented tuple

$$\text{CRC} = (\mathbf{R}, \Gamma, \sigma, \mathbf{B}, \text{Term}, \mathbf{H})$$

where  $\mathbf{R}$  is a typed reticulum,  $\Gamma$  a configuration space,  $\sigma$  an effective scheduler,  $\mathbf{B}$  a resource policy,  $\text{Term}$  a tagged terminal sum type, and  $\mathbf{H}$  the de-

clared horizon. A configuration records the current node, typed local value, evidence and governance state, immutable ledger prefix, policy identifier, and remaining budget. Compactness is operational: the evaluator has a finite or effectively enumerable description. It is not a claim that the represented domain is finite, complete, or metaphysically closed. The contract is continuous with operational semantics, type soundness, computability, termination, and model checking rather than a new computability class (Plotkin, 1981/2004; Wright and Felleisen, 1994; Clarke, Grumberg and Peled, 1999; Baier and Katoen, 2008). The terminal type is explicit:

$$\begin{aligned} \textit{Term}(Y) = & \textit{Answer}(Y) + \textit{Unknown} + \textit{Con-} \\ & \textit{tradiction} + \\ & \textit{OutOfHorizon} + \textit{Review} + \textit{Timeout} + \textit{Fail-} \\ & \textit{ure}. \end{aligned}$$

A CRC evaluator is **horizon-closed** when every in-scope execution either advances according to the declared transition relation or returns one of these terminals. This is routing closure, not truth

closure: the evaluator can finish honestly by saying that it cannot answer.

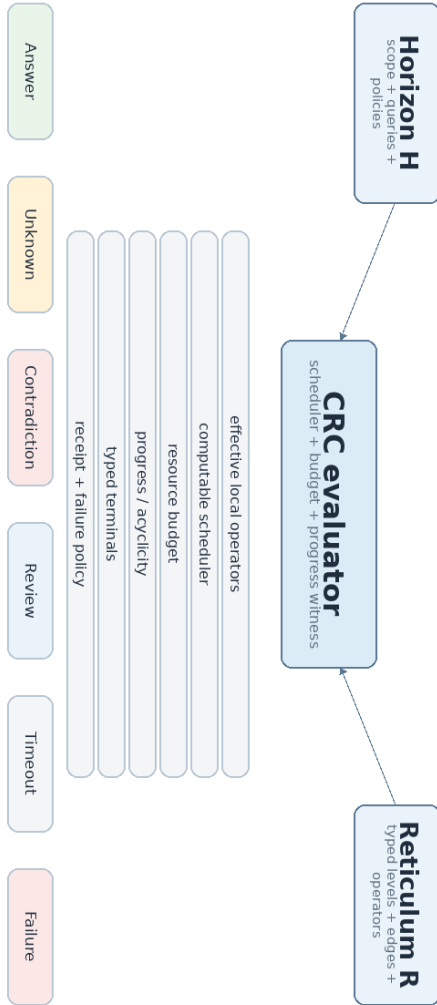


Figure 7. Compact Reticular Computability acceptance contract. A reticulum becomes an executable bounded evaluator only when operators, scheduling, resources, progress, typed terminals, receipts, and failure policy are explicit.





## CHAPTER 16

# Why Local Computability Does Not Guarantee Global Termination

## Original Compact Reticular Computability Criteria and Their Revised Status

Historical correction. The founding IOA / EC / TC triplet is retained only as genealogy: internal autonomy becomes explicit type/horizon closure; epistemic closure becomes typed terminal completeness; “Turing computability” becomes effectivity plus scheduling and resource obligations. The operational contract, not the labels, is the scientific object.

*Formal reminder. Under the typed transition contract, progress and preservation are local type properties; effective local operators plus a computable scheduler yield a partial evaluator; total terminalization additionally requires a budgeted dispatcher or another explicit well-founded progress argument. Full statements and proofs are F05–F06 in Part IX.*

## **Global Termination as a Separate Proof Obligation**

For finite acyclic ECU graphs, local termination plus finite scheduling is sufficient. Cyclic graphs require a global well-founded rank or an explicit timeout/budget terminal; local termination alone does not compose into global termination.

*Countermodel. Two individually terminating units that alternately re-enable each other can still generate an infinite global run.*

## CHAPTER 17

# What Formal Limits Survive Across Levels

Formal reminder — two boundaries. If an undecidable predicate factors through a computable transmission, the induced target predicate remains undecidable (F07). If a transmission maps two source states with different declared query values to the same target state, no target-only decoder can answer that query correctly on both (F08). Part IX carries the formal statements and proofs.

Interpretation. Collapse does not decide a hard problem; it can remove the distinctions needed to pose it from the collapsed representation. Re-expansion may restore both the distinction and the original limit. Mandatory abstention is justified only when the requested decision actually reduces to the preserved undecidable or information-blocked predicate under the declared horizon.

*Boundary. Undecidability propagates only when the relevant predicate is preserved; information loss blocks a query rather than solving it.*

## **Interlude - The Bryophyte Laboratory: Where Compilation Exposes Debt**

CRC becomes scientifically interesting only when it is forced to compile something that resists clean closure. The bryophyte case does that work. The source model assembles twenty abstraction levels and forty-two parameters from plant physiology and ecology, with update equations, causal relations, scenarios, thresholds, and a finite execution loop. That construction witnesses that a literature-derived reticulum can be made explicit and executable; it does not establish that the plant has been reproduced, that every parameter is identifiable, or that the selected literature is complete.

The important event is what happens at the seams. Scientific literature does not arrive as one interoperable program. A needed relation may be species-specific and directly measured; it may be reported only as a range; it may be known qualitatively but not as a calibrated function; it may have to cross taxon or scale; or it may be absent. A compiler - human or AI-assisted - must therefore

choose among leaving the gap open, transferring a relation, approximating a function, synthesizing a candidate relation, or narrowing the horizon. Each choice is epistemically legitimate only when its status travels with the resulting object.

*Approximation and Transfer Ledger. Every relation used by a compiled scientific reticulum should carry a profile rather than one mutually exclusive status: source and exact locator; source organism or population; original scale and conditions; mathematical form as reported; source\_basis  $\in \{\text{DIRECT}, \text{TRANSFERRED}, \text{SYNTHESIZED}, \text{UNRESOLVED}\}$ ; compiler\_ops = the finite set of declared transformations actually applied, which may include APPROXIMATED; calibration\_state; intended observable family; admitted use; and the test or evidence that would discharge the remaining obligation. UNRESOLVED belongs to this compilation ledger; UNKNOWN remains a typed epistemic terminal, and U remains the distinct unknown code of the DebtVector. The ledger makes a familiar modelling obligation*

*machine-addressable; it is a candidate instrument, not a new law of scientific method.*

*Qualitative obligation map. These fields do not impose uncalibrated severity floors. DIRECT supplies lineage information but never implies  $\text{debt}=0$  or validation. TRANSFERRED activates transfer, horizon/scale, provenance, and local-validation questions. APPROXIMATED in `compiler_ops` activates transformation, calibration, and validation questions. SYNTHESIZED activates trace, transformation, reification, and validation questions. UNRESOLVED records that the required scientific relation is not presently resolved by the admitted evidence and therefore cannot be silently compiled as established fact. The `DebtVector` is coded separately from these lineage/operation facts.*

This is the precise place where one class of AI hallucination becomes intelligible. A plausible but unsourced equation is not made safe by being numerically reasonable, and a transferred relation is not made bryophyte-specific by fluent prose. The defect is hidden status: a candidate bridge has



been reified as established science. Declared approximation remains a model choice with open debt; undeclared approximation is a provenance, transformation, label, and validation failure at once. The empirical target remains external. Synthetic trajectories may be compared with empirical trajectories or blind expert judgement only under preregistered observables, sampling conditions, thresholds, and baselines. Indistinguishability on one declared observable family would support adequacy only for that family; it would not establish biological equivalence, total model closure, or a privileged ontology. The compilation exercise has now exposed the key failure mode: missing science can be converted into a plausible candidate unless the transformation state survives. The next mandatory move in the genealogy is therefore epistemic convolution - a CNN-inspired semantic-transduction method for scanning, combining, collapsing, and stabilizing candidate objects while preserving typed non-answer and debt. The simulation bridge that follows is optional and may be skipped unless a cross-formalism claim is actually being made.



## CHAPTER 18

# Optional Simulation Bridge: What Must Be Proven Across Formalisms

A PCE bridge is not a resemblance claim. A one-way simulation contract declares source and target transition systems, an encoding, a simulation relation, a time/step-overhead map, source and target observables, a distortion bound, a complexity/overhead bound, and a finite horizon.

The obligation is observable preservation along related finite traces within the declared distortion and overhead. One-way simulation is not equivalence: equivalence additionally requires a reverse certificate or another explicit observational quotient, together with decoder adequacy and the relevant robustness/query obligations. The canonical finite-trace result is F10 in Part IX (Park 1981; Milner 1989; Sangiorgi 2011; Cook 2004).

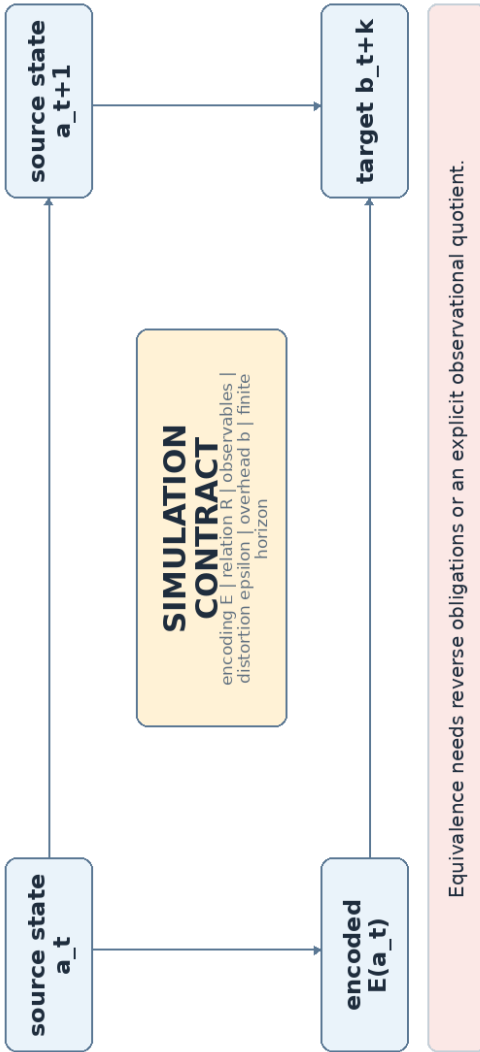


Figure 8. Bounded simulation certificate. One-way simulation requires encoding, relation, observables, distortion, overhead, and a finite trace horizon; equivalence requires additional reverse obligations.

## **Bounded Simulation Certificate: Encoding, Trace, Distortion, and Overhead**

Canonical certificate.  
 $C=(A,B,E,R,O\_A,O\_B,\varepsilon,b,H)$  records the models, encoding, relation, observables, distortion, overhead, and horizon. It certifies only the declared one-way finite-trace claim; it does not infer universal computational or physical equivalence.

*Falsifier. Find an admitted state, intervention, or time at which the relation fails, observable error exceeds  $\varepsilon$ , or overhead exceeds  $b$ .*





**PART V**

# **Epistemic Convolution and the Bounded Ob- server**

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## CHAPTER 19

# Epistemic Computational Units as Local Semantic Operators

### Epistemic Matrices, Deterministic Regimes, and Self-Audit

The ECU specification used the term epistemic matrix for the local schema of admissible inputs, transformations, terminal states, evidence requirements, and authority constraints. In this monograph the matrix is not a learned truth table; it is a versioned policy and type object. “Deterministic ECU regime” now refers only to the certified control plane: fixed input hashes, versioned operators, fixed scheduler, schema validation, policy evaluation, and receipt construction. A generative model call remains a candidate-producing operation unless bit-level replay has actually been established; temperature zero, a fixed seed, or a named checkpoint is not treated as a proof of deterministic output, and empirical nondeterminism is recorded rather than defined away

(Summers and Dinneen, 2021). Self-audit means replaying the unit against declared invariants, receipts, and challenge suites; it does not grant the unit authority to certify itself.

***What is the smallest governed semantic transducer?***

An Epistemic Computational Unit (ECU, or UCE in Italian) is a typed partial transducer. It consumes a local value, horizon fragment, evidence state, policy state, and resource budget; it emits a typed value plus an immutable receipt, or one of the declared terminal constructors. The unit is local: it does not own the global truth of the reticulum. Its contract states input and output types, admitted transformations, validation predicates, failure modes, and which evidence or debt components it may change. Its inherited formal neighbours are labelled transition systems, partial transducers, typed effects, provenance, and assurance arguments; the proposed delta is the receipt-bearing epistemic contract and its non-answer terminals (Plotkin, 1981; Belnap, 1977; W3C, 2013a–c; Bishop and Bloomfield, 1998).

This design separates model computation from epistemic standing. A language model may be an operator inside an ECU, but the ECU does not derive authority from the model. It derives its standing from the contract and the external objects it reads: source manifests, receipts, validations, policies, and review states. The unit is therefore implementable as a function, service, workflow, rule engine, checklist, or human-machine step.



## CHAPTER 20

# Epistemic Convolution: From Local Operators to a Navigable Semantic Re- ticulum

*Is an ECNN necessarily a neural network?*

An ECNN is a typed reticulum of ECUs, representation operators, a scheduler, an epistemic head, and an explicit state register. Early documents use a CNN analogy: convolution supplies local receptive fields, pooling supplies collapse, and an epistemic layer supplies unknown and contradiction. That analogy is historical and non-constitutive. The mature object is a deterministic navigation and validation substrate over a semantic reticulum whose nodes may be human-certified, AI-candidate, disputed, superseded, unknown, or contradictory. Neural models can propose nodes, links, labels, and summaries; they do not acquire certification authority by being inside the graph. Selective prediction, conformal risk

control, provenance graphs, and human review already provide major ingredients; the candidate contribution, if any, lies in their composition as deterministic semantic navigation with candidate/certified separation and cross-level query contracts (Chow, 1970; El-Yaniv and Wiener, 2010; Angelopoulos and Bates, 2023; Moreau and Groth, 2013). A conventional CNN is recovered when the reticulum is a deterministic feed-forward chain, all local operators are neural layers, and no epistemic terminal or governance channel is present. This special-case relation preserves the historical intuition without requiring the architecture to be neural. It also prevents a category error: an epistemic network cannot be validated by classification accuracy alone because its distinctive outputs include abstention, contradiction, and review routing.

## **Deterministic Semantic Navigation and Certification**

*How does the logical movement become a reproducible tool?*

Let  $N = \langle V, E, \lambda, \kappa, \sigma, P \rangle$ .  $V$  is a finite or effectively enumerable set of semantic objects;  $E$  is a set of

typed transmissions;  $\lambda$  assigns an abstraction level;  $\kappa$  assigns the epistemic state candidate\_AI, certified\_human, disputed, superseded, unknown, contradiction, or review-required;  $\sigma$  is a fixed scheduler; and P is the versioned policy and validation bundle. A navigation query returns a path, terminal, and receipt, not merely a string.

The logical movement is progressive. At level 0, sources and atoms enter with lineage. At level 1, AI or humans may propose candidate relations and abstractions. At level 2, deterministic checks evaluate type, schema, permitted transformation, query fidelity, contradiction, and required receipts. At level 3, an authorized human accepts, rejects, narrows, or requests evidence. At level 4, certified and candidate objects remain jointly navigable without being conflated. The reticulum can therefore grow while preserving the difference between what a machine suggested and what an accountable person signed.

Deterministic replay is conditional, not magical. If the source vector, node versions, deterministic operators, scheduler, budgets, and policy bundle are fixed, then the control-plane path and receipt



are reproducible. A stochastic proposal may still vary, but it enters as a new candidate with a new hash. This separation turns model variability from hidden drift into an observable event.

Scientific boundary. Knowledge graphs, workflow engines, typed transition systems, assurance cases, and human-in-the-loop review already provide most ingredients. The candidate ECNN contribution is their use as a multi-level epistemic navigation contract with non-answer terminals, candidate/certified state separation, query-relative preservation, and receipts on every transition. ECNN is only relabeling if those relations add no diagnostic or empirical value beyond a simpler typed graph. Agentic systems sharpen this boundary rather than creating a new epistemic layer: ReAct-style reasoning and action make environment interaction part of the execution surface (Yao et al., 2023); long-horizon context engineering shows that selective retention and summarization can improve tool-agent reliability while changing which state survives (Lodha et al., 2026); and safety-focused shadow memory targets threats that emerge only across extended trajectories (Wang, Jiang, et al., 2026). In ROA terms, context selection is a typed transmission or collapse, a tool call is an ECU transition with receipt and failure semantics, and persistent memory is a governed dependency whose omissions, poisoning, staleness, and authority remain inside the threat model.

## **Multi-reader insight — ECNN**

Computer scientist: a typed labelled transition system with a fixed replay contract. Engineer: a deterministic control plane around probabilistic proposal. Philosopher: an institutional distinction between conjecture and warranted acceptance. General reader: the map shows both pencil marks and signed roads, and never pretends they are the same.

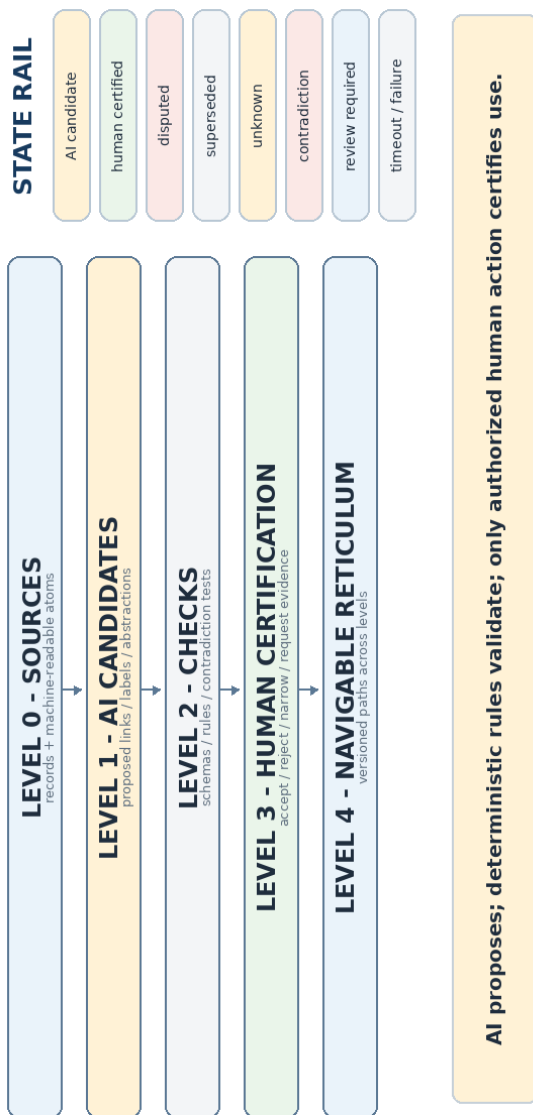


Figure 9. ECNN deterministic semantic navigation. AI may propose candidates, while deterministic checks, human certification, and a visible state rail preserve the distinction between proposal and accepted semantic structure.

## **Compositional Soundness of Epistemic Computational Unit Networks**

Let each ECU implement a typed partial relation from inputs and local context to a terminal plus receipt. Composition preserves type safety when every edge matches the producer codomain to the consumer domain and the scheduler never treats UNKNOWN, CONTRADICTION, REVIEW, TIMEOUT, or FAILURE as an ANSWER. It preserves evidential soundness only if receipts and dependencies are complete enough for the downstream use. The first property is local and machine-checkable; the second depends on dependency soundness and remains an empirical-governance problem.

*Hostile test. Delete or relabel a non-answer terminal on one edge. If a downstream unit can still emit an authoritative answer, the network has an epistemic type error even when ordinary program types pass.*



## CHAPTER 21

# Unknown, Contradiction, Review, Timeout, and Failure as Typed Outcomes

*Can not-knowing be a successful computation?*

The construction has a destination, and naming it clarifies the value of the whole programme. If knowledge is accumulated as a typed semantic reticulum with provenance, validation state, candidate/certified status, and debt attached to every object, then a system can answer a query by navigating that reticulum rather than regenerating an answer. The governed responder returns a path, terminal, receipt, and the status of every node traversed. Human-certified nodes can support uses allowed by their gate; AI-candidate nodes may guide search, hypothesis generation, or review but cannot silently inherit certification. Asked whether a supplier-payment control covers every excep-

tion path, the system reaches the control object, exposes the uncertified branch and open debt, and returns review-required rather than inventing closure. The model ceases to be the knowledge and becomes a replaceable navigator and proposer over a durable, inspectable semantic field.

## BRIDGE — THE FRAMEWORK APPLIED TO ITSELF

Selective prediction provides a formal ancestor for abstention: the system trades coverage for lower risk (Chow, 1970; El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017). ROA adds two dimensions. First, the reason for abstention is typed: insufficient evidence, contradiction, horizon failure, authority, resource exhaustion, or an undecidable promise problem. Second, the terminal travels with a receipt describing what would change it. This is not a claim that all uncertainty can be resolved into a finite taxonomy. It is a claim that the system must not collapse every unresolved condition into an answer token.

## CHAPTER 22

# Compiling the Bounded Observer

### Modal Hygiene, Epistemic History, and Metacognitive Control

Modal hygiene is the rule that constructible, observed, supported, warranted, and necessary claims remain different types. The observer compiler therefore persists an epistemic history: a versioned sequence of horizons, inputs, transformations, terminals, reviews, waivers, and revocations. Path dependence is operational when two identical current inputs can legitimately yield different outcomes because their histories contain different receipts, authority states, or unresolved contradictions. Metacognitive monitoring is used only in this engineering sense: the system measures its own resource state, coverage, conflicts, and policy triggers. It is neither phenomenal awareness nor privileged self-knowledge.



### ***How is a field turned into a governed reticulum?***

Observer compilation is the construction process that turns a bounded field into an explicit observer. The compiler identifies source classes, levels, state types, query families, transmissions, local operators, receipts, validation predicates, terminal states, and governance routes. It emits not only an executable graph but also a source manifest, type environment, invariant registry, claim-to-evidence map, failure policy, and rollback plan. The compiler is itself inside the audit horizon: its prompts, transformations, and human interventions are candidate artefacts, not invisible authorship. The selected established comparators are assurance cases, runtime monitors, provenance systems, knowledge representation, and end-to-end system contracts; ROA must outperform or clarify these neighbours rather than merely rename them (Bishop and Bloomfield, 1998; Bloomfield and Bishop, 2010; Saltzer, Reed and Clark, 1984; W3C, 2013a–c; Fagin et al., 1995).

The central move is to treat intermediate artefacts as first-class. Definitions, taxonomies, formu-

las, issue chains, graph fragments, and review notes are persisted and linked rather than discarded after a final answer. Cumulative intelligence then resides in the governed substrate and the human decisions that shape it, not in one transient model call. This is the difference between navigating knowledge and repeatedly improvising it.

What is gained by navigation over generation?

A governed oracle is not an omniscient oracle. It is a responder over a bounded reticulum. Given a query, it traverses typed relations, evaluates evidence and debt, invokes relevant ECUs, and returns a terminal plus a reconstructable path. Its distinctive value is persistence: the same source, receipt, transformation, and human decision can support later queries without being recreated. Its distinctive restraint is typed non-knowledge: when no admissible path supports a use, the response is unknown, contradiction, review, or out-of-horizon rather than invented closure.

Navigation does not eliminate generation. Models can still propose queries, candidate links, summaries, and hypotheses. The architecture changes their role. Generated material is a candi-

date input to controlled reification, not self-authenticating evidence. The model becomes a replaceable operator over a durable semantic field.

## CHAPTER 23

# Reticular Observer Architectures: Governance, Memory, Monitoring, and Human Authority

*Where does the system stop and hand the pen back?*

In normative domains the architecture adds meta-control over normative boundaries, modelled over a world model (facts, evidence, obligations), a self model (limits, calibration, drift, known gaps) and a normative kernel (rules, policies, constraints), with an epistemic history that makes the process path-dependent. The meta-controller consumes the governance signals the reification chain and debt propagation produce: it is where unsupported claims, missing evidence, modal drift and unresolved blocking debt are escalated to human authority rather than silently accepted. It does not decide moral truth and does not authorise itself —

its function is to make the boundary between the system's reach and human authority visible, not to move it. Read operationally, this is the sober core of what the broader work calls proto-consciousness: epistemic self-boundedness, the capacity to compute when a query crosses what the observer's ontology and horizon can support, and to emit structured non-knowledge. (Meta-controller: Annex G. A second, explicitly speculative pillar — topological proto-consciousness, prominent enough to name the main source paper's subtitle — is developed in the source corpus and Annex D; ROA deliberately excludes it from its claims while acknowledging its place in the programme.)

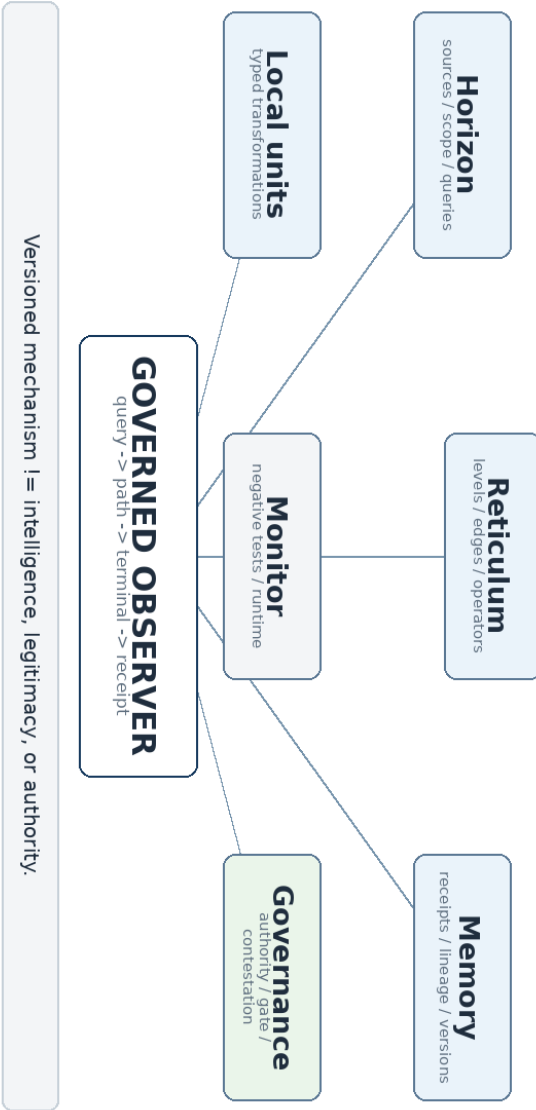


Figure 10. Reticular Observer Architecture as a versioned bounded observer. Horizon, reticulum, memory, local units, monitoring, and governance converge on a reconstructable query-path-terminal-receipt mechanism.

The iKant pattern compares a world model, a self-model, a normative kernel, and an epistemic history. The name is not a claim of moral agency. It denotes a circuit breaker that detects when a proposed operation exceeds evidence, scope, authority, or policy and routes the case to an accountable human. Normative rules remain authored, contestable, versioned, and institution-specific. The system can enforce a declared boundary; it cannot establish the moral legitimacy of the boundary by enforcing it.

Can the framework build and criticize its own artefacts?

There is an implementation witness that is neither pure theory nor an external validation: the framework was used to organise the construction of A-OSP through a staged AI-assisted workflow. The stages capture intent, produce engineerable blueprints, check them against declared constraints, decompose them into coherent subblueprints, generate a dependency-aware roadmap, and write issue bodies with definitions of done and review checks. Outputs at each stage are candidates until a human or deterministic control ac-

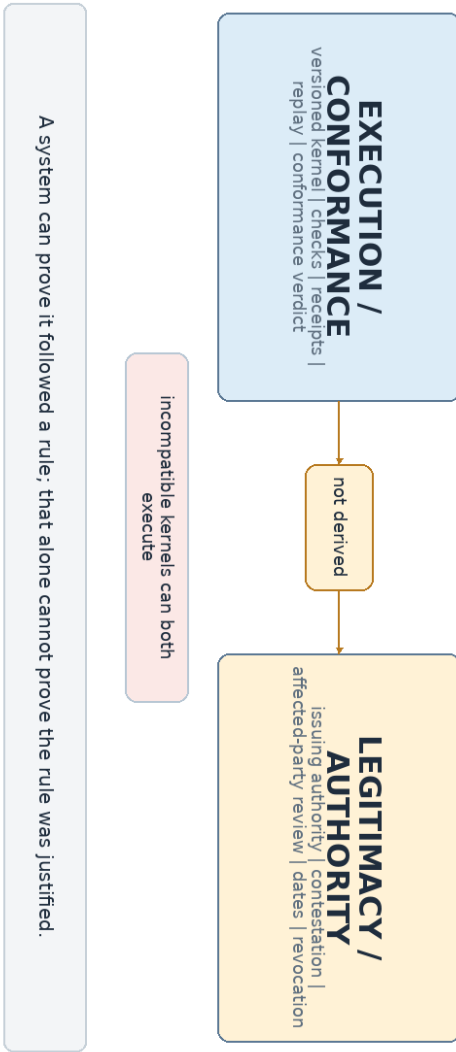
cepts them under the workflow. This witnesses method feasibility and the constructibility of typed intermediate artefacts; it does not establish effectiveness, efficiency, independence of error, or correctness of the underlying theory.

What this does and does not prove (efficiency caveat) It does not claim A-OSP was built in the most efficient way, nor that A-OSP will be effective in itself. The hypothesis is asymmetric and stated openly: robustness and enterprise-MVP effectiveness are candidate — A-OSP may attain them; efficiency is, by design, not the goal. Corollary, stated plainly: more efficient software suites may exist that do the same thing as A-OSP. The value claimed is governability, reconstructability and proof — not speed or economy. The self-application is therefore a witness of method feasibility — the framework is operable enough to have driven the construction of a non-trivial system — not a benchmark of effectiveness or efficiency.

Self-application is a feasibility witness and a threat. It shows that typed intermediate artefacts, checks, and issue chains can coordinate a large AI-



assisted build. It also creates correlated error: the same conceptual assumptions can shape the theory, implementation, and audit. ROA therefore treats self-application as evidence of constructibility only. Independent review, external baselines, negative tests, and empirical outcomes remain required.



*Figure 11. Execution does not establish legitimacy. Runtime conformance to a versioned normative kernel is distinct from external authority, contestability, affected-party review, effective dates, and revocation.*

## **Execution Does Not Establish Legitimacy**

A ROA can enforce a versioned normative kernel and prove that a run conformed to it. It cannot infer from conformance that the kernel was morally, legally, or democratically legitimate. Issuing authority, contestation, affected-party review, effective dates, and revocation are therefore external governance premises carried in the observer contract. This separation is essential for philosophers of science and engineers alike: an executable norm is a fact about a system; a justified norm is a claim in a social and legal order.

*Falsifier of overclaim. Construct two internally consistent kernels with incompatible prescriptions. Execution alone cannot choose which is legitimate without importing an authority relation not derived from the runtime.*



**PART VI**

# Epistemic Debt: Accounting for Unpaid Warrant

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## CHAPTER 24

# Controlled Reification: When Patterns Become Objects

*When does a pattern become a dangerous object?*

Reification is the promotion of a pattern, label, relation or abstraction into a manipulable object. In ordinary work this happens constantly: a repeated payment practice becomes a control, a cluster of symptoms becomes a case flag, a recurring code smell becomes a technical-debt issue, a narrative motif becomes a character trait. Reification is not wrong — knowledge advances by making objects that can be inspected, tested and reused.

The danger is uncontrolled reification: when a candidate object is treated as validated, when the conditions of its construction disappear, or when downstream systems rely on it without knowing its status. Controlled reification keeps the promotion traceable, status-bearing and reversible: it

records the source pattern, the operator, the new object, the distinctions lost, the status carried, the allowed and refused uses, and a rollback or discharge path.

Controlled reification means that the system is allowed to promote a sentence into an object only if the promotion carries its papers. The CFO sentence may become a control object, but it must carry its source, scope, extraction method, review state and allowed uses. Without those fields, the object becomes easier to reuse precisely because it has become harder to challenge.

Controlled reification is also a permission to work. The monograph is not asking institutions to stop naming things. It is asking them to name things with enough metadata that the name remains revisable. A provisional object is useful precisely because it can be reused without pretending to be final.



## CHAPTER 25

# Epistemic Debt: The Unpaid Remainder of Semantic Transformation

*Is debt the same as being wrong?*

Epistemic Debt is the unresolved warrant left by a semantic transformation and carried by the resulting object whether or not a use has yet been named. A debt-bearing object is not automatically false and not automatically useless. Its vector records source, trace, transformation, collapse, label, reification, validation, authority, and horizon-or-scale obligations under the current horizon; a later gate determines whether those obligations are tolerable, review-requiring, or blocking for a named use. A draft risk factor may therefore be usable for brainstorming and blocked for board reporting without its debt severity changing. The asset side records what was conserved and supported: source, scope, receipt, readback, validation, reviewer authority. The liability side records what

was lost or remains unjustified: missing source, unvalidated inference, overbroad label, scale transfer, modality drift, or premature reification. A transformation is honest when both sides are booked; it becomes dangerous when only the asset side is shown. A hallucination is a severe special case—a debt falsely presented as paid—but the theory also covers silent gaps and valid claims used outside their horizon. The claimed delta is a typed, componentwise, use-independent liability combined with a separate use-sensitive gate (Cunningham, 1992; Kruchten, Nord and Ozkaya, 2012; Sculley et al., 2015; IAASB, 2018; Moreau and Groth, 2013).

The liability is institutional because downstream users often meet only the polished object, not the missing warrant. A dashboard tile, a legal issue list, a scientific variable or a product risk score can all borrow authority from presentation. The ledger counters this by making the liability travel with the object until discharged, or until a gate logs a waiver for a named use.

Asset/liability reading of one transformation

Liability here is not moral blame but an unresolved obligation attached to the object, recorded before any use is named. That matters because organizations can manage obligations: they can assign owners, due dates, waivers, discharge evidence and escalation routes. The same cannot be done with a vague suspicion that “the AI might be wrong.”



## CHAPTER 26

# The Debt Vector: Why One Score Is Not Enough

*Why is one red flag not enough?*

Debt is not one thing. A claim can be well sourced but poorly scoped, well scoped but unvalidated, validated in one domain but transferred to another, or supported as a possibility but written as a certainty. A single red flag hides these distinctions; a vector preserves them. The proposed nine-component vector is a candidate coding instrument, not yet a validated or universal metric. Its ontology is part of the research hypothesis until independent componentwise reliability and incremental utility over simpler provenance/missing-data baselines are established. Its nine components are source, trace, transformation, collapse, label, reification, validation, authority, and horizon-or-scale debt. Each component should be coded with examples, thresholds and discharge conditions, so a reviewer can say not only that a claim is risky but why —

missing source, broken trace, invalid label, unreviewed authority, unsupported transfer, or overcompressed summary. Different debts demand different remedies, and the same total risk can hide very different repair paths. The vector also disciplines performative caution: “needs review” is too vague; the vector says which review, which dependency, which missing receipt, and which use is blocked until the problem is paid.

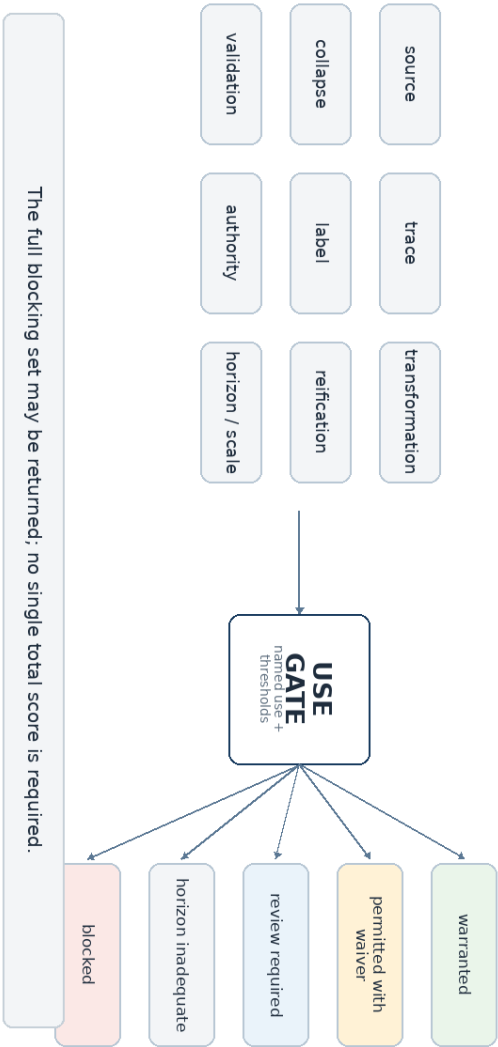


Figure 12. Debt vector and named-use gate. The nine debt components preserve distinct repair paths; the gate may return a full blocking set rather than collapsing the vector into one score.

The vector is learned by use rather than by list, and each component carries its own diagnostic question. Source debt asks the oldest question: can the origin be identified and admitted under the horizon? A board memo resting on “internal analysis” that no one can produce is not wrong yet; it is unpayable, which is worse. Trace debt asks whether the path from source to claim can be reconstructed. A true claim with a broken trace is a liability in a fact’s clothes: it survives every challenge except the first serious one.

Transformation debt asks whether meaning was changed by a declared, allowed operation. Summaries, translations, extractions and classifications all move meaning; undocumented movement is debt even when the destination happens to be right, because rightness can no longer be shown. Collapse debt asks whether distinctions were compressed away that a later use will need — the dashboard question, priced in full in the chapter on information collapse, compression, and emergence. Label debt asks whether the name exceeds its criteria: the five letters of “fixed” on a sandbox-



tested patch; the two words “control exists” over a partial practice in two departments.

Reification debt asks whether a candidate was promoted early — an object minted before its papers, easier to reuse precisely because it is harder to challenge. Validation debt asks the bluntest question in the set: has the required check actually run? Not “is it plausible”, not “does it read well” — did the sample get drawn, did the test execute, did it pass. Authority debt asks which authority is documented for the reviewer or source; the gate compares it with what a use requires. A brilliant unauthorized reading is still unauthorized, and in law, audit and medicine this component alone can decide the gate. Horizon-or-scale debt asks whether warrant earned in one place is being spent in another: one interview billed as a population, a lab constant billed as a field prediction, last year’s scope billed as this year’s.

Nine, and deliberately not one, because the components fail independently and the remedies do not transfer: a missing source is paid at a different counter than an overreaching label, and a validation gap cannot be discharged by a signature —

only postponed by one. This is also why the vector resists collapsing into a single score: the pooling would erase exactly the information that makes the debt payable — which counter. When a single decision is nonetheless required, the named-use gate defined in the chapter on proof chains, use gates, discharge, and waiver refuses to elect a worst component at all: it reports the blocking set, every component over its threshold, none privileged. A one-point exceedance in authority is not commensurable with a one-point exceedance in collapse, and electing a winner between them would be a collapse the ledger had not declared.

#### Debt vector — CFO approval example

The example below codes the running claim against one intended use — assurance of operating effectiveness — using the manual of Appendix B, which carries the severity scale, the counterexamples and the discharge conditions this chapter deliberately leaves to it: the walk above is for learning the components, the manual is for coding them. The Code column carries the declared 0-3 and U scale, not an ad-hoc adjective: a table that says partial where the manual says 2 is a reading, not a

coding. Every component appears, including the quiet ones: a vector that hides its zeros teaches the wrong habit.

The vector also improves disagreement. Reviewers may disagree about whether a claim is usable, but the vector lets them locate the disagreement: source sufficiency, scope, inference, validation, authority or scale. A disagreement that is located can be governed; an unevidenced yes/no fight usually cannot.

One red flag is not enough because debt is not an amount but an anatomy. The same total can conceal a missing source in one claim and an overreaching label in another, and the two are paid at different counters. The vector is the difference between knowing that a claim is risky and knowing what it owes, and only the second can be governed.



## CHAPTER 27

# Proof Chains and Use Gates: Discharge, Waiver, and Authorization

*What turns an output into something an institution may rely on?*

Within this architecture, an output becomes institutionally usable only by crossing the declared proof chain. A source becomes an atom; an atom receives a receipt; a receipt supports an artefact; an artefact becomes a witness; a witness enters governance. These transitions are proof-relevant boundaries: some transform or validate evidence, while only the final named-use gate determines permission. A generated paragraph is not proof, a log is not automatically a receipt, an export is not automatically a witness, a green interface is not automatically warrant, and model memory is not an epistemic authority. These false equivalences are common ways debt launders into authority.

A proof chain keeps the trust boundary outside the proposing model. The model may transform,

summarize, classify, or propose; the governed evidence substrate consists of inspectable sources, receipts, readback, validation predicates, dependency records, and authorization events. Those objects can still be incomplete, stale, forged, or institutionally illegitimate, so the architecture treats them as reviewable evidence rather than truth. The proof chain extends provenance and assurance-case disciplines by making evidence status, residual debt, named use, and waiver or discharge state explicit rather than treating lineage itself as permission (Buneman, Khanna and Tan, 2001; Cheney, Chiticariu and Tan, 2009; W3C, 2013a–c; Kelly and Weaver, 2004; Necula, 1997).

The chain should be simple enough for humans to inspect and strict enough for machines to enforce. Its positive side is proof capital: a source atom with scope, receipt, readback, validation and witness can serve many later registers — audit, legal review, compliance, environmental-social-governance review, and scientific provenance — so the ledger is not only a brake but a memory system for earned warrant.

A proof chain is a chain of typed, proof-bearing transformations that culminates in a named-use decision. Earlier links establish source, transformation, validation, or review facts; only the gate grants or refuses standing for the intended use.

### Minimal proof-chain object

Linear and reticular reading: the chain is linear when a single claim travels from source to decision. It becomes reticular when several claims share sources, receipts, transformations or governance decisions. That reticular reuse is proof capital, but only if each reuse preserves the status boundary of the original object.

### When is conditional use legitimate?

A gate compares proof status and debt against an intended use. The same object can pass a brainstorming gate, fail a board-reporting gate, and require a waiver for a deadline-sensitive preliminary use. That is not inconsistency; it is use-sensitive governance. Thresholds should be explicit: some debts are tolerable for low-risk exploration, some require review, and some block authoritative use. Authority debt is especially delicate: a model may propose a legal, medical, safety, or audit claim, but

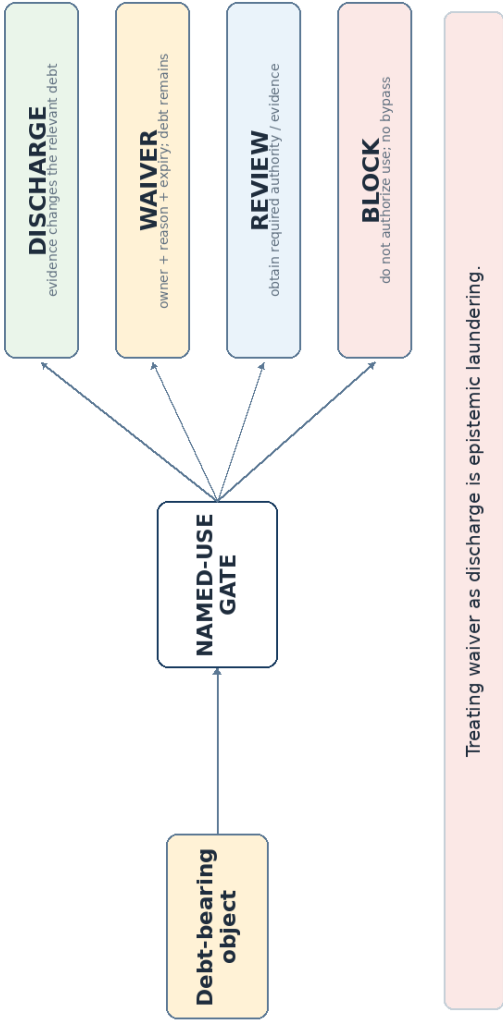
domain authority may be required before institutional use. Waiver and discharge must not be confused. Discharge pays a specified debt component only when the new event is admissible for that component and scope: evidence may pay source debt, a completed check may pay validation debt, contradiction resolution may pay a contradiction-related obligation, and an authorized review may pay authority or validation debt only where the adequacy contract makes that review the required evidence. A signature cannot discharge unrelated source, transformation, collapse, or label debt. Waiver does not pay; it records a bounded decision to proceed despite residual debt, with scope, reason, owner, expiry, and residual risk.

A gate is where the ledger stops being descriptive and becomes operational. A gate does not ask whether a claim sounds plausible. It asks whether the intended use is allowed under current status. If not, the path branches: discharge the debt, log a waiver with owner and expiry, narrow the use, or block the downstream action.

A waiver is not discharge. Discharge pays the debt by adding evidence or review. A waiver per-



mits a constrained use despite remaining debt, and therefore must carry owner, reason, expiry and residual risk. Treating waiver as discharge is one of the cleanest forms of laundering.



*Figure 13. Gate outcomes. Discharge changes the relevant debt component, waiver changes permission without paying debt, review seeks missing authority or evidence, and blocking refuses the named use.*

## CHAPTER 28

# Anti-False-Green: Preventing Unsupported Success

*Can the system fail honestly?*

The non-laundering invariant is simple: no unresolved threshold-level debt may become authoritative use without discharge or logged waiver. It is not mathematically deep; its importance is architectural. It prevents a system from converting unknowns, gaps and unverified transformations into institutional certainty by style, formatting or workflow convenience.

False green is the interface version of laundering: a modal says complete, a dashboard turns green, an issue is closed, a document is exported, or a model emits confident prose, but the proof predicates have not passed. It is dangerous because it creates the social appearance of payment without the epistemic payment. Anti-false-green design therefore requires negative tests — the system must demonstrate that it refuses green when a re-

ceipt is missing, when readback fails, when scope is absent, or when a critical debt touches the intended use. A proof-sensitive system should be willing to block itself for the right reason.

False green is the system-level form of epistemic laundering. It occurs when an interface, export, log, confidence score, or waiver is presented as if it were warranted use without the receipts and review state that would justify that status. The anti-false-green rule is strict: warranted green may be derived only from a current WARRANTED gate outcome and its receipts.

PERMITTED\_WITH\_WAIVER is a distinct, visibly qualified state carrying residual debt, owner, scope, reason, and expiry; it may permit action but must never visually or semantically masquerade as warrant.

The practical test is adversarial. A mature system should include negative tests that deliberately remove receipt, readback, source or reviewer state and verify that the interface cannot still present an authoritative green. Anti-false-green is a property of failure behavior, not of the happy path.

## **Epistemic Non-Laundering Under Complete Monitored Traces**

Formal reminder — F14. Under complete gate mediation, current evidence/debt/adequacy inputs, tamper-evident records, and a presentation rule that derives green only from WARRANTED, unresolved above-threshold debt cannot appear as warranted green. The result is conditional on complete mediation and trace integrity; missing dependency edges or bypasses violate the premises.

*Negative test. Remove a receipt, reviewer, adequacy contract, or dependency edge and verify that authoritative green disappears or becomes REVIEW.*

## **Reader checkpoint — from formal state to human consequence**

Computer science asks whether the transition was permitted; engineering asks whether the receipt can be replayed; philosophy asks whether the rule was justified; governance asks who accepted the residual risk. Epistemic Debt keeps these questions adjacent without pretending they are identical.

**PART VII**

# Propagation, Transfer, and Scientific Applica- tion

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## CHAPTER 29

# Three Kinds of Propagation: Formal Limits, Dependencies, and Policy

*What travels by necessity, what travels by dependency, and what changes by choice?*

The preceding sections built the ledger: a claim has status only under a horizon; a reification creates an object; debt is unpaid warrant; a gate prevents unresolved debt from becoming authority. Part VII asks a different question: once a debt or limit exists, how does it move? The answer is not one mechanism but three, and confusing them is one of the fastest ways to overclaim the theory.

Some things travel by necessity. Where a critical channel is preserved, a formal limit at one level induces a formal limit at another. No review, signature or governance ritual pays this kind of No. The only honest action is to expose the limit, change the channel, or narrow the question. This is the formal side of propagation and it is the part that remains owed as a stronger theorem wherever

the reticular conditions are stronger than the monograph proves here.

One term must be paid for locally, because the whole necessary tier hangs on it. A channel is critical for a question when the distinctions that question turns on survive transmission across it — when answering upstairs still requires telling apart states the channel keeps apart. Keep the distinctions and the limit below rides through with them; merge them and the question upstairs is not answered but blocked — a different fate with a different price, and the chapter on information collapse, compression, and emergence keeps its price sheet.

In miniature: whether one specific payment was approved survives a per-transaction log, and so do the hard cases beneath the question; it does not survive a monthly aggregate, where the question cannot even be asked. The aggregate solved nothing. It merged the states the question needed.

Other things travel by dependency. A conclusion depends on sources, labels, summaries, scores, reviewers and prior models. If an upstream dependency carries debt that touches a downstream

use, the debt travels with it unless it is discharged, waived, or made irrelevant by a recorded transformation. This is the payable kind of travel, and it is the subject of the chapter on debt propagation across dependencies. A third class travels by choice. A parameter, pattern, clause, risk score or model rule earned warrant under one horizon and is moved into another. The object moves; the warrant does not automatically move with it. This is transfer, and it creates the local debt treated in the chapter on warrant transfer across domains and scales.

Collapse sits between these classes. A non-injective transformation can block a formal question by merging distinctions; it can also create macro-regularities and usable abstractions. But a collapse is not amnesty. It is a price sheet: if the lost distinctions are needed later, the debt returns. The chapter on information collapse, compression, and emergence gives the everyday accounting of that price. The practical rule is simple: name the kind of travel before choosing the remedy. Necessity calls for boundary honesty; dependency calls for propagation and discharge; transfer calls

for a passport; collapse calls for a record of what was merged and what would be needed to drill back down. One word, propagation, cannot do all this work without laundering the differences between the cases.

This chapter is therefore a bridge, not a boast. It adopts a working distinction strong enough to guide the ledger and explicit enough to be tested, while leaving the deeper formal reduction as an owed item in Appendix D. The duality is useful only if it reduces confusion. Limits travel by critical structure; debt travels by dependency; transfer debt travels by choice. The first is a formal problem, the second a provenance problem, and the third a governance problem. Mixing them produces bad remedies: a waiver cannot dissolve a formal limit, and a theorem cannot repair a missing dependency edge. The policy tier is made explicit later as `PropagationPolicy` in the formal core: a declared join, unknown-value treatment, relevance rule, and receipt discipline rather than an implicit universal algebra.

So — what travels, and how? Limits travel by necessity where critical structure is preserved; debt

travels by dependency, and can be paid; warrant does not travel at all — objects move, and standing is re-earned at the destination or borrowed against it. Name the traveller before choosing the remedy: that is the operational content of the duality, and everything it still owes as a theorem is booked, unsoftened, in Appendix D.



## CHAPTER 30

# How Debt Travels - and Where the Model Can Miss It

*Can a clean-looking artefact be dirty upstream?*

The preceding chapter on necessary, dependency-based, and policy-driven propagation established what travels by necessity: where a critical channel is preserved, an undecidable question below induces an undecidable question above, and no evidence, review or signature can change that. This chapter descends to what travels by dependency — and the descent changes everything about the remedy. Necessity is irreducible; dependency is contingent: what it carries can be discharged, waived, or made irrelevant by the transformation itself. The previous chapter told the gate which No it was facing; this chapter tells the ledger how the payable kind moves.

Semantic artefacts are rarely alone. A conclusion depends on sources, intermediate atoms, la-

bels, summaries, scores, prior models, reviewer decisions — a lattice of borrowings, most of them invisible in the final prose. The propagation rule is one sentence: if an upstream object carries unresolved debt that touches a downstream use, the debt travels — unless it is discharged, waived, or made irrelevant by the transformation between them. All three exits are real and all three are recorded acts, never assumptions. Discharge pays upstream and the whole subtree breathes. A waiver crosses with papers, and the residual risk rides along, visible. And “made irrelevant” is the subtle one: a transformation can genuinely sever a dependency — a summary that drops the contested paragraph no longer inherits its source debt for uses the paragraph touched — but irrelevance is a claim about the transformation, and claims about transformations are booked in this ledger like any other, with the burden on the claimant. Say it plainly, because the instinct runs the other way: propagation is not the pathology. Propagation is the honesty. A summary that inherits its source’s open debt and shows it is a healthy object — usable, priced, repairable at the root. The pathology is



the opposite flow: the artefact that arrives downstream clean-looking, its upstream debts filtered out not by payment but by distance. Inheritance made visible is the ledger working; inheritance made invisible is laundering by hop-count. Which brings this chapter to its honest centre, and the monograph to one of its most consequential admissions. The propagation rule has a hidden premise: along dependencies. The ledger can only propagate debt along dependencies that are marked or discoverable. If a real dependency is missing from the record — the analyst reused last year's exclusion list and nobody wrote that down; the model was fine-tuned on the very corpus it now evaluates — then the downstream object is treated as clean because the road the debt would have travelled does not exist on the map. This is dependency soundness, and it is hereby declared what it is: the central open problem of the whole accounting layer — not solved by this monograph, not solvable by vocabulary, and not honest to hide behind the elegance of the rule it undermines. Provenance and dependency modelling are therefore not decorative metadata on this architec-

ture. They are its nervous system: the propagation rule is exactly as trustworthy as the dependency graph beneath it, and not one edge more.

What can be done, while the problem stays open, is to fail in the right direction. Four disciplines, none sufficient, all compounding. Conservative defaults: where dependency information is weak, the gate must not warrant a high-authority use when dependency information is weak — an artefact from a poorly-mapped region of the graph does not earn a heavy use merely because no debt is visible; invisibility of debt is not absence of debt, and the default says so. Included here, imported from the preceding chapter's third tier: cyclic arcs are debt-preserving by default — no loop may launder its own inheritance, no dashboard that steers a process may treat itself as cleaner than the process it steers. Reviewer correction: humans discover unmarked dependencies constantly — “wait, isn't this the same interview as...” — and the system's job is to make adding the edge one gesture, not a workflow. Graph inspection: dependency structure exposed as a first-class view, because an inspectable graph invites repair

and a buried one guarantees rot. And negative tests for missing edges: seed the corpus with known dependencies deliberately unmarked, and measure whether the workflow surfaces them — the empirical programme of Part VIII carries exactly this family.

So — can a clean-looking artefact be dirty upstream? Yes, in two ways that must never be confused: honestly, when its inherited debt is visible and priced, which is the system working; and dangerously, when an unrecorded edge let the debt vanish in transit — and since the second case is exactly as clean-looking as innocence, the ledger’s rule is that a map’s silence is a reason for caution, never a certificate of solvency.

### **Monotonic Debt Propagation and Its Premises**

Formal reminder — F12. Under the declared strict conservative profile, unresolved debt is monotone along a recorded use-relevant dependency path unless a typed discharge pays the component or a recorded transformation establishes irrelevance. The result is conditional on the relevance set and dependency graph; missing relevant edges can cre-

ate false cleanliness. Challenge sensitivity is not global dependency coverage.

*Falsifier. Seed a hidden relevant dependency and show that the workflow still warrants the downstream claim without surfacing uncertainty.*

## CHAPTER 31

# Transfer Across Domains and Scales: Objects Move, Warrant Does Not

*When is similarity not inheritance of warrant?*

The third traveller. The preceding chapter's limits move by necessity; the debt-propagation chapter's debts move by dependency; this chapter's objects move by choice — the deliberate carrying of an object from the horizon where it earned its standing to a horizon where it has not. Transfer is among the easiest ways to create hidden debt, precisely because it feels like the opposite: reuse reads as thrift, analogy reads as insight, and the object arrives with its old confidence still attached, like furniture shipped with the tags of the previous house.

The mechanism deserves one exact sentence: transfer moves the object but not the warrant, because warrant was never a property of the object

— it was a relation between the object and a horizon, and the horizon stayed home. A parameter, a legal clause, a risk score, a software pattern, a model rule: each earned its status under declared sources, a declared scope, declared validations. Move it, and the assumptions that funded the warrant may quietly fail in the target: the related species differs in exactly the trait the parameter compresses; the neighbouring jurisdiction shares the words but not the doctrine; the enterprise system shares the architecture but not the load. Nothing about the object changed. Everything about its backing did — and similarity of surface is precisely what hides it, because similarity is a claim about the objects, and warrant was never about the objects.

The special case with its own name is scale-transfer debt: warrant earned at one scale, spent at another. One interview supports an observation; it does not support a population claim — the sample was sized for the first sentence and is being billed for the second. A lab model supports a mechanism; it does not support field behaviour — the controls that made the mechanism visible are

exactly what the field removes. A unit test supports a function; it does not support system safety — composition, as the preceding propagation chapter already showed, is where the guarantees of the parts stop being guarantees of the whole. In each pair the first claim is paid and the second is on credit, and the drift between them is usually one adverb wide.

Now the ruling, stated with both hands, because this chapter is where a censorious misreading of the whole monograph would take root if allowed. The ledger does not forbid transfer. It must not: analogy is central to human and machine reasoning alike; the corpus itself distinguishes morphological analogy — surface resemblance, cheap and fertile — from structural transfer — the disciplined carrying of load-bearing structure — and builds with both. Science advances by borrowed parameters; law reasons by imported clauses; engineering is reuse with a conscience. A framework that taxed analogy into silence would be sterile scepticism with paperwork, the exact failure the chapter on epistemic status without ontological overreach explicitly rejects.

What the ledger demands is that transfer be visible, and visibility has a fixed anatomy — six fields, the passport of a travelling object: the origin horizon (where the warrant was earned, under what); the target horizon (where the object now proposes to work); the preserved structure (what genuinely carries over, and why one believes so); the changed assumptions (what the target does not share — the honest column, and the hardest to fill); the validation needed (what test, review or evidence would earn local standing); and the allowed use in the target, now (what the object may already support while that validation is pending).

The discipline then compresses into the two sentences this monograph asks every transferring analyst to be able to write — one of permission, one of debt: this object may be used in the target for X; it may not be used for Y until Z is validated. Anyone who can complete both sentences has done a controlled transfer. Anyone who can complete only the first has done an optimistic one — and the second sentence is where the ledger lives.

So — when is similarity not inheritance of warrant? Always: similarity is a fact about objects and



warrant is a relation to a horizon, so nothing inherits by resembling — an object earns its standing where it lands or it travels on credit; and the entire difference between borrowed insight and imported error is whether its passport's second sentence — not for Y until Z — was written before Y happened.

*Transfer passport. Every transfer records origin horizon, target horizon, preserved structure, changed assumptions, required validation, allowed use now, and blocked use until validation. For scientific-model compilation, the passport is paired with the Approximation and Transfer Ledger Contract in Appendix C so source basis, compiler operations, calibration state, and target-use restrictions remain independently inspectable.*





## CHAPTER 32

# Scientific Knowledge as a Reticular Warrant Structure

*How can AI help science without pretending to be science?*

Science is the hardest register for a book like this to enter without overclaiming, and so the entry is made under a strict rule, stated first: the observer is not a knower of the world; it is a governed structure for making the status of representations explicit. With that stake in the ground, the question becomes answerable. AI can help science not by doing science — not by paying the empirical debt only experiment can pay — but by compiling it: reading a domain reticularly, as levels, variables, equations, parameter ranges, validation criteria, failure modes and admitted sources, and making that structure computable, inspectable and falsifiable. The warrant stays human, empirical, revisable and domain-bound. What the ob-

server adds is not knowledge but bookkeeping of obligation.

Read this way, a scientific corpus becomes a governed evidence node: a bounded set of published claims, models, parameters, and validation obligations compiled into the observer's horizon. It is an evidence and model substrate rather than an epistemic authority, and it does not replace the field

it represents. Compiling it makes debts visible that fluent synthesis hides: literature-selection debt; parameter-harmonisation debt; equation-transfer debt; scale debt; model-closure debt; and empirical-validation debt, which only observation or experiment can discharge.

Everything in this chapter turns on one distinction, and it is the distinction most often lost precisely where it matters most: computability is not validation. A model can be internally computable — it runs, it terminates, it produces coherent, even beautiful outputs — and remain entirely unvalidated for the scientific purpose that matters. A simulation that halts is not a simulation that is true. The peril is that mathematical elegance reads

as warrant: a clean model, a smooth curve, an equation that closes, all wear the appearance of having earned something they have not. The observer's contribution is to keep that gap from being papered over by the aesthetics of the formalism — to hold, beside every computed trajectory, the open ledger of what it still owes the world. Put positively, and this is the sentence the chapter exists to deliver: a compiled reticulum does not answer the scientific question; it tells the experiment where to bite — it locates, precisely, the parameter that most needs measuring, the transfer that most needs testing, the closure that most needs relaxing. The observer is not a rival to the laboratory. It is the laboratory's map of its own most valuable ignorance.

So — how can AI help science without pretending to be science? By compiling its claims and obligations into a governed, falsifiable structure and then stopping — leaving the warrant exactly where it belongs, with evidence, with expertise, with the world — so that the machine's real gift to science is not an answer but a sharpened question, and the honesty to keep the two apart.

How can a scientific field become a computable object without becoming an ontology?

The central method treats a selected scientific corpus as the material from which a bounded observer is constructed. Published levels, variables, equations, parameter ranges, feedbacks, and validation criteria are harmonized into a compact reticulum that is explicit for the declared inventory, horizon-closed only in the routing sense that it answers or returns a typed non-answer, and executable only when its local rules and scheduler satisfy the CRC conditions. The product is not the domain and not the scientific community's total knowledge; it is an epistemic model of the selected literature whose synthetic consequences can be compared with observations and expert judgment. AI contributes candidate structure—taxonomies, reticula, hypotheses, and gap analyses—not truth extraction. The governed reticulum is therefore a durable evidence and model substrate, while sources, experiments, and accountable expert review remain external authorities. The horizon is negotiated from the corpus

and intended task; it is never simply “read off” nature.

The method is deliberately domain-general: select a bounded corpus, declare the horizon and admissible questions, type the scientific objects and transformations, preserve source and transformation receipts, expose unresolved relations rather than silently closing them, and generate explicit validation obligations for the laboratory or field. Constructibility shows only that this compilation can be represented and executed under the stated contract. The bryophyte case is used elsewhere for two narrower jobs: the laboratory interlude exposes approximation and transfer debt, while “External Validation of the Literature-Compiled Bryophyte Model” specifies the separate empirical burden.





## CHAPTER 33

# External Validation of the Literature-Compiled Bryophyte Model

*Can a literature-rich domain become an object to falsify rather than to trust?*

The earlier bryophyte interlude used the case for one purpose only: to show that compiling published science exposes transfers, approximations, and missing warrant. This chapter asks the separate empirical question. A literature-compiled bryophyte reticulum is a candidate scientific model, not the plant, not a synchronized surrogate, and not a live-coupled engineering digital twin. Its executability is already established by construction; its scientific adequacy remains entirely open until compared with observations under an external validation design.

The right claim is deliberately modest. The reticulum can generate synthetic trajectories, and those trajectories can in principle be compared with empirical observations or blind expert

judgement. The preregistered target is epistemic indistinguishability: under a fixed set of observables, sampling conditions, thresholds, and expert tasks, blind reviewers may fail to distinguish selected synthetic trajectories from empirical ones. This is a target, not a result. It is scientifically informative only if the observables and threshold are fixed before inspection, trivial baselines are included, experts are appropriately sampled, and failure modes such as mode collapse, data leakage, and selective reporting are tested. A favourable result on one observable family would not establish biological equivalence or validity outside that family.

The gloss is a warning against two seductions: a reticulum is not true because the literature was mined, and not valid because a machine helped organise it. Neither the provenance of the inputs nor the sophistication of the assembly discharges empirical debt. The model's warrant depends on source quality, honest parameter reconciliation, explicit equations, expert review, and empirical discrimination against the world. Its value is not that it might be indistinguishable by construction;

it is that it makes the conditions of discrimination explicit and testable, converting “the literature says roughly this” into a structure a blind expert can try to break.

So — can a literature-rich domain become an object to falsify rather than to trust? That is the wager of the case: not a moss to believe, but a moss-shaped structure built so that an expert can try to catch it failing. Until that test is run, the correct status of the literature-compiled model is neither validated nor refuted; it is a falsifiable candidate awaiting external discrimination.

Bryophyte witness boundary



## CHAPTER 34

# Domain Translation: Compliance, Software, Law, and Creative Work

*Why is compliance the natural first home for the ledger?*

The scene that opened this monograph was a compliance scene, and not by accident. Of all the worlds the ledger might enter, compliance already thinks in its grammar: it cares, natively and professionally, about evidence, scope, responsibility and review. An auditor who has never heard the word reification nonetheless refuses to accept a control on its say-so, asks which document supports it, notes who signed off, and flags what was never tested. The ledger does not teach compliance a new morality. It gives an existing morality a type system — and that is why compliance is the wedge: the least translation, the fastest recognition, the shortest distance between this monograph’s vocabulary and a working desk.

Watch a single control claim decompose, because the decomposition is the argument. “Payments above the threshold require CFO approval” is not a sentence in this world; it is a chain: a source interview, a policy document, an extracted atom, a control object, a gap where the exception procedure should be, a score, a reviewer’s judgement, an audit memo, a governance decision. The proof-chain chapter’s stations are not imposed on compliance from outside — they are what a competent control assessment already is, drawn out and made explicit. And the claim’s status is nakedly horizon-relative, exactly as the declared-horizon analysis requires. Under a narrow horizon — two interviews admitted — the claim may sit at supported-not-validated. Under a stronger one — a horizon that admits only written policy, sampled transactions, exception handling and reviewer acceptance — the very same sentence stays supported-not-validated with three components unpaid, and the gate refuses the use called assurance of operating effectiveness. One sentence, two contracts, two statuses, no change in the world: the auditor’s instinct, formalised.

The value that makes the ledger worth its friction, here, is reuse — proof capital from the proof-chain chapter, spending itself across a landscape of overlapping regimes. One well-made evidence object — the sampled-transactions receipt, the reviewer's logged acceptance — can serve internal audit this quarter, the organisational-liability review next, the privacy-accountability file, the cyber-resilience assessment, the AI-governance register, the supplier-assurance questionnaire. It should not be recreated for each: what travels is the evidence object — sources, receipts, transformations, debt vector. Its status is recomputed under each regime's horizon and its standing re-evaluated at that regime's gate. The object moves; the warrant is re-earned where it lands. This is the compliance officer's chronic grief — the same evidence rebuilt six times for six frameworks — answered by the monograph's central mechanism: warrant that carries its own papers is warrant that can be presented, unaltered, to every desk that shares the standard.

And one reframing that changes the emotional weather of the whole function. In a compliance



culture, a gap is an embarrassment to be managed; in the ledger, a gap is knowledge. A missing exception procedure is not a failure to be smoothed in the memo — it is a located, typed, addressable object: a reification debt open on one branch, awaiting a document, a sample, a signature. It can be governed, remediated, waived with an owner and an expiry, or escalated. What it can no longer do is hide inside a green tile. The ledger does not make compliance harder; it makes compliance's findings honest — and a found gap, openly booked, is worth more to an institution than a smooth surface concealing it.

So — why is compliance the natural first home for the ledger? Because compliance was already keeping the ledger's books by hand, in prose, under deadline, and losing the papers between frameworks; the ledger only asks it to keep them in type — and hands back, in exchange, evidence that no longer evaporates at the border of the next audit.

Compliance translation table

Does the ledger generalize, or impose one formula on every domain?

A framework that fit only compliance and science would be a specialism with pretensions. This chapter tests the ledger's reach by turning it on three worlds that price warrant in visibly different currencies — and the finding is the one the monograph needs: the structure generalises; the meaning of warrant is local, and stays local. The ledger supplies the common grammar — source, scope, transformation, receipt, review, use — and each register fills that grammar with its own evidence and its own authority. The lawyer still decides legal adequacy; the engineer, release readiness; the scientist, validation; the author, canon. The ledger does not annex expertise. It gives experts of every kind a shared way to say what stands behind this, and for what use — and then gets out of the way.

Walk the three, and watch the same six-station chain speak three languages. In software, warrant comes from tests, review, runtime evidence, security checks and rollback paths; a receipt is a passing CI run with a readback of what was actually deployed, and false green has a literal cousin — the pipeline that reports success while a check was skipped. In law, warrant comes from jurisdiction,

authority, facts, qualified review and procedural posture; a source atom is a statute or a holding, and authority debt is not a nuance but the whole ballgame — an unreviewed reading is not advice, however fluent, and the eighth component of the vector governs the register. In creative work, warrant shifts register entirely and does not thereby vanish: it may mean internal coherence, declared draft status, attribution, or consistency with an established canon; a “fact” about a fictional world is warranted by the author’s acceptance, and a generated lore summary is a draft until the pen that owns the world says otherwise. That these are so different is the point: warrant is not one thing wearing four hats; it is four things sharing one skeleton.

One caution closes the chapter, aimed at a failure the framework could commit rather than one it detects. The ledger’s generality is a genuine asset and a standing temptation: because the grammar fits everywhere, it can be imposed everywhere, flattening the expert judgement it was meant to serve — a compliance-shaped template stamped onto a poem, a software gate bolted onto a diag-

nosis. The discipline is the reverse: the ledger adapts its evidence and authority to the register while holding its structure constant. Same skeleton, native flesh. A framework that forgot this would have committed, at the scale of whole domains, the exact sin of uncontrolled reification it spends the monograph policing — promoting one register’s notion of warrant into the notion, and using it downstream as if the promotion had been earned.

The generality of the ledger depends on localizing authority. In software, authority may mean tests, review and deployment evidence. In law, it may mean jurisdiction, facts and qualified advice. In creative work, it may mean canon acceptance by the author. The ledger carries the same skeleton while refusing to impose one register’s warrant on another.

So — does the ledger generalise, or impose one formula? It generalises by holding the structure fixed and letting the substance stay native — a common grammar for source, scope and use, spoken in each world’s own evidence — because a discipline that imposed one register’s warrant on

every domain would not have unified anything; it would merely have reified compliance, and called the flattening a theory.



**PART VIII**

# **Falsification, Comparators, and Open Scientific Debt**

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## CHAPTER 35

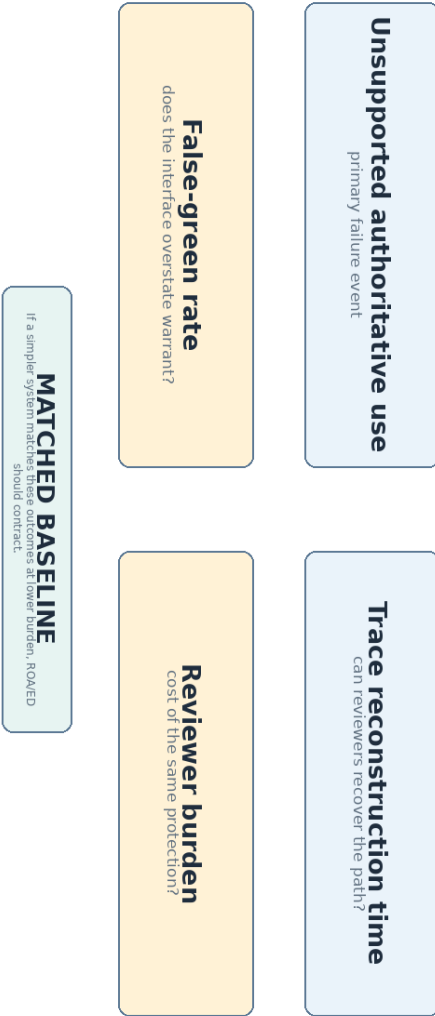
# How to Attack the Architecture: Severe Challenge Suites

*How can an epistemic architecture be attacked rather than admired?*

A Popper-style challenge suite should contain contradictory pairs, unanswerable or out-of-horizon queries, semantic adversaries, missing-receipt cases, stale-policy cases, authority conflicts, hidden dependencies, and deceptive green interfaces. Classical accuracy remains relevant but insufficient. Evaluation should report coverage, selective risk, contradiction recall, abstention precision and recall, debt-detection agreement, trace reconstruction, reviewer effort, and bypass behaviour.



Figure 14. Severe-test programme. Workflow comparison, reticular fidelity, typed terminals, and governance are attacked with explicit failure probes; simpler baselines are allowed to win.



*Figure 15. Empirical comparison endpoints. Unsupported authoritative use, reconstruction time, false-green rate, and reviewer burden are matched-baseline outcomes that can support or defeat the practical claim.*



## CHAPTER 36

# What Would Count as Empirical Improvement?

*What would show that the ledger changes real work?*

The theory becomes stronger when it says how it could be wrong. If a debt ledger does not improve detection of unsupported use, reduce false green, cut reconstruction time, improve reviewer agreement, or improve downstream decisions, then its practical value is weaker than claimed. Hallucination rate is relevant but insufficient as the primary metric: a system can avoid lexical hallucination while still producing unsupported authoritative use, and it can produce a factually wrong but properly blocked candidate without creating an authorized failure. The primary endpoint therefore targets the event the architecture claims to prevent: unsupported authoritative use under a declared horizon and governance contract.

Hallucination rate deserves one more sentence of demotion, because it is the metric everyone

reaches for first. A workflow can drive hallucination to zero and launder happily: every sentence sourced, every source real, and the whole assembled into an authority the evidence never earned. Conversely, a workflow can hallucinate occasionally and stay honest, if the ledger catches the fabrication before use. The target variable was never falsehood in the output; it is unsupported authority in the use — and only the second is what institutions actually pay for.

A publication-ready ledger therefore carries its own readiness states: implemented, witnessed, target, open, and not-a-claim. These states do not make the work weaker. They make the strongest claims easier to see because they prevent every weaker claim from being silently promoted.

Measurement begins with separation. A study should not ask whether the framework is “good” in the abstract. It should ask whether typed horizons, receipts and debt gates reduce specific failure modes: unsupported downstream use, reconstruction time, false-green events, unreviewed authority upgrades and hidden waivers.

Measurement targets

A negative result would still be valuable. If the ledger does not reduce unsupported downstream use, if it merely increases review burden, or if users bypass it under deadline, the framework learns exactly where its operational debt lies. Falsifiability is not an ornament of Part VIII; it is the price of making the monograph more than a vocabulary.

So — what would show that the ledger changes real work? Fewer claims used past their status, cheaper reconstruction, greens that always have receipts behind them, and disagreements that land on nameable components — measured against simpler workflows that are allowed to win. Anything less specific is the theory grading its own homework.

What is the smallest experiment the theory could lose?

The first serious test should not try to validate the whole philosophy. It should compare workflows on a bounded task: prompt-only, retrieval-based, manual graph, and typed proof-chain with a debt ledger. A preregistered protocol freezes the corpus, tasks, reviewers, metrics and failure criteria before results are seen, and includes adversarial



cases: missing evidence, conflicting sources, scope shifts, misleading summaries, invalid labels, unreviewed authority and false-green temptations.

For bookkeeping across the corpus, the empirical programme runs as two declared tracks. Track A is the workflow comparison of this chapter: bounded tasks, preregistered endpoints, the ledger against three baselines. Track B is the scientific case developed in the chapter on the bryophyte model as a falsifiable literature-compiled epistemic model: the bryophyte literature-compiled model's blind expert comparison. The tracks share the discipline and nothing else — A tests whether the accounting pays; B tests whether a compiled reticulum survives an expert trying to break it.

The ledger can win in several ways — fewer unsupported authoritative claims, better trace completeness, lower reconstruction time, more correct abstention, clearer reviewer agreement on why a claim is blocked. It can also lose, through excessive friction, false blocks or confusing classifications. The point of preregistration is not to protect the theory from failure but to make failure informa-

tive: if the ledger fails, the debt vector, thresholds, interface or coding manual must be revised.

A credible comparison should include at least three baselines: prompt-only workflow, retrieval-augmented workflow and untyped evidence workflow. The ledger condition wins only if it reduces unsupported authoritative use or reconstruction cost without merely shifting all work into excessive manual review.

### Minimal preregistration sketch

The comparison should also measure human trust calibration. A system that blocks too often may be safe but unusable; a system that allows too often may be fluent but dangerous. The useful region is between those failures: fewer unsupported authoritative uses, acceptable review time, and clearer escalation.

Abstention must be audited in both directions. Report ordinary error, selective risk, coverage, erroneous-abstention rate, subgroup-conditional calibration, deferral burden, and the downstream fate of deferred cases. A workflow fails the no-laundering objective when it merely moves difficult cases into an opaque review queue, or when a

globally acceptable coverage-risk curve hides a subgroup that is systematically over-deferred or under-protected.

What is the smallest experiment the architecture can lose?

Use matched high-stakes knowledge tasks in at least three heterogeneous domains, for example scientific synthesis, software assurance, and compliance. Randomize cases or teams to four conditions: prompt-only; provenance and documentation; calibrated/selective abstention; and ROA/ED with typed status, dual-order debt, readback, and gate. Preregister tasks, exclusion rules, power or precision targets, adjudication, and cost weights. Reviewers blind to condition code the claim, evidence envelope, debt components, gate outcome, and actual use event.

The primary endpoint is unsupported authoritative use, adjudicated independently of the intervention debt ontology: an authoritative-use event judged by a preregistered adequacy charter to exceed the evidence, scope, validation, authority, waiver, or transfer conditions required for that use. The adequacy charter is authored separately

from the intervention ontology; the source pack is fixed independently of model condition; adjudicators are blinded to condition; disagreement is reported; and sensitivity to alternative adequacy contracts is analysed. Framework debt codes may be predictors or explanatory variables, but they do not define the endpoint they are supposed to improve. Secondary endpoints include false-green rate, reconstruction time and completeness, componentwise inter-rater reliability, selective risk and coverage, review burden, latency, bypass, rollback cost, privacy burden, dependency-coverage error, and cross-domain transfer error.

A cost-sensitive net benefit is

$$\text{NB} = \text{Delta\_Harm} - (\text{C\_reconstruction} + \text{C\_governance} + \text{C\_latency} + \text{C\_staffing} + \text{C\_privacy} + \text{C\_bypass})$$

ROA/ED is not operationally supported when  $\text{NB} \leq 0$ , when debt coding is not sufficiently reliable for the declared use, or when a simpler baseline dominates the coverage-risk-cost frontier.

H1. ROA/ED reduces unsupported authoritative use relative to all baselines.

H2. Debt components achieve preregistered inter-rater reliability after bounded training, or components with poor reliability are removed or redefined.

H3. Proof-chain plus readback reduces reconstruction time and missing derivation hops.

H4. Typed review and abstention reduce high-impact errors without unacceptable coverage loss.

H5. Debt state adds predictive value for review, rework, rollback, bypass, or false-green events beyond missing-data, provenance, and confidence indicators.

H6. Controlled reification reduces label drift and correction cost without unacceptable operational burden.

H7. Net benefit remains positive under realistic staffing, privacy, throughput, and latency constraints; safety benefit does not count as net benefit when achieved by unstable queue growth, unacceptable latency, systematic bypass, or review load that scales faster than governed artefact volume.

Reviewer-capacity measurements. Report arrival rate, reviewer service time, queue length, review latency, bypass rate, false-block rate, unsafe-pass rate, and total human minutes per governed artefact. These are failure dimensions, not optional usability metrics.

H8. Transfer passports reduce cross-domain authority errors without eliminating useful reuse.

The obligations are stated because stating them is part of the discipline. The reduction theorem F07 already establishes propagation of an undecidable predicate when it factors through a computable transmission; what remains open are richer extensions involving feedback, changing horizons, stochastic channels, or partially observed dependencies whenever the architecture wishes to claim them. The nine-component debt vector is specified, but its coding reliability, thresholds, predictive validity, and component interactions require separate empirical protocols and matched baselines. The bryophyte model remains a design target awaiting preregistered blind expert comparison. Complementarity with neural-symbolic systems is argued rather than demon-

strated: a reproducible bounded comparison should report task performance, selective risk, abstention, contradiction detection, provenance preservation, debt-coding reliability, reconstruction effort, and review burden. The Augmented Ontological Semantic Platform remains an engineering witness; its D1 receipt/readback seam can supply negative-test evidence, while the broader Epistemic Query Language proof seam and end-to-end proof spine remain open in the corpus evidence.

Falsification must be level-specific. The architecture-level practical claim is materially weakened or rejected if repeated preregistered, matched studies across heterogeneous domains show no net reduction in unsupported authoritative use or reconstructability relative to simpler baselines, or if the operational burden is consistently worse on the coverage-risk-cost frontier. The debt instrument is weakened if independent coders cannot apply its components reliably or if the vector adds no predictive or diagnostic value beyond simpler provenance and missing-data indicators. The bryophyte claim is falsified locally if qualified experts

distinguish synthetic from empirical trajectories beyond the preregistered criterion on the declared observables. Failure of one domain target does not logically refute every formal result; it refutes the claim at the level where the evidence was promised.

The primary comparison should pit prompt-only, retrieval-augmented, untyped evidence-graph, and ROA/ED workflows against the same bounded tasks and source packs. Independent reviewers should assess unsupported authoritative use, false-green rate, reconstruction time, abstention correctness, contradiction detection, debt-coding agreement, reviewer effort, and bypass. The architecture loses if a simpler workflow achieves equal or better epistemic outcomes at lower burden, or if users routinely bypass the machinery under realistic deadlines.

Frontier stress condition. On genuinely agentic tasks, add a matched tool-using baseline with ordinary logs and memory, plus a state/context-managed agent without Epistemic Debt or a named-use gate, so any ROA/ED advantage cannot be attributed merely to better orchestration.



Where model-side uncertainty is available, include semantic-uncertainty or selective-prediction baselines; where deployment assurance is the target, include an explicit safety-case workflow. Agent-trace and execution-provenance systems should also be treated as direct reconstructability baselines rather than as evidence for ROA by resemblance (Wang, Zhang, et al., 2026). The architecture loses when these matched alternatives reproduce its reconstruction, abstention, error localization, and authorization protection at lower burden.

## CHAPTER 37

# Differential Contribution Against Selected Research Traditions

*Which established traditions already solve parts of the problem?*

Scientific representations are selective and purpose-relative rather than literal duplicates of their targets (Giere, 1988; Hacking, 1983; Frigg & Nguyen, 2020; Weisberg, 2013). Information theory and information-flow semantics distinguish message capacity from meaning and distributed constraint (Shannon, 1948; Barwise & Seligman, 1997; Floridi, 2011). Abstract interpretation gives sound approximation relations between concrete and abstract program domains (Cousot & Cousot, 1977). Blackwell comparison orders experiments by decision-relevant informativeness (Blackwell, 1953), information bottleneck methods preserve predictive relevance under compression (Tishby et al., 1999), and causal ab-

straction studies when low- and high-level causal models agree under interventions (Pearl, 2009; Rubenstein et al., 2017; Beckers & Halpern, 2019; Beckers et al., 2020). Work on emergence, coarse-graining, and universality also shows that macro-level usefulness, critical behavior, and computational simulation require stronger premises than many-to-one encoding (Anderson, 1972; Wilson, 1971; Batterman, 2002; Humphreys, 2016; Wolfram, 1984, 2002; Cook, 2004). These traditions justify query- or intervention-relative fidelity. They do not determine whether a produced semantic object has adequate evidence, governance authority, or a valid downstream use.

Database provenance distinguishes why-, where-, and how-provenance and supplies formal derivation semantics (Buneman et al., 2001; Cheney et al., 2009; Moreau & Groth, 2013). W3C PROV standardizes entities, activities, and agents (W3C, 2013a, 2013b). Reproducible research requires executable methods, inspectable data, and rerunnable artifacts (Donoho, 2010; Nosek et al., 2015; Munafò et al., 2017). Distributed-systems design further distinguishes an

acknowledgement from the end-to-end property that the user actually requires (Saltzer et al., 1984). These results motivate lineage, receipts, and read-back. They do not by themselves establish that a representation preserved the distinctions required by a question or that an institution may rely on it.

Selective classification formalizes reject options and coverage-risk tradeoffs (Chow, 1970; El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017). Conformal prediction provides finite-sample coverage under explicit exchangeability assumptions (Vovk et al., 2005; Angelopoulos & Bates, 2023). Calibration, ensembles, semantic entropy, and self-consistency estimate different uncertainty properties (Guo et al., 2017; Lakshminarayanan et al., 2017; Farquhar et al., 2024; Manakul et al., 2023). Surveys of hallucination and abstention underline that detection, uncertainty, and refusal are distinct engineering problems rather than interchangeable confidence thresholds (Ji et al., 2023; Wen et al., 2025). These methods can guide abstention. None turns confidence into provenance, validation, authority, or warrant.

Hoare logic, type soundness, model checking, verified compilation, proof-carrying code, process semantics, and temporal specification establish properties relative to formal models and trusted assumptions (Hoare, 1969; Wright & Felleisen, 1994; Clarke et al., 1999; Leroy, 2009; Nacula, 1997; Milner, 1989; Lamport, 2002). Belief revision, paraconsistent logic, and abstract argumentation make contradiction and defeasible support explicit rather than forcing premature consistency (Alchourrón et al., 1985; Belnap, 1977; Priest, 2002; Dung, 1995). Assurance cases connect claims, arguments, and evidence for safety-critical systems (Bishop & Bloomfield, 1998; Kelly & Weaver, 2004; Bloomfield & Bishop, 2010; Leveson, 2011). These traditions support explicit proof contracts and trusted boundaries. They do not automatically track representational loss or the residual warrant created when semantic objects are reified and transferred.

Measurement theory warns that numerical operations require justified scales and representation theorems (Stevens, 1946; Krantz et al., 1971). Classification and commensuration reshape social

worlds rather than merely describe them (Porter, 1995; Bowker & Star, 1999; Espeland & Stevens, 1998). Social epistemology and epistemic-injustice research show that authority, exclusion, and credibility are institutional variables (Longino, 1990; Fricker, 2007). Human-factors research shows that automation can redistribute error and induce inappropriate trust (Bainbridge, 1983; Parasuraman & Riley, 1997; Lee & See, 2004). These results motivate controlled reification, contestability, and explicit authority.

Datasheets, model cards, factsheets, internal algorithmic auditing, and AI risk frameworks improve documentation and accountability (Mitchell et al., 2019; Arnold et al., 2019; Gebru et al., 2021; Raji et al., 2020; NIST, 2023). Work on sociotechnical abstraction and data cascades shows why system boundaries and organizational context matter (Selbst et al., 2019; Sambasivan et al., 2021; Suresh & Gutttag, 2021). Software-engineering studies and the technical-debt literature show how hidden dependencies, deferred obligations, and AI-assisted development create long-lived operational cost (Cunningham, 1992;

Kruchten et al., 2012; Sculley et al., 2015; Amershi et al., 2019). Neural-network practice supplies powerful transformation mechanisms but no automatic warrant bridge from learned representation to authorized use (LeCun et al., 1998; Goodfellow et al., 2016). These practices are compatible with ROA/ED, but they usually do not supply a single typed semantics connecting query preservation, evidence status, debt propagation, and use authorization.

*Comparative map | Adjacent traditions close different proof obligations. Filled circles indicate a primary strength, not exclusivity. ROA/ED contributes the typed seam across preservation, derivation, uncertainty, assurance, and authorized use; it does not replace the underlying fields.*

The closest alternatives can each model part of the pipeline. Abstract interpretation can certify a sound approximation but not institutional authority. Provenance can reconstruct derivation but not query adequacy. Selective prediction can reject uncertain predictions but not distinguish waiver from discharge. Assurance cases can structure a claim and evidence argument but do not ordinarily propagate representation-relative debt through semantic transformations. Governance

frameworks can specify roles and controls but rarely give a query-factorization semantics or a typed non-launders invariant. The comparison is anchored in abstract interpretation (Cousot & Cousot, 1977), data provenance (Buneman et al., 2001; W3C, 2013), selective prediction (Chow, 1970; El-Yaniv & Wiener, 2010), assurance cases (Bloomfield & Bishop, 2010; Kelly & Weaver, 2004), and AI risk governance (NIST, 2023; Autio et al., 2024).

ROA/ED has a candidate differential contribution only to the extent that the following conjunction is absent from a comparator: (i) query-relative transmission contracts; (ii) evidence polarity and validation separated from confidence; (iii) residual warrant represented as a non-scalar, partially assessed ledger; (iv) append-only reification and readback; and (v) a versioned, use-specific authorization gate. A future system that already satisfies this conjunction would be an implementation of the same architectural class, whether or not it uses the ROA/ED vocabulary.

The 2024 NIST Generative AI Profile strengthens the lifecycle and risk-management context.



Contemporary abstention research shows that refusal remains difficult even for strong reasoning models, while semantic entropy and related uncertainty measures remain model-side signals rather than provenance or authorization. Advanced-AI safety-case work is now a direct governance comparator: structured deployment arguments make claims, evidence, defeaters, and confidence explicit, while confidence in top-level frontier-AI safety claims remains difficult to quantify reproducibly (Clymer et al., 2024; Barrett et al., 2025). Tool-using and long-horizon agents add a second current comparator because planning, memory, context selection, tool outputs, and actions move epistemic state through an environment; recent execution-provenance work increasingly treats evidence tracing, memory lineage, tool-use provenance, audit, and recovery as first-class concerns (Wang, Zhang, et al., 2026). Work on artificial epistemic agents likewise foregrounds competence, falsifiability, provenance, and socio-epistemic infrastructure (Marchal et al., 2026). ROA does not replace these traditions. A safety case can be a governance artefact consumed by a named-use gate;

an uncertainty estimator can be an ECU signal; an agent framework can instantiate the scheduler and tools; a provenance system can supply dependency evidence. The residual ROA/ED claim survives only where typed preservation, residual warrant, debt-preserving waiver/discharge, and use authorization add discriminating value beyond those stronger neighbours.

What is the narrowest defensible contribution?

No individual ingredient is new. The contribution is a typed composition in which query fidelity, evidence, debt, reification, and use authority retain separate semantics while participating in one artifact and one decision procedure. The strongest mathematical contribution is the use of factorization and recoverability as the exact meaning of purpose-relative preservation. The strongest architectural contribution is a trace safety invariant with explicit trusted assumptions. The strongest methodological contribution is to make operational superiority falsifiable against simpler baselines.

Source gaps, transformation gaps, validation gaps, and authority gaps have different repair

routes and governance owners. A universal sum would imply exchange rates that have not been justified. The dual-order interval product records both severity bounds and assessment precision. A domain may validate a local scalar utility or threshold policy, but that is a derived instrument under a versioned governance contract, not a universal measure of knowledge.

A horizon states which sources, transformations, queries, validation rules, and governance contracts support an internal status. This limits the claim; it does not imply that all horizons are equally adequate or that external truth is created by declaration. Horizons can be criticized by empirical failure, contradiction, missing sources, invalid scope, unethical authority, or comparison with a stronger observer. Observer relativity is therefore a condition of representation, not a denial of reality.

Waivers, discharge authority, adequacy contracts, and consequential uses are institutional acts. They can encode power and exclusion. A conforming system must therefore preserve author, scope, purpose, expiry, contestation route,

supersession, and residual risk. Formal transparency does not itself confer legitimacy; it makes the locus of human judgment visible and auditable.

The strongest candidate contribution is architectural and compositional. RLA supplies query-relative transmission contracts across multiple representational levels. CRC requires typed, terminating or honestly failing execution over the reticulum. ECU/ECNN makes unknown, contradiction, review, timeout, and failure local operational outcomes. ROA compiles these into a versioned observer. Epistemic Debt adds a persistent unpaid-warrant state and a use gate. The PCE bridge prevents universality language from becoming metaphor. None of these ingredients is wholly unprecedented; the claim is the interface discipline that prevents one register from impersonating another.



## CHAPTER 38

# Open Scientific Debt and Conditions of Defeat

*What must be true outside the formal core before ROA earns scientific or operational standing?*

The indexed formal statements are heterogeneous and premise-sensitive. Their premises define a trusted computational envelope; they do not certify sources, policies, monitors, institutions or the external world. A severe review must therefore distinguish proof inside the calculus from validity of the calculus for a deployment. ROA fails scientifically if it hides that bridge.

Internal validity is threatened when a theorem encodes the desired conclusion in a transition rule, mediation premise, or monitor assumption. The by-construction safety results F12–F14 and the terminal-preservation result F24 are useful conformance guarantees, but they are not independent evidence that a policy is adequate, that dependencies are complete, or that every real use

surface is mediated. Construct validity is threatened when query fidelity, evidence status, debt, confidence, and truth are measured by one proxy. External validity is threatened when a reticulum, threshold, or debt ontology is moved across domains without a transfer passport and local validation.

Provenance completeness is the central information-science risk: debt propagates only along recorded or discoverable dependencies. Security is the central systems risk: an unmonitored bypass can invalidate the non-laundering guarantee while leaving the internal trace perfectly safe. Human governance is the central institutional risk: a transparent waiver can still be illegitimate, coercive or systematically biased. Performance and adoption are the central engineering risks: a system that is formally honest but too costly or cumbersome may be bypassed in practice.

A conforming implementation must therefore ship more than code. It requires a versioned schema, a reference interpreter or executable semantics, a frozen conformance corpus, positive and negative tests, a threat model, monitored use sur-

faces, receipt readback, policy and horizon identifiers, a reproducible build, and a benchmark that measures both epistemic benefit and operational burden. Absence of any one item must be visible as an open obligation rather than inferred away from a green interface.

Severe-review verdict.

The ROA/Epistemic Debt architecture is formally coherent relative to its declared signatures and assumptions, but its scientific standing remains conditional on public conformance artefacts, independent mathematical review, and empirical evaluation. This monograph is therefore releasable as a theoretical and formal-methodological research monograph, not as evidence of empirical superiority, production assurance, calibrated governance, or universal observer completeness.

What does the monograph itself still owe?

The debt ontology may be too coarse, too fine, or domain-dependent. Ordinal scales may not support stable coding. The evidence-status rules can be poor even when applied consistently. Append-only lineage can conflict with privacy, eras-



ure, and confidentiality. The trusted boundary may omit a use surface. A proof-rich interface can induce false confidence. Any concrete PCE-related bridge remains unestablished until a specific source/target simulation contract, observables, distortion, and complexity/overhead analysis are supplied; the generic F10 contract does not itself establish such a bridge. The empirical protocol has not yet been executed. Finally, the monograph gives a sufficient architecture for the stated safety property, not a uniqueness theorem.

Constructs are classified by burden of proof rather than judged by one evidential standard. Computable semantics is an organizing thesis about bounded status computation, not truth. Science-as-node and the literature-compiled epistemic model are methods for compiling a selected corpus into a testable model, not demonstrations of domain adequacy. Epistemic indistinguishability is a preregisterable prediction. The governed oracle is a design vision with an implementation witness. Self-application is a feasibility witness, not a benchmark. Reification and debt propagation are the central architectural contribution.

Propagation of undecidability is accepted only under explicit reduction hypotheses. Mandatory abstention is a typed terminal policy; blocking debt is a governance outcome. Each construct is therefore assessed against its own stated proof, test, or falsifier.

Four scope clarifications remain explicit. Epistemic convolution is a CNN-inspired local-operator analogy over semantic fields, not a requirement for a classical convolutional neural network. An Epistemic Computational Unit is a bounded typed transducer operating under a declared matrix, horizon fragment, and execution envelope. Speculative proto-consciousness or topological-consciousness claims are excluded from the accepted scientific theory; operational self-boundedness means only that the system can represent when a query exceeds its declared supportable horizon. The Augmented Ontological Semantic Platform is an implementation witness: its documented proof seams can test constructibility and negative failure behaviour, but they are not independent validation of the architecture.

Does the monograph apply its own rule to itself?

The ledger should be applied to the monograph. Some claims are argued, some are indexed formal statements of different classes, some are inherited from the ROA corpus, some are empirical targets, and some remain open. Maturity honesty is not a weakness of the project; it is the project following its own rule. The principal formal debt is now proof-assistant mechanisation and extension of selected invariants to richer dynamic or partially observed dependency structures where such claims are desired. The principal scientific debt is empirical and authority-sensitive: dependency completeness, debt-component reliability, horizon and threshold calibration, benchmark benefit, and cross-domain validity. The principal implementation debt concerns monitored mediation, security, privacy, and whether receipts, readback, validation, and governance remain usable at scale. The principal governance debt concerns legitimacy, authority independence, waiver abuse, contestation, and adoption.

A theory of debt gains credibility precisely when it refuses to launder its own debt into certainty. The reader should never have to guess whether a sentence is a result, a proposal, a witness, a target or an open problem — and Appendix D states, without softening, which parts of the theory and evidence programme remain only partially paid.

The theory owes what it asks others to owe: evidence. It owes calibration of the debt vector, independent reliability testing of the approximation/transfer ledger, tests of dependency and relevance-set soundness, empirical comparisons of workflow outcomes, independent review of A-OSP's proof seams, and scientific trials of the bryophyte literature-compiled model. Naming these debts is not a weakness of the monograph; it is the monograph obeying itself.

Open debts of the monograph

This chapter is the internal audit. It prevents the monograph from presenting its own architecture as if architecture were evidence. The monograph earns credibility when it distinguishes argued thesis, design invariant, engineering wit-

ness, empirical target and open problem - the same distinctions it asks every downstream system to carry.

Each item is open for its own reason, and the reasons carry more instruction than the list. Dependency soundness is hard because it is a claim about what is not recorded — no amount of care inside the graph can certify the edges that never entered it, which is why its discharge path runs through negative tests and conservative defaults rather than through diligence. Debt-vector calibration is hard because inter-rater agreement is an empirical property, not a design property: the coding manual can be sharpened at a desk; whether two reviewers land on the same vector can only be measured. The workflow payoff is hard for the most uncomfortable reason: the ledger must beat simpler disciplines that are allowed to win, and the preregistration design in the chapter on measurement, baselines, and preregistered comparison exists precisely so that a loss would be legible.

The bryophyte comparison is hard because it needs qualified experts willing to be fooled on the record, under a threshold declared in advance.

The proof-grade seam of the engineering witness is hard because proof-grade is a phrase this corpus refuses to spend before the receipts exist — the phrase is the debt. And horizon adequacy, the register's newest entry, is hard because it asks for thresholds — which uses require which minimum horizons — and thresholds are exactly the kind of object that must be calibrated in the field, not declared in a book.

So — does the monograph apply its own rule to itself? It tries to, in public and incompletely: representative load-bearing claims and every programme component carry a state, Appendix D says which — the register is a maturity register at component level, not yet a claim ledger sentence by sentence, and saying otherwise would be the first thing this chapter is against — and the answer to this chapter's question is itself typed — argued, yes; demonstrated, not yet; the difference booked where a reader can check it.



## CHAPTER 39

# A-OSP as an Engineering Witness, Not Scientific Validation

## Engineering Vocabulary and Evidential Limits of the Augmented Ontological Semantic Platform

The Augmented Ontological Semantic Platform (A-OSP) calls its append-only text events stripes, its scoped retrieval surface the Epistemic Query Language (EQL), and its five-stage intake-to-kernel pipeline D1–D5. An Artifact Witness is a reader-facing map from an artefact to its source, receipt, readback, and governance state. Error Truth is the interface discipline that preserves failure, timeout, fallback, and partial-capability states instead of converting them into empty success. These names are implementation vocabulary, not additional laws of the observer architecture. Their scientific relevance is as a constructibility witness and as a source of negative tests: remove a receipt, corrupt readback, revoke authority, or



force a fallback, and verify that no authoritative green survives.

***What does a running system prove — and what can it never prove?***

A theory of warrant that never touched a running system would be right to be suspicious of itself. But a theory that pointed at a running system and said “see, it works, therefore the theory is true” would be committing, in public and at the very end, the exact fallacy the preceding sections named. This chapter walks the narrow path between those two failures. The source corpus contains a reference build — an append-only, text-first, browser-native environment for AI-assisted work — and the whole task here is to say, with the monograph’s own discipline turned inward, precisely what that build does and does not entitle the monograph to claim.

The mechanism is one exact distinction: a running implementation can witness constructibility of a specified mechanism, but it does not establish the truth, usefulness, optimality, or adoption of the surrounding theory. The corpus evidence documents a text-first source-of-truth architecture

and, for D1 intake, a dedicated proof route with receipt persistence, read-after-write verification, canonical hashing, and negative tests. That is evidence that these separations can be engineered on the documented seam. The same corpus keeps the Epistemic Query Language proof receipt/routing and the broader proof spine open; those planned surfaces cannot be spent as completed evidence. Constructibility is therefore local to the implemented configuration, and every claim beyond it carries a separate evidential burden.

What the witness actually shows is not a feature list but a refusal. In ordinary workflows interface state, logs, exports, and model memory can be mistaken for proof. The documented D1 seam holds a narrower set of distinctions apart: generic append is not proof PASS; a proof-sensitive success requires a typed receipt, readback confirmation, and explicit scope; readback failure emits failure rather than green. This is receipt-backed provenance and fail-closed proof-state handling, not a claim of cryptographic provenance for every surface. The Epistemic Query Language seam remains a target in the corpus, so the witness is

strongest when it reports exactly which separations are runtime-wired and which remain open.

There is a second, subtler witness, and it must be reported with its limits welded on or not reported at all. The framework was used to build its own reference implementation — a multi-stage agent pipeline that turned blueprints into issues into working code under a documented method. This witnesses method feasibility: that a non-traditional, AI-orchestrated construction path can produce a real system of real scale. It witnesses nothing about efficiency and nothing about effectiveness. The honest corollary is the corpus's own, repeated here without softening: more efficient suites almost certainly exist that would do the same thing, and the pipeline's existence is a proof of feasibility, never a benchmark of optimality. To let feasibility read as superiority would be to reify a build into a boast.

The strongest formulation is also the narrowest: the system witnesses that the separations can be engineered. It does not prove that users will adopt them, that costs will fall, that the architecture is optimal, or that the theory is true. A running sys-

tem is a witness to constructibility and a temptation to overclaim, and this chapter exists to keep both facts visible.

So — what does a running system prove, and what can it never prove? It proves that the thing is buildable, and that the conflation the monograph spent itself forbidding can be held apart in real code — and it can never, merely by running, prove that the discipline pays, that the design is best, or that the theory behind it is true. The reference build is the monograph's existence proof and its standing temptation in a single object: the moment it is cited as evidence rather than as witness, the ledger has laundered its own foundations. Which is why the discipline that keeps the build honest is precisely the discipline it was built to demonstrate — applied, last of all and hardest of all, to the demonstration itself.



## CHAPTER 40

# Human Authority and the Final Authorization of Use

*Who signs, when the machine can write everything?*

Return to where the monograph began: a civilization filling with confident pictures it did not pause to judge. The preceding chapters of typed objects, gates, debts, and receipts can look, from the far end, like an elaborate machine for tidiness. They are not. They serve a declared normative governance commitment: consequential standing should remain attributable to an accountable human or institutionally constituted human authority, even when computation prepares the evidence and executes pre-authorized policy. That commitment is not a theorem of computability; its legitimacy must be defended in the affected institution and remains contestable.

State the point at its highest level, one last time. Under the governance stance adopted here, the

architecture leaves the consequential act of warrant attributable to accountable human or institutional authority. An automated system may execute a policy that was validly delegated, but execution does not create the authority, legitimacy, or accountability of that delegation. The declared horizon, visible debt, and abstention machinery prepare a legible decision object; they do not manufacture the standing to bind people or institutions.

The corpus gives this human-in-command pattern a name — iKant — and the name is a discipline against two opposite errors. It is not a moral subject: it holds no values of its own, only a normative kernel that some human wrote. It is not a metaphysical oracle: it does not know what is right; it detects where a case has turned hard, norm-laden, or self-contradictory, and escalates it. Mechanically it is a circuit-breaker over four things kept deliberately apart — a model of the world, a model of itself, a normative kernel, and an epistemic history — and its highest function is not to answer but to recognise the answers it must not give alone. A meta-controller that quietly de-

cided the hard cases would be exactly the silent crossing the monograph forbids, wearing a governance badge. iKant's dignity is the opposite: it is a mechanism whose finest act is to stop and pass the pen.

Recall the little word that has haunted the argument from the chapter on epistemic status without ontological overreach — “must” — and the four claims it hides stacked in a line: that something is constructible, that it is stable for an observer, that it is scientifically objective, and that it exists necessarily in itself. Both an ambitious metaphysics and a confident machine can slide from the first of these to the last without ever marking the passage, and the monograph has been, in one long sense, a grammar for marking it: for forbidding the silent crossing and demanding, at the last and most tempting border, an explicit warrant or an honest abstention. Keeping the pen is what that grammar is for. It is the insistence that the signature under the final step — this claim may responsibly support this use — is a human signature, applied in daylight, answerable to those it binds.



So — who signs, when the machine can write everything? Under this monograph's governance commitment, an accountable human or institutionally constituted human authority remains answerable for consequential authorization, even when machines generate the candidate artefact or execute a pre-authorized rule. The claim is not that a person declares truth by signature; it is that the authorization of a claim for a named use must remain attributable, reviewable, and contestable. Machines may fill pages and execute workflows. The authority relation that makes consequential use legitimate is an external governance premise, not an output of computation.



**PART IX**

# **Formal Core: Definitions, Results, and Proof Boundaries**

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## CHAPTER 41

# Typed Domains, Epistemic Horizons, and Reticula

The formal core is intentionally local. It does not attempt a universal semantics of science or natural language. It fixes a finite or effectively presented operational domain and proves only what follows inside that domain.

A **level** is a triple

$$L_i = (X_i, \Sigma_i, Q_i),$$

where  $X_i$  is a state set,  $\Sigma_i$  is a family of admitted partial effective operations or transition relations, and  $Q_i$  is a declared family of queries. A deterministic transmission is a typed map  $\tau: X \rightarrow Y$ ; a stochastic transmission is a Markov kernel  $K: X \rightsquigarrow Y$ . Side information, tolerances, and randomness policies are part of the transmission contract and may not be inferred silently.

A **horizon** is the versioned tuple

$$H = (S, \Omega, Q, T, R, M, V, \Pi)$$

where  $S$  is the admitted source manifest,  $\Omega$  the scope,  $Q$  the query family,  $T$  the allowed trans-

formations, R the required receipts, M the support and contradiction rules, V the validation procedures, and  $\Pi$  the priority and conflict policy. Resource budgets and scheduling belong to the evaluator contract rather than to H. The tuple is a contract for a computation, not a claim about all reality.

For a query family  $Q$  on  $X$ , define

$$x \equiv_Q x' \Leftrightarrow \forall q \in Q, q(x) = q(x').$$

For approximate queries, equality is replaced by a declared metric and tolerance. A deterministic transmission  $\tau: X \rightarrow Y$  is  **$Q$ -faithful** when

$$\tau(x) = \tau(x') \Rightarrow x \equiv_Q x'.$$

This term replaces the historically useful but ambiguous phrase *quasi-injective*. It states exactly which distinctions are preserved and does not suggest global recoverability.

## CHAPTER 42

# Representation Theorems and Recovery Conditions

### Credited standard result — Query Factorization Theorem (F01)

Let  $\tau: X \rightarrow Y$  and let  $Q$  be a family of queries  $q: X \rightarrow A\_q$ . Define  $\tau$  to be  $Q$ -faithful when  $\tau(x) = \tau(x')$  implies  $q(x) = q(x')$  for every  $q \in Q$ . The following are equivalent: (1)  $\tau$  is  $Q$ -faithful; (2) for every  $q \in Q$  there exists a unique decoder  $\bar{q}\_q: \text{im}(\tau) \rightarrow A\_q$  such that  $q = \bar{q}\_q \circ \tau$ .

Proof. Assume  $Q$ -faithfulness. For  $y \in \text{im}(\tau)$ , choose  $x$  with  $\tau(x)=y$  and set  $\bar{q}\_q(y)=q(x)$ . If  $x'$  is another preimage,  $Q$ -faithfulness gives  $q(x)=q(x')$ , so  $\bar{q}\_q$  is well defined; uniqueness follows because every  $y$  in  $\text{im}(\tau)$  has a preimage. Conversely, if  $q=\bar{q}\_q \circ \tau$  and  $\tau(x)=\tau(x')$ , then  $q(x)=\bar{q}\_q(\tau(x))=\bar{q}\_q(\tau(x'))=q(x')$ . ■

Boundary. The canonical decoder is defined on  $\text{im}(\tau)$ , not automatically on all of  $Y$ . A total extension  $Y \rightarrow A\_q$  requires an additional extension assumption, such as a chosen default when  $A\_q$  is

inhabited. This is a standard factorization fact; the RLA contribution claimed here is the explicit query family, domain, side information, tolerance, and loss receipt at every transmission.

### **Credited standard result — Full Losslessness Corollary (F02)**

If  $Q$  separates points of  $X$ , then  $Q$ -faithfulness implies that  $\tau$  is injective.

**Proof.** If  $\tau(x) = \tau(x')$ , faithfulness gives  $q(x) = q(x')$  for every  $q$ . Point separation yields  $x = x'$ . ■

### **Credited standard result — Compositional Preservation Theorem (F03)**

Let  $\tau: X \rightarrow Y$  be  $Q$ -faithful and let  $Q_{\tau} = \{q_{\tau} \mid q \in Q\}$  be the induced query family on  $\text{im}(\tau)$  from Theorem F01. Let  $\sigma: Y \rightarrow Z$ . If the restriction  $\sigma|_{\text{im}(\tau)}: \text{im}(\tau) \rightarrow Z$  is  $Q_{\tau}$ -faithful, then  $\sigma \circ \tau$  is  $Q$ -faithful.

Proof. If  $\sigma(\tau(x)) = \sigma(\tau(x'))$ ,  $Q_{\tau}$ -faithfulness of the restricted map gives  $q_{\tau}(\tau(x)) = q_{\tau}(\tau(x'))$  for every  $q$ . Since  $q = q_{\tau} \circ \tau$ ,  $q(x) = q(x')$  for every  $q$ , so  $\sigma \circ \tau$  is  $Q$ -faithful. Values or extensions of induced



queries outside  $\text{im}(\tau)$  are irrelevant to this composition. ■

### **Credited standard result — Recovery and Query-Recovery Theorem (F04)**

A left inverse  $r: \text{im}(\tau) \rightarrow X$  satisfying  $r \circ \tau = \text{id}_X$  exists if and only if  $\tau: X \rightarrow Y$  is injective; no finiteness assumption is required. For a query family  $Q$ , a decoder  $\bar{q}_q: \text{im}(\tau) \rightarrow A_q$  exists for every  $q \in Q$  if and only if  $\tau$  is  $Q$ -faithful.

Proof. A left inverse implies injectivity:  $\tau(x)=\tau(x')$  gives  $x=r(\tau(x))=r(\tau(x'))=x'$ . If  $\tau$  is injective, define  $r(\tau(x))=x$ ; injectivity makes  $r$  well defined on  $\text{im}(\tau)$ . The query-decoder statement is Theorem F01. A total left inverse  $Y \rightarrow X$  is a stronger request and requires an extension from  $\text{im}(\tau)$ , which can fail for empty  $X$  and nonempty  $Y$  or otherwise needs a declared default. ■

### **Credited standard boundary result — Non-Injectivity Does Not Establish Emergence (F09)**

Non-injectivity alone does not imply an emergent macro-property. Let  $|X|>1$  and let  $\tau: X \rightarrow Y$  be constant. Then  $\tau$  is non-injective, but  $\text{im}(\tau)$  is a

singleton, so every predicate that factors through  $\tau$  is constant on  $X$ . Non-injectivity by itself therefore yields no non-trivial macro-property on the represented image. Any emergence claim requires an additional language-relative novelty criterion, macro-dynamic, intervention structure, or other explicit condition beyond non-injectivity. ■

## CHAPTER 43

# Computability, Termination, and Formal Limits

A **compact reticulum** is a finite or effectively enumerable directed graph whose nodes are levels, whose edges are typed transmissions, and whose evaluation contract declares an effective scheduler, resource policy, terminal sum type, and horizon-relative closure. “Closure” means that every in-scope query is routed to an answer or a typed non-answer; it does not mean that the reticulum contains every truth about its domain.

### Local derived result — Bounded Reticular Evaluator (F05)

Let  $C$  be an effectively encoded configuration space. Suppose the one-step transition dispatcher is partial recursive, the scheduler is computable, and every local operation or transmission is invoked through that dispatcher. Then the evaluator obtained by iterating scheduled steps until a terminal is reached is a partial recursive function

from initial configurations to terminals and receipts.

Proof. Encode configurations, scheduler state, and receipt prefixes as natural numbers. One scheduled step is partial recursive by hypothesis. Finite iteration is partial recursive by composition and primitive recursion; unbounded search for the first terminal is partial recursive by minimization. If a local operation diverges, the evaluator is undefined, which is why F06 requires a stronger terminalized dispatcher. The result is a standard closure fact instantiated for CRC. ■

### **Local derived result — Budgeted Totality into Typed Terminal States (F06)**

Let  $\text{Result}(Y) = \text{Answer}(Y) + \text{Unknown} + \text{Contradiction} + \text{OutOfHorizon} + \text{Review} + \text{Timeout} + \text{Failure}$ . Assume a total computable dispatcher  $\delta: C \times \{0, \dots, B\} \rightarrow \text{Continue}(C) + \text{Result}(Y)$  such that  $\delta(c, 0) = \text{Timeout}$  and every  $\text{Continue}(c')$  returned at budget  $b > 0$  is followed only with budget  $b - 1$ . Then the budgeted evaluator  $\text{eval\_B}: C \rightarrow \text{Result}(Y)$  is total computable.

Proof. Induct on  $b$ . At  $b = 0$  the dispatcher returns Timeout. At  $b > 0$ , totality of  $\delta$  yields either a

terminal or Continue( $c'$ ); in the second case the induction hypothesis applies at  $b-1$ . Computability follows from bounded recursion. The premise is deliberately stronger than “local functions are computable”: every scheduled step must itself terminate or be wrapped by a terminating timeout or failure constructor. ■

### **Credited standard reduction instantiated here — Propagation of Undecidability Under Query Factorization (F07)**

Let  $C$  be an effectively encoded promise domain and let  $P:C\rightarrow\{0,1\}$  be undecidable as a promise problem. Let  $\tau:C\rightarrow Y$  be total computable. If there is a predicate  $P$  on the promise set  $\tau(C)$  such that  $P=P\circ\tau$ , then  $P$  is undecidable on  $\tau(C)$  relative to the induced representation.

Proof. A decider for  $P$  on the promised inputs  $\tau(C)$  would decide  $P$ : on input  $x\in C$  compute  $\tau(x)$  and return  $P(\tau(x))$ . This contradicts undecidability of  $P$ . The result says that a hard question propagates only when it factors through the transmission; it does not say that every property or every encoding preserves the difficulty. ■

## **Credited standard factorization obstruction — Information Collapse Blocks Exact Queries (F08)**

If  $P: X \rightarrow \{0,1\}$  is not constant on the fibers of  $\tau: X \rightarrow Y$ , no target predicate  $\acute{P}: Y \rightarrow \{0,1\}$  satisfies  $P = \acute{P} \circ \tau$ .

**Proof.** Choose  $x, x'$  in one fiber with  $P(x) \neq P(x')$ . Any  $\acute{P}$  would give  $P(x) = \acute{P}(\tau(x)) = \acute{P}(\tau(x')) = P(x')$ , contradiction. ■

**Interpretation.** Collapse does not solve the source question. It removes the distinctions needed to state or answer it from the target representation alone. This corrects the stronger and generally false claim that a non-injective map makes an undecidable property decidable.

## CHAPTER 44

# Optional Simulation Bridge as a Bounded Con- tract

*This formal section is optional. No claim about RLA, CRC, ROA, Epistemic Debt, abstention, reification, or use gating depends on the Principle of Computational Equivalence. The section is retained only to specify what would have to be shown if a cross-formalism simulation or equivalence claim is invoked.*

Let  $S=(X, \rightarrow_S)$  and  $T=(Y, \rightarrow_T)$  be source and target transition systems. A one-way simulation contract contains an encoding  $e:X \rightarrow Y$ , a relation  $R \subseteq X \times Y$ , paired observables  $(o_S, o_T)$  with a declared metric  $d_o$ , a distortion bound  $\varepsilon_o$  for each observable, and a per-source-step overhead bound  $p:X \rightarrow \mathbb{N}$ .

## **Credited standard simulation result instantiated here — Finite Trace Preservation (F10)**

Assume (i)  $(x_0, e(x_0)) \in R$ ; (ii) whenever  $(x, y) \in R$  and  $x \rightarrow_S x'$ , there exists a target path  $y \rightarrow_{T^{\wedge\{ \leq p(x) \}}} y'$  with  $(x', y') \in R$ ; and (iii) for every declared observable pair,  $d_o(o_S(x), o_T(y)) \leq \epsilon_o$  whenever  $(x, y) \in R$ . Then every finite source trace  $x_0 \rightarrow_S \dots \rightarrow_S x_n$  has a related target trace with at most  $\sum_{i=0}^{n-1} p(x_i)$  target steps, and every paired state satisfies its declared pointwise distortion bound.

**Proof.** Induct on source-trace length. The base is the initialization relation. The step condition extends any related target prefix by a bounded target path and re-establishes  $R$  at the next source state. Summing the per-step bounds gives the total overhead; the observable condition holds at each paired state. No cumulative error claim follows unless a separate composition rule is supplied. ■

**Boundary.** One-way simulation is not equivalence. Bisimulation, robust decoding, perturbation stability, and complexity preservation are separate obligations.



## CHAPTER 45

# Formal Statements, Derivations, and Conformance Invariants for Epistemic Units, Provenance, Debt, and Use Gates

An ECU is a partial function from local state, horizon fragment, evidence state, governance state, and budget to a result plus receipt. A composite execution is a set of immutable events

$$e = \left( \begin{array}{l} id, parents, input, operator, output, \\ status, version \end{array} \right).$$

### Local derived proposition — Provenance Directed Acyclic Graph and Admissible Linearization (F11)

If every event has a fresh identifier, every parent reference names an earlier event, and the recorded parent relation is complete for the dependencies

the execution claims to expose, then the finite event-provenance graph is acyclic. A topological sort yields at least one admissible linearization of the recorded partial order.

Proof. A directed cycle would imply, by transitivity of the “earlier than” relation, that an event precedes itself. Finite acyclic graphs admit a topological ordering. The ordering need not be unique and does not reconstruct real-time order among concurrent events; omitted parent edges remain an external dependency-soundness failure. ■

For each component let  $D_i = \{0, 1, 2, 3, U\}$ . A PropagationPolicy  $P$  declares (i) a join operator  $\sqcup_P$ , For each transformation  $t$ , a typed relevance set  $Rel_t$  identifies which recorded parents are semantically relevant to that transformation. Excluding a recorded parent is itself a relevance claim and must be justified by a replayable receipt when the exclusion can affect an authority-sensitive use. DependencyCoverage is a separate gate input: challenge sensitivity is not coverage; a bounded-estimate state requires an explicit sampling frame and estimator; an audited state requires the declared audit evidence; otherwise coverage remains

unknown. Unknown coverage cannot silently satisfy complete mediation and routes the gate to REVIEW or BLOCKED according to the use contract.

$$d(t(a_1, \dots, a_n)) = \text{discharge}_{\{\Delta_t\}}(\text{join}^{\wedge\{P\}}_{\{i \in \text{Rel}_t\}} d(a_i) \sqcup_P \delta_t).$$

### **Conditional conformance invariant — Monotone Debt Propagation Under the Strict Conservative Profile (F12)**

If  $\Delta_t$  is empty, then each known output component is at least the maximum known severity of the corresponding relevant input components, and any U input propagates as U.

Derivation from the strict conservative profile. With empty discharge, the update is the policy-indexed componentwise join over  $\text{Rel}_t$ . For  $P=\text{strict}$ , max is above each known operand and U is absorbing. This establishes monotonicity only for the declared relevant inputs under the strict profile; it is not evidence that  $\text{Rel}_t$  is complete; an exclusion from  $\text{Rel}_t$  that affects authority-sensitive use requires a justified receipt, that dependency coverage is known, that the coding scale

is reliable, or that this profile is operationally optimal. ■

### **Definition-level consequence — Waiver Does Not Discharge Epistemic Debt (F13)**

Why this follows from the contract. A `WaiverRecord` is a governed record with at least `{claim_or_object, named_use, authorizing_authority, reason, scope, issued_at, expiry, residual_debt_reference}`. It is consumed by the `named-use` gate under the applicable waiver policy; it is not itself a permission function and it does not occur in the `debt-update` function. Therefore a valid waiver may cause the gate to return `PERMITTED_WITH_WAIVER` while leaving the debt vector unchanged. ■

### **Conditional conformance invariant — Non-Laundering Under Complete Mediation (F14)**

This is a reference-monitor conformance property. It does not estimate whether mediation is complete, whether hidden dependencies were discovered, or whether the policy is adequate. Let `Safe(s)` mean that every state rendered as warranted green in `s` is backed by a valid current

WARRANTED gate receipt, while PERMITTED\_WITH\_WAIVER and bypass states are represented by distinct constructors that retain residual debt and governance metadata. Assume  $\text{Safe}(s_0)$ , complete mediation of authoritative-use transitions, integrity and freshness of gate inputs, measured or otherwise explicitly bounded DependencyCoverage appropriate to the use contract, and a transition relation that preserves Safe or emits a visible waiver/bypass constructor rather than warranted green. Then every state reachable by a finite monitored trace satisfies Safe.

Proof under the stated conformance premises. Induct on trace length. Safe holds initially. Each monitored transition either preserves warranted green with a current WARRANTED receipt or records an exceptional permission through a distinct waiver/bypass constructor; neither exceptional constructor satisfies the warranted-green predicate. The result is a safety invariant conditional on complete mediation and trustworthy inputs; it is not evidence that the monitor is complete, the policy is legitimate, the waiver is justified, or the sources are true. ■

Trusted-boundary warning. The theorem is conditional on monitor completeness, receipt integrity, version freshness, and correct policy. It proves no institutional legitimacy and no security property outside those assumptions. Complete mediation is therefore an assumption to be measured or challenged, never an output of F14. The archived finite enumeration formerly associated with this invariant is not severity evidence because its stored safety predicate was vacuous; future stress testing must be mutation-sensitive and must demonstrate that seeded invariant-breaking mutations are detected.

**Non-entailment boundary proposition —  
Warrant Does Not Transfer by Object Identity (F15)**

Let  $W(a, H, u, G)$  mean that artefact  $a$  is warranted under horizon  $H$ , use  $u$ , and governance context  $G$ . From  $W(a, H, u, G)$  alone one cannot infer  $W(a, H', u', G')$ .

**Proof by countermodel.** Let  $a$  be a parameter measured for species A,  $H$  admit those measurements, and  $u$  be sensitivity analysis. Let  $H'$  concern species B and  $u'$  be field prediction. The

antecedent may be true while the consequent is false because the target validation is absent. Thus no logical entailment exists without a transfer bridge. ■

### **Contract-level consequence — Epistemic Horizon Adequacy Precedes Claim Gating (F16)**

If an adequacy contract requires source or validation classes absent from  $H$  for use  $u$ , then a conforming gate cannot return warranted for  $(a, H, u, G)$ .

Why this follows from the gate contract. Adequacy is a declared precondition of the gate. Returning warranted when the precondition is false violates the gate specification. ■

### **Non-entailment boundary proposition — Limits of Implementation Witnesses (F17)**

The existence of a running implementation entails constructibility of that implementation, but not empirical benefit, optimality, adoption, or truth of the theory.

**Proof by countermodels.** A build can run while producing no benefit, costing more than

alternatives, being rejected by users, or implementing a false theory. Therefore none of those conclusions follows from constructibility alone. ■

**Non-entailment boundary proposition — Execution Does Not Entail Authorization (F18)**

Let  $E(a)$  mean that a machine execution produced artefact  $a$ , and  $A(a,u)$  mean that an actor with the required external authority approved use  $u$ . Without an explicit governance premise connecting these predicates,  $E(a)$  does not entail  $A(a,u)$ .

Proof by countermodel. There is a model in which  $E(a)$  is true and no authorized actor approves  $u$ ; hence  $A(a,u)$  is false. An implementation can enforce a rule requiring approval, but execution alone cannot manufacture the authority relation that makes the rule legitimate. ■



# Recovery, Replay, and Certification Boundaries

**Credited standard information result instantiated here — Zero Conditional Query Entropy and Almost-Sure Recovery (F19)**

Let  $X$  be a finite-valued random variable,  $\tau$  a deterministic transmission, and  $Q(X)$  the finite vector of answers to a declared query family. Then  $H(Q(X) \mid \tau(X)) = 0$  if and only if there exists a measurable decoder  $\bar{q}$  such that  $Q(X) = \bar{q}(\tau(X))$  almost surely.

Proof. For finite variables, zero conditional entropy means that  $Q(X)$  is almost surely a function of  $\tau(X)$ ; define  $\bar{q}$  on each positive-probability fiber by that unique value. Conversely, if such  $\bar{q}$  exists,  $Q(X)$  is determined by  $\tau(X)$ , so the conditional entropy is zero. Boundary. This is distribution-relative and almost-sure; it neither guarantees pointwise recovery outside the support nor converts Shannon information into warrant.

## **Local derived result — Deterministic Replay of the Bounded Epistemic Computational Network Control Plane (F20)**

Let  $N$  be a finite acyclic ECNN, or a cyclic ECNN evaluated under a fixed finite schedule and budget. Assume every control-plane ECU is a deterministic terminalized function; the scheduler is fixed; and the versioned input bundle contains every value that can affect execution, including source hashes, policy and code identifiers, concrete outputs of stochastic proposal calls, clock or locale inputs, external-service replies, random seeds where used, and the declared numerical environment when floating-point variation is material. Then the terminal, traversed path, state transitions, and receipt are deterministic functions of that bundle.

Proof. Topologically order the acyclic graph or follow the fixed bounded schedule. Induct on evaluation steps. At each step identical complete inputs and deterministic local operators produce the same output and receipt fragment, which fixes the next transition. A missing exogenous input is a failure of the premise, not evidence of replay. The theorem excludes unconstrained generative calls unless their realized outputs enter the bundle by content hash. ■

### **Conditional conformance invariant — Certification as an External State Transition (F21)**

Suppose  $\kappa(v) \in \{\text{candidate\_AI}, \text{certified\_human}, \text{disputed}, \text{superseded}\}$  and the transition  $\text{candidate\_AI} \rightarrow \text{certified\_human}$  requires a signature from an authority relation  $A$  stored outside the proposing operator. Then proposal generation alone cannot establish  $\text{certified\_human}$ .

Derivation from the certification contract. The certification transition is undefined without a valid A-witness. A proposer may create or modify  $v$  but cannot satisfy the transition premise merely by asserting it. Boundary. Compromise or capture of A is a threat-model failure, not a proof that authorization is intrinsic to the model.

### **Contract corollary — No Self-Certification Under Separated Authority (F22)**

Under the hypotheses of F21, any run in which the same unreviewed model output both proposes  $v$  and supplies the only certification evidence must terminate in candidate, review-required, or failure—not certified\_human.

Derivation from F21 and the authority-separation contract. The required external authority witness is absent; by the transition contract the certification edge is disabled. Falsifier. A deployed implementation that reaches certified\_human without an independently verifiable authority event violates the model.

## **Local derived replacement result — Layer Removal Under Interface Reconstructability (F23)**

Let a layer  $B$  implement an exported interface  $f_B: I \rightarrow E_B$  under horizon  $H$ , and suppose all admitted downstream computations observe  $B$  only through  $E_B$ . If an effective replacement  $r: I \rightarrow E_B$  is extensionally equal to  $f_B$  on every admitted input, uses no undeclared side information, preserves the terminal, receipt, debt, threat-model, and resource contracts, and leaves every  $Q$ -indexed downstream observation unchanged, then replacing  $B$  by  $r$  is observationally equivalent relative to  $H$  and  $Q$ .

Proof. Substitute  $r$  for  $f_B$  at every downstream occurrence. Extensional equality preserves the complete exported object on admitted inputs; the factorization premise ensures downstream transitions and observations depend on no hidden internal state of  $B$ . Induction over the finite or budgeted trace therefore yields the same declared observations and receipts. The lemma does not show that  $r$  is simpler, cheaper, more interpreta-

ble, or globally minimal; those require independent cost and threat criteria. ■

**Conditional conformance invariant —  
Preservation of Typed Non-Answer Terminal  
States (F24)**

Let  $t$  be a terminal in  $\{\text{UNKNOWN}, \text{CONTRADICTION}, \text{OUT\_OF\_HORIZON}, \text{REVIEW}, \text{TIMEOUT}, \text{FAILURE}\}$  on a required dependency edge of a conforming ECU/ECNN composition. Assume the composition rules provide no eliminator from that terminal to an unqualified ANSWER, and no independent evidence path performs a separately receipted re-evaluation that discharges or replaces the dependency. Then an output that claims to depend on that edge cannot be an unqualified ANSWER.

Derivation from the composition contract. Induct along the required dependency path. At each conforming edge the constructor is preserved, refined to another non-answer, or routed to review or failure. An ANSWER can arise only from a separately typed re-evaluation whose receipt identifies new evidence, scope, policy, and result. A downstream unit may answer a different query

from independent evidence; that is not coercion of  $t$ . A waiver may change permission for a named use but cannot change the evidence terminal. ■







**PART X**

# Reference      Semantics, Threat Model, and Con- formance

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## CHAPTER 47

# Reference Semantics for Epistemic Use Gates

The gate is a deterministic policy interpreter over versioned inputs; it is not a learned confidence threshold. Its precedence is:

1. verify horizon and policy versions;
2. test horizon adequacy for the named use;
3. reject out-of-scope, uncomputed, or unsupported evidence states;
4. route contradiction and unknown debt components to review;
5. compare the debt vector componentwise with the use threshold vector;
6. accept a typed discharge only for the component, authority, scope, and version it names;
7. evaluate waiver authority, scope, expiry, and residual risk;
8. emit an integrity-protected receipt containing every input identifier and the complete blocking set.

The same precedence is the reference semantics: verify versions; test horizon adequacy; compute evidence status and debt; route unsupported, contradictory, or unknown states; compare every debt component with the named-use threshold; then accept only a typed discharge or a valid bounded waiver. The emitted receipt records all input identifiers, the policy version, the complete blocking set, and the terminal outcome.

The algorithm does not establish that the policy is legitimate or that the sources are true. It makes the policy application reconstructable and prevents an evidence status from silently becoming a use authorization.



## CHAPTER 48

# Threat Model and Trusted Boundary

The non-laundersing guarantee depends on assets and attack surfaces that must be collected in one threat model. Assets are source identity, receipt integrity, dependency completeness, event order, policy and horizon identifiers, reviewer authority, waiver scope and expiry, monitor coverage, and user-visible status. Threats include forged or stale receipts, source-selection bias, omitted dependencies, policy capture, privilege escalation, collusion between proposer and certifier, bypass of monitored surfaces, optimistic defaults on unknown components, replay under superseded policy, selective disclosure of blocking components, and interface relabelling of review as approval.

Minimum controls are separation of roles, immutable event identifiers, signed or integrity-protected receipts, version pinning, revocation, complete blocking-set reporting, negative tests, fail-closed unknown semantics, contestation, and

an explicit bypass event. None of these controls proves source truth or institutional justice. They define the computational envelope within which the local safety theorem applies.



## CHAPTER 49

# Result Classes and Conformance Criteria

Credited standard mathematics instantiated here. Items: F01–F04, F07–F10, F19. Status: ordinary factorization, recovery, reduction, obstruction, simulation, and finite-variable information results; no originality is claimed for the mathematics itself.

Local computability and replay results. Items: F05, F06, F20. Status: proved under effective-dispatcher, complete-input, scheduler, and budget assumptions; they establish computability or deterministic replay, not truth, tractability, robustness, or empirical benefit.

Conformance invariants, replacement lemmas, and non-entailments. Items: F11–F18, F21–F24. Status: hold for implementations satisfying the declared event, debt, mediation, authority, interface, and typing contracts. Several are intentionally negative or by-construction results; none

establishes institutional legitimacy or complete dependency capture.

Architectural invariants. Items: I01–I30. Status: design obligations for conforming implementations and monitored use surfaces; they are falsified by a counterexample implementation or an unmediated path.

Empirical targets. Items: C25–C30 and C35, together with debt reliability, horizon calibration, workflow net benefit, and the bryophyte comparison. Status: require preregistered external tests and are allowed to fail.

A conforming implementation must publish a machine-readable signature, reference evaluator, frozen conformance corpus, negative tests, threat model, reproducible build, and governance record. A document that merely repeats the vocabulary is not a conforming implementation.

## CHAPTER 50

# Layer Acceptance Tests and Proof Dependencies

RLA. Minimum acceptance evidence: typed domains, query family, transmission, side information, fidelity or distortion test Refusal condition: preserved query fails, domain drifts, or side information is undeclared

CRC. Minimum acceptance evidence: effective operators, scheduler, rank or budget, terminal policy and reproducible evaluator Refusal condition: execution can diverge or fail without a typed terminal

PCE. Minimum acceptance evidence: encoding, simulation relation, observables, error, overhead and finite trace tests Refusal condition: analogy is asserted without a satisfying relation

ECU / ECNN. Minimum acceptance evidence: typed input/output, receipt, failure semantics, channel compatibility, non-answer preservation, candidate/certified separation, and terminal tests Refusal condition: untyped output, silent failure,

terminal collapse into confidence, non-answer-to-answer coercion, or certification without external authority

ROA. Minimum acceptance evidence: source manifest, horizon, reticulum, memory, monitor, policy versions and human-authority boundary  
Refusal condition: path cannot be reconstructed or monitor/gate can be bypassed silently

Epistemic Debt. Minimum acceptance evidence: coding manual, evidence status, debt vector, adequacy contract, gate receipt and negative false-green tests  
Refusal condition: debt disappears without discharge or authoritative green survives a missing predicate

### Proof Dependency Matrix

F01–F04. Depends on: ordinary set-theoretic factorization and declared query families  
What it establishes: exact conditions for query recovery and full recovery  
What it does not establish: empirical usefulness of a chosen query family

F05–F06. Depends on: effective encoding, a computable scheduler, a partial-recursive one-step dispatcher for F05, and a total budgeted dispatch-

er for F06. What they establish: partial computability and total terminalization under the stronger budget premise. What they do not establish: tractability, source correctness, semantic truth, or that a timeout wrapper exists for every external operation.

F07–F08. Depends on: computable transmission and exact factorization or fiber counterexample What it establishes: propagation or blocking of a predicate What it does not establish: that every hard question is preserved or that collapse solves it

F10. Depends on: explicit one-way simulation contract What it establishes: finite trace preservation within error and overhead What it does not establish: bisimulation, universality, physical equivalence, or robustness

F11–F14. Depends on: fresh event identity; complete recorded parent relations for the declared relevance set; an explicit `PropagationPolicy`; measured, bounded-estimate, or explicitly unknown `DependencyCoverage`; current receipts; complete mediation for F14; and trusted monitor inputs. What they establish: an acyclic recorded provenance order, profile-relative propagation

properties, waiver/discharge type separation, and a monitored safety invariant. What they do not establish: unique historical order, hidden-edge completeness, source truth, monitor completeness, policy adequacy, reviewer capacity, or legitimacy.

F15–F18. Relational warrant and authorization establish non-transfer and non-entailment; empirical benefit and moral legitimacy remain open.

F19 depends on finite-valued information theory and is distribution-relative. F20 depends on a complete versioned input bundle and deterministic composition; it excludes unrecorded exogenous inputs and unconstrained generation. F21–F22 depend on a separated external authority relation. F23 is an interface replacement lemma and proves neither lower cost nor unique minimality. F24 applies only to required dependency edges and permits an answer from an independent evidence path or a separately receipted re-evaluation.

# Appendix A — Registry of Formal Primitives and Invariants

## Primitives

| ID  | Definition / contract                                                                                                                                                                                                                                   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P01 | Statement: observer. A versioned bundle of representational, operational, evidential, and governance commitments.                                                                                                                                       |
| P02 | Statement: horizon.<br>$H=(S,\Omega,Q,T,R,M,V,\Pi)$ : a declared perimeter of sources, scope, queries, transforms, receipts, support rules, validation, and conflict priority. Scheduler and resource budget are evaluator objects, not horizon fields. |
| P03 | Statement: level. A state domain with an admitted language of effective operations.                                                                                                                                                                     |
| P04 | Statement: reticulum. A directed multi-level graph of levels, transmissions, local operators, and contracts.                                                                                                                                            |

| ID  | Definition / contract                                                                                                                       |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------|
| P05 | Statement: transmission. A typed deterministic map or stochastic kernel between level representations.                                      |
| P06 | Statement: query family. The declared set of distinctions an analysis must preserve.                                                        |
| P07 | Statement: query equivalence. Two states are equivalent when every declared query gives the same answer within tolerance.                   |
| P08 | Statement: collapse fiber. The set of source states mapped to one target representation.                                                    |
| P09 | Statement: compact reticulum. An effectively presented reticulum closed relative to a declared horizon and executable with typed terminals. |
| P10 | Statement: ECU. A typed partial transducer that emits a value or terminal plus a receipt.                                                   |
| P11 | Statement: ECNN. A reticulum of ECUs, channels, scheduler, memory, and epistemic head; neural implementation is optional.                   |



| ID  | Definition / contract                                                                                                                                                                                   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P12 | Statement: simulation contract. A source system, target system, encoding, relation, time map, observables, distortion, and overhead.                                                                    |
| P13 | Statement: observer compiler. The process that constructs a bounded observer and its source manifest, type environment, invariants, tests, and rollback plan.                                           |
| P14 | Statement: governed oracle. A responder that navigates a bounded reticulum and returns a terminal plus a reconstructable path.                                                                          |
| P15 | Statement: reified artefact. A promoted pattern, label, relation, or claim that can be re-used downstream.                                                                                              |
| P16 | Statement: epistemic debt. A use-independent vector of unresolved warrant attached to a reified semantic artefact under a declared horizon; a separate gate determines the consequence for a named use. |

| ID  | Definition / contract                                                                                                                                                                                                                                                                                              |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P17 | Statement: debt vector. Nine typed obligations: source, trace, transformation, collapse, label, reification, validation, authority, and horizon-or-scale. Each component is coded 0–3 or U; U propagates conservatively and routes the gate to review rather than being treated as zero or as a numerical maximum. |
| P18 | Statement: proof chain. The typed path source $\rightarrow$ atom $\rightarrow$ receipt $\rightarrow$ artefact $\rightarrow$ witness $\rightarrow$ governance.                                                                                                                                                      |
| P19 | Statement: discharge. A typed operation that pays a specified debt component within scope.                                                                                                                                                                                                                         |
| P20 | Statement: waiver. A logged permission to proceed despite residual debt; it does not pay debt.                                                                                                                                                                                                                     |
| P21 | Statement: gate. A use-sensitive decision function over horizon adequacy, evidence status, debt, authority, and policy.                                                                                                                                                                                            |
| P22 | Statement: false green. An authoritative-looking state not derived from the receipts and gate predicates that would warrant it.                                                                                                                                                                                    |

| ID  | Definition / contract                                                                                                                                                                                 |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P23 | Statement: iKant pattern. Normative meta-control that escalates hard cases to accountable human authority without claiming moral agency.                                                              |
| P24 | Statement: severe test. A test that a claim would probably not pass if it were false or materially defective.                                                                                         |
| P25 | Candidate/certified state register $\kappa$ : every semantic object carries a state in {candidate_AI, certified_human, disputed, superseded, unknown, contradiction, review-required}.                |
| P26 | Certified control plane: versioned deterministic operators, scheduler, budgets, schemas, policies, and receipt construction; stochastic proposals enter only as hashed candidate inputs.              |
| P27 | Interface export object. Statement: the complete typed bundle a layer exposes downstream: value or terminal, receipt, debt update, policy version, and every metadata field used by admitted queries. |



## Invariants

| ID  | Invariant / boundary                                                                                    |
|-----|---------------------------------------------------------------------------------------------------------|
| I01 | Every material claim is indexed by a horizon and source mode.                                           |
| I02 | Observer relativity does not exempt an observer from cross-observer criticism or empirical tests.       |
| I03 | A transmission claim declares domain, query family, side information, tolerance, and randomness policy. |
| I04 | Losslessness is always relative to a declared family of distinctions unless injectivity is proved.      |
| I05 | Non-injectivity alone does not establish emergence.                                                     |
| I06 | Classical computability is not changed by CRC; CRC packages conditions for a multi-level evaluator.     |
| I07 | Unknown, contradiction, out-of-horizon, review, timeout, and failure are valid terminal constructors.   |

| ID  | Invariant / boundary                                                                                      |
|-----|-----------------------------------------------------------------------------------------------------------|
| I08 | Evidence status, confidence, debt, authority, runtime result, and gate outcome are disjoint tagged types. |
| I09 | Every material ECU transformation emits a receipt or an explicit failure terminal.                        |
| I10 | PCE language requires an explicit simulation contract; resemblance is not equivalence.                    |
| I11 | Generated material is a candidate input to controlled reification, not self-authenticating evidence.      |
| I12 | The observer may prepare but may not self-authorize a use.                                                |
| I13 | A waiver changes permission and never debt.                                                               |
| I14 | No risk decrease is valid without a witnessed discharge operation.                                        |
| I15 | Controlled reification is append-only and rollback preserves genealogy.                                   |
| I16 | A proof-chain link supports only its declared claim set and scope.                                        |
| I17 | Authoritative use requires a current gate receipt or a recorded bypass event.                             |

| <b>ID</b> | <b>Invariant / boundary</b>                                                                                                                 |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| I18       | Changed policy, horizon, source, code, or model versions invalidate unqualified reuse.                                                      |
| I19       | Interface state, export, log, and model confidence are not proof by themselves.                                                             |
| I20       | Transfer moves an object; warrant must be re-earned at the destination.                                                                     |
| I21       | Collapse is not amnesty: later re-expansion can revive blocked questions and debts.                                                         |
| I22       | Constructibility is not evidence of empirical efficacy or comparative superiority.                                                          |
| I23       | Every empirical value proposition has a pre-registerable endpoint and a declared falsifier.                                                 |
| I24       | Every load-bearing claim is tagged as standard result, derived theorem, architecture invariant, empirical target, witness, or open problem. |
| I27       | Certification separation: no candidate becomes human-certified without an external authority event recorded in the receipt chain.           |

| ID  | Invariant / boundary                                                                                                                                                                                                 |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I28 | Limit hygiene: Gödel incompleteness, Church–Turing undecidability, Rice extensional undecidability, and Shannon information are never used outside their stated hypotheses or as universal metaphors.                |
| I29 | Layer minimality is query-relative: a layer is removable only when its complete export can be reconstructed without hidden side information, new debt, lost queries, weakened receipts, or exceeded resource bounds. |
| I30 | Typed non-answer preservation: UNKNOWN, CONTRADICTION, OUT_OF_HORIZON, REVIEW, TIMEOUT, and FAILURE cannot become ANSWER without a new evidence bundle and a valid re-evaluation receipt.                            |



## Appendix B — Epistemic Debt Vector Coding and Claim Falsification

The nine debt components form a candidate coding instrument and are coded independently pending external reliability evidence. Severity is use-independent; use enters only through the gate threshold vector. The code set is 0 = obligation absent, 1 = documented and bounded caveat, 2 = material unresolved obligation, 3 = critical unresolved structural obligation, and U = unknown because the dependency record is insufficient. Under the strict conservative PropagationPolicy, U is absorbing on the declared relevance set; other profiles must be declared and receipted. At the gate, U is not compared numerically; it routes according to the use contract, while DependencyCoverage remains a separate input.

**Source. Diagnostic question:** Can the origin be identified and admitted under the horizon?

**Typed repair:** Attach or exclude the source; recompute status.

**Trace. Diagnostic question:** Can the path from source to claim be reconstructed? **Typed repair:** Add dependency edges, hashes, receipts or source maps.

**Transformation. Diagnostic question:** Was meaning changed by an admitted operation? **Typed repair:** Record operator, inputs, outputs and validation.

**Collapse. Diagnostic question:** Were distinctions needed later compressed away? **Typed repair:** Expose pooling, preserve drill-down or narrow the use.

**Label. Diagnostic question:** Does the name exceed its declared criteria? **Typed repair:** Define criteria, relabel or narrow the claim.

**Reification. Diagnostic question:** Was a candidate promoted before its obligations were attached? **Typed repair:** Return to candidate state; add review and rollback.

**Validation. Diagnostic question:** Did the required check actually run and pass? **Typed repair:** Run the test, sample, experiment or review.

**Authority. Diagnostic question:** Is the source or reviewer authorized for this use? **Typed**

**repair:** Obtain qualified review or downgrade the use.

**Horizon / scale. Diagnostic question:** Is warrant earned in one domain, population or scale being spent in another? **Typed repair:** Write a transfer passport and validate locally.

## **Load-Bearing Claims and Their Falsifiers**

Each load-bearing claim has a type, its current support, and a condition that would defeat or materially weaken it. “Formal” means proved only under the declared finite/effective assumptions; “architecture” means guaranteed only by conformance; “empirical” means unvalidated until the named study is run.

C01 — formal. Claim: RLA levels are epistemic modelling slices, not asserted ontological strata. Support: definition plus scope boundary Falsifier: Counterexample: the framework requires a level to exist mind-independently.

C02 — formal. Claim: Query-relative fidelity is equivalent to factorization through transmission

fibers. Support: F01 Falsifier: A query differs inside a fiber while claimed preserved.

C03 — formal. Claim: Full losslessness follows only when the query family separates points. Support: F02 Falsifier: A non-injective map preserves a point-separating family.

C04 — formal. Claim: Non-injectivity alone does not prove emergence. Support: F09 Falsifier: A general theorem derives emergence from non-injection without extra assumptions.

C05 — formal. Claim: CRC induces a partial computable evaluator under effective local conditions. Support: F05 Falsifier: A required operation is not effectively computable.

C06 — formal. Claim: A finite budget makes the evaluator total into a typed terminal set. Support: F06 Falsifier: A run exceeds the budget without emitting a terminal.

C07 — formal. Claim: Undecidability propagates only when the predicate factors through the transmission. Support: F07 Falsifier: A decider exists on the image under the stated assumptions.

C08 — formal. Claim: Collapse can make a source question unanswerable rather than solved.

Support: F08 Falsifier: An exact target predicate recovers a non-fiber-constant source predicate without side information.

C09 — formal. Claim: PCE claims require explicit bounded simulation obligations. Support: F10 Falsifier: A claimed bridge lacks encoding, relation, observables, distortion, or overhead.

C10 — architecture. Claim: An ECU is a typed partial transducer with a receipt-bearing terminal. Support: P10/I09 Falsifier: A conforming ECU silently returns an untyped output.

C11 — architecture. Claim: epistemic convolution is a CNN-inspired semantic-transduction method, and an ECNN is one reticular realization whose distinctive epistemic properties do not require classical convolutional neural layers. Support: P11 Falsifier: the framework's distinctive properties require a conventional CNN substrate.

C12 — architecture. Claim: Unknown and contradiction are successful terminal computations. Support: I07 Falsifier: Matched tasks show typed terminals add no diagnostic or safety value.

C13 — architecture. Claim: Observer compilation persists intermediate artefacts and source

contracts. Support: P13 Falsifier: A compiler can reproduce results without manifests, types, or lineage.

C14 — architecture. Claim: ROA is a bounded observer bundle, not a consciousness claim. Support: P01/P14 Falsifier: The architecture requires phenomenal or moral agency.

C15 — formal. Claim: Machine success does not entail authorization. Support: F18 Falsifier: Authorization is derivable from execution alone in the declared governance model.

C16 — theoretical. Claim: Epistemic Debt is unpaid warrant, not probability of falsity. Support: P16 Falsifier: Debt reduces without loss to a calibrated truth probability.

C17 — architecture. Claim: Debt is vector-valued because obligations have non-substitutable remedies. Support: P17 Falsifier: A scalar preserves all blocking and discharge information on representative tasks.

C18 — formal. Claim: Debt cannot decrease along declared lineage without discharge. Support: F12 Falsifier: A conforming unreceipted transform decreases debt.

C19 — formal. Claim: A waiver permits but does not discharge. Support: F13 Falsifier: A waiver alone changes the debt vector.

C20 — architecture. Claim: Authoritative use requires a gate outcome, not evidence status alone. Support: P21/I17 Falsifier: A safe use policy is fully determined by evidence status across uses.

C21 — conformance. Claim: Under complete mediation, current gate receipts, and adequate declared dependency coverage, a monitored proof chain preserves non-laundering over finite traces. Support: F14. Falsifier: a valid conforming trace reaches warranted green without a current gate receipt, or a seeded hidden authoritative-use path bypasses mediation.

C22 — theoretical. Claim: Transfer moves objects, not warrant. Support: F15 Falsifier: Warrant follows from object identity across materially different horizons.

C23 — architecture. Claim: Horizon adequacy is a precondition on the contract, not the claim. Support: F16 Falsifier: A use is warranted under a horizon missing its mandatory evidence classes.

C24 — witness. Claim: A-OSP witnesses constructibility of separations, not theory truth. Support: F17 Falsifier: A running implementation logically entails empirical superiority.

C25 — empirical. Claim: Typed proof chains should reduce unsupported authoritative use. Support: preregistered workflow comparison Falsifier: No improvement or excessive burden relative to simpler baselines.

C26 — empirical. Claim: Debt coding should localize reviewer disagreement. Support: interrater study Falsifier: Agreement and diagnosis are no better than an untyped rubric.

C27 — empirical. Claim: Reticular navigation should reduce reconstruction time. Support: matched reconstruction benchmark Falsifier: No reduction relative to retrieval/manual graph.

C28 — empirical. Claim: Negative proof-seam tests should reduce false green. Support: adversarial interface test Falsifier: Green survives missing receipt/readback/reviewer.

C29 — empirical. Claim: Transfer passports should reduce invalid cross-domain reuse. Sup-



port: cross-domain trial Falsifier: No predictive or preventive value.

C30 — programme. Claim: literature-compiled epistemic models are objects to falsify, not substitutes for measurements. Support: blind expert and empirical comparison Falsifier: The programme treats executability as validation.

C31 — candidate contribution. Claim: The candidate differential contribution is compositional: reticular abstraction, typed terminals, provenance, debt, and use gates in one architecture. Support: differential literature matrix Falsifier: A prior framework already contains the full composition.

C32 — limitation. Claim: The theory is closed only relative to a frozen horizon, type system, and finite proof domain. Support: scope declaration Falsifier: The manuscript claims universal ontological or semantic completeness.

C33. Claim: the framework is compatible with Gödel, Church–Turing, and Rice because it computes only horizon-declared predicates and terminalizes unresolved runs. Class: architectural consequence of standard results. Defeater: a pas-

sage that claims universal semantic decision, complete self-grounding, or removal of undecidability.

C34. Claim: ECNN supports deterministic replay of the certified control plane while permitting stochastic candidate generation. Class: local theorem plus engineering target. Falsifier: identical versioned bundles yield different control-plane paths or receipts, or model output silently becomes certified.

C35. Claim: candidate/certified separation improves traceability and prevents silent authority upgrade. Class: empirical hypothesis. Comparator: ordinary knowledge graph or workflow with review. Lose when: no reduction in unsupported authoritative use, reviewer confusion, or reconstruction cost.

C36. Claim:  $H(Q(X)|\tau(X)) = 0$  characterizes almost-sure recovery of finite query answers under a fixed distribution. Class: standard information-theoretic result localized to RLA. Boundary: no pointwise or warrant inference.

C37. Claim: a layer is observationally removable relative to a horizon and query family when its complete exported interface is effectively recon-

structable from adjacent interfaces and all downstream queries, receipts, debt updates, and resource bounds are preserved. Class: local theorem F23. Falsifier: removal changes any admitted terminal, query answer, receipt predicate, debt component, or bound, or requires hidden side information.

C38. Claim: a conforming ECNN cannot silently convert a typed non-answer terminal into an unqualified answer. Class: local conformance theorem F24. Falsifier: an admitted trace reaches ANSWER from UNKNOWN, CONTRADICTION, OUT\_OF\_HORIZON, REVIEW, TIMEOUT, or FAILURE without a new receipted evidence evaluation.

C39 — candidate instrument / empirical. Claim: the multi-axis Approximation and Transfer Ledger can be applied with useful independent agreement and can discriminate undeclared promotion of transferred, approximated, synthesized, or unresolved scientific relations from explicitly declared compilation choices. Support: proposed coding contract only. Falsifier: independent coders cannot apply the profile reliably, provenance

cards do not reduce disagreement, or a simpler free-form rubric performs equivalently with lower burden.

## **Evidence Required for the Next Falsification Test**

A falsifier says what can defeat a claim; the next-evidence column says what work should be done next even when the claim survives. This prevents a manuscript from treating non-refutation as completion.

C01. Apply the level schema in a domain with rival ontologies and test whether disagreements are localized rather than hidden.

C02. Property-based tests over finite transmissions and query families; mechanize the factorization lemma.

C03. Construct point-separating and non-separating query suites and verify the injectivity boundary.

C04. Compare non-injective maps with and without predictive macro-dynamics.

C05. Publish an executable CRC reference evaluator and termination traces.

C06. Exhaustively test budget exhaustion and every terminal constructor.

C07. Formalize a concrete reduction with a declared critical domain.

C08. Generate fiber counterexamples and verify mandatory refusal at the target level.

C09. Provide a worked bridge with encoding, relation, error and overhead tests.

C10. Conformance suite for typed ECU inputs, outputs, receipts and failures.

C11. Implement a conventional CNN-backed realization and a non-CNN semantic-transduction realization under one typed epistemic interface; compare terminal behaviour, traceability, reconstruction cost, and burden.

C12. Matched study against calibrated selective prediction on diagnosis, safety and burden.

C13. Rebuild an observer from manifest and receipts without conversational context.

C14. Independent philosophical review of the observer/consciousness boundary.

C15. Audit executions where authorization is withheld, revoked or contested.

C16. Test whether debt components predict use failure beyond calibrated confidence.

C17. Inter-rater study comparing vector and scalar debt representations.

C18. Mutation tests that attempt to reduce inherited debt without a discharge receipt.

C19. Waiver-expiry and residual-debt tests under policy changes.

C20. Cross-use experiments showing identical evidence status with different gate outcomes.

C21. Adversarial proof-seam tests for receipt deletion, replay, staleness and bypass.

C22. Transfer experiments across species, jurisdictions, scales or deployments.

C23. Calibrate minimum horizons for at least two use classes and test self-serving horizons.

C24. Independent code audit limited to constructibility and separation-by-design.

C25. Preregistered workflow comparison with burden and unsupported-use endpoints.

C26. Multi-reviewer coding study with agreement and error localization measures.

C27. Matched reconstruction benchmark against retrieval and manual graphs.

C28. UI mutation suite that removes each proof predicate independently.

C29. Prospective cross-domain trial of transfer passports.

C30. Blind expert and empirical comparison of the literature-compiled epistemic model.

C31. Systematic review and feature-by-feature selected-neighbour comparison.

C32. External review aimed specifically at finding undeclared universal claims.

C33. Mechanize the horizon-declared decision boundary and mutation-test passages for illicit universalization of Gödel, Church–Turing, Rice, or Shannon.

C34. Replay identical versioned control-plane bundles across independent runtimes and compare paths, terminals, and receipt hashes.

C35. Compare candidate/certified separation against an ordinary reviewed knowledge graph on unsupported authority upgrades and reviewer reconstruction.

C36. Property-test zero conditional query entropy against decoder existence on finite distributions, including zero-probability fibers.



C37. Remove each candidate layer in finite reference reticula and test whether terminals, query answers, receipts, debt vectors, and bounds remain observationally equivalent.

C38. Mutation-test every channel by coercing each non-answer constructor toward ANSWER; require the conformance suite and interface to reject the mutation.

C39. Run a preregistered inter-rater pilot on independently selected scientific relations using source provenance cards, blinded coding, and an adjudicator who did not author the candidate labels; report agreement by axis, confusion patterns, burden, and comparison with a simpler free-form rubric. Numeric sample sizes and decision thresholds belong to the preregistered study design, not to this monograph.



## Appendix C — Proof-Chain Adequacy and Cross-Layer Interface Contracts

**source\_id.** Identifier, origin, scope, date, authority and admissibility.

**claim\_atom.** Minimal statement carried forward and its source span.

**transformation.** Operator, inputs, outputs, model or human version, and caveats.

**receipt.** Integrity identifier, readback, timestamp, scope and result.

**artifact / witness.** Composition, dependencies, reader-facing proof map and reviewer state.

**evidence\_status.** One member of EvidenceStatus, before a use is named.

**debt\_vector.** Nine components, each coded 0–3 or U.

**governance context.** Policy, adequacy contract, threshold profile and reviewer authority versions.

**gate use / outcome.** Named use, complete blocking set, unknown components and terminal outcome.

**waiver / discharge.** Typed component, authority, scope, evidence, owner, expiry and residual debt.

## **Approximation and Transfer Ledger Contract**

ATL-08. register separation. UNRESOLVED is a ledger state about scientific compilation; UNKNOWN is a terminal action/state; U is the DebtVector code for insufficient dependency information.

ATL-07. qualitative debt obligations. TRANSFERRED raises transfer/horizon-scale/provenance questions; APPROXIMATED raises transformation/calibration/validation questions; SYNTHESIZED raises trace/transformation/reification/validation questions; UNRESOLVED raises source/validation and non-answer obligations. No state assigns a numeric DebtVector floor.

ATL-06. use\_boundary. Admitted use now, blocked or review-requiring use, and the evidence that would change that boundary.

ATL-05. calibration\_state / observable\_family. What has actually been calibrated, over which observables, conditions, and domain.

ATL-04. `compiler_ops`. Finite, ordered set of declared transformations applied during compilation; APPROXIMATED may coexist with any `source_basis`.

ATL-03. `source_basis`. DIRECT, TRANSFERRED, SYNTHESIZED, or UNRESOLVED; the values describe lineage and may not be read as validation grades.

ATL-02. `provenance`. Exact source locator, source organism or population, reported conditions, scale, and mathematical form as published.

ATL-01. `relation_id` / `claim_id`. Stable identifier for the scientific relation consumed by the compiled model.

ATL-09. `test hook`. The severe-test protocol in Appendix E treats ledger coding as an inter-rater instrument problem and tests undeclared promotion, register confusion, burden, and incremental value against a simpler rubric.

## Proof-Chain Adequacy Contract

A deploying institution must populate and sign the following candidate schema; this monograph does not calibrate its values.

**contract\_id.** Versioned identifier

**use\_class.** Class of intended use

**issuing\_authority.** Who sets the minimum horizon

**minimum\_sources.** Source classes that must be admissible

**minimum\_validation.** Predicates that must be runnable or completed

**reviewer\_class.** Authorized reviewer role

**effective\_date / expiry.** Validity interval supersedes. Identifier of the superseded contract

**contestation.** How decisions are challenged

**exception\_authority.** Who may authorize a bounded exception

**signature.** Accountable approval





## Layer Interface Contract

B01. Interface:  $RLA \rightarrow CRC$  Obligation: A reticulum becomes a bounded computational object only after effectiveness, scheduling, closure, and terminal policies are declared.

B02. Interface:  $CRC \rightarrow ECU$  Obligation: Local effective operations are packaged as typed partial transducers with receipts and explicit failure.

B03. Interface:  $ECU \rightarrow ECNN$  Obligation: Units compose through typed channels, scheduler, memory, and an epistemic head.

B04 — optional side bridge. Interface:  $CRC \leftrightarrow$  bounded simulation certificate. Obligation: when a cross-formalism simulation claim is invoked, it is converted into an explicit encoding, observable, relation, distortion, overhead, and bounded trace contract. No downstream ROA/ED claim requires this bridge unless that simulation claim is actually used.

B05. Interface:  $ECNN \rightarrow ROA$  Obligation: The network is embedded in a horizon, source manifest, memory, monitor, and governance context.

B06. Interface:  $\text{ROA} \rightarrow \text{ED}$  Obligation: Every reified object and transformation receives an evidence status, debt vector, and use-sensitive gate path.

B07. Interface:  $\text{ED} \rightarrow \text{A-OSP}$  witness Obligation: The accounting rules become enforceable storage, readback, receipt, witness, and UI constraints.

B08. Interface:  $\text{Theory} \rightarrow \text{experiment}$  Obligation: Every architecture claim is mapped to an observable endpoint, baseline, burden metric, and falsifier.

B09. Interface:  $\text{Layer} \rightarrow \text{compression/removal}$  Obligation: The complete exported interface object must be reconstructable from adjacent bundles; every admitted query, terminal, receipt predicate, debt update, threat assumption, and resource bound must remain unchanged. Otherwise the layer is not removable.

## Appendix D — Register of Open Epistemic Debt

O01. Open debt: Dependency soundness and coverage. Status: Debt propagation is only as complete as the recorded or discoverable dependency graph. Challenge sensitivity on seeded hidden edges is not global coverage. A bounded coverage estimate requires a declared sampling frame and estimator; audited coverage requires the declared independent reconstruction or audit evidence; otherwise `DependencyCoverage` remains unknown. Next evidence: hidden-edge challenge suites, explicit sampling-frame estimators where defensible, dependency-recall studies against independently reconstructed graphs, and conservative gating when coverage is unknown.

O02. Open debt: Debt-vector calibration. Status: Severity codes and thresholds require inter-rater and outcome validation. Next evidence: blinded multi-rater coding, componentwise reliability estimates, outcome association, and preregistered threshold calibration by use class.

O03. Open debt: Horizon adequacy calibration. Status: Minimum source and validation requirements for each use class are domain-specific and uncalibrated. Next evidence: domain panels, retrospective failure analysis, prospective threshold validation, and documented contestation of adequacy contracts.

O04. Open debt: Principle-of-Computational-Equivalence bridge strength. Status: The accepted formal core supplies only a generic one-way bounded simulation contract, not bisimulation or universality.

Next evidence: specify concrete source and target systems, observables, encoding, relation, distortion, overhead, reverse obligations where claimed, and falsifying traces.

O05. Open debt: Observer completeness. Status: No finite reticulum is claimed complete beyond its declared horizon. Next evidence: coverage audits, out-of-horizon challenge sets, source-gap discovery, and explicit horizon expansion rather than any claim of global completion.

O06. Open debt: Workflow payoff. Status: Comparative benefit, cost, and adoption remain

empirical targets. Next evidence: preregistered matched workflow trials reporting unsupported authoritative use, reconstruction time, selective risk, review burden, latency, bypass, and cost-sensitive net benefit.

O07. Open debt: Literature-compiled model validity. Status: Literature compilation and executability do not discharge empirical validation. Next evidence: preregistered observables and thresholds, blind expert discrimination, empirical trajectory comparison, leakage controls, and independent replication.

O08. Open debt: Institutional legitimacy. Status: Transparent governance can still be coercive, biased, or captured. Next evidence: affected-party review, authority and conflict mapping, contestation and appeal procedures, distributional impact analysis, and independent governance evaluation.

O09. Open debt: Security and bypass. Status: Monitor completeness, revocation, and mediation remain deployment-specific assumptions. Next evidence: threat-model review, bypass and tamper testing, stale/revoked receipt tests, monitoring-

coverage measurement, and independent security assessment.

O10. Open debt: Systematic review. Status: The literature comparison is broad and differential, not a completed systematic review. Next evidence: preregistered search protocol, databases and date bounds, inclusion/exclusion criteria, dual screening, extraction table, and reproducible synthesis.

O11. Open debt: Formal mechanisation. Status: The local proofs have been stress-tested and reviewed but not encoded in a proof assistant. Next evidence: first mechanisation lot F01-F04 and F08; second lot F06 after the evaluator and budget interfaces are fixed; retain conformance invariants outside the first lot unless a concrete monitor model is supplied. The manuscript is tool-agnostic; proof-assistant choice belongs to the execution project. Mutation-test assumptions and theorem interfaces rather than treating successful encoding as empirical validation.

O12. Open debt: Observable selection in the literature-compiled model. Status: Epistemic indistinguishability can be made trivial by favourable observables or thresholds. Next evidence:

preregister observables, baselines, experts, leakage controls, and failure criteria before generating trajectories.

O13. Open debt: Policy capture and authority independence Status: Separation of proposer and certifier is architectural, not guaranteed socially. Next evidence: define role independence, conflict-of-interest rules, contestation, revocation, and adversarial tests for collusion.

O14. Open debt: Independent peer review. Status: The manuscript has not yet undergone external anonymous peer review. Next evidence: independent review by formal-methods, information-science, AI-evaluation, domain-science, and governance reviewers, with public response-to-review or revision ledger.

O15. Open debt: Minimality and operational burden. Status: F23 supplies a relative removability criterion, not a proof that the architecture is globally minimal or cost-optimal. Next evidence: ablation by layer, matched simpler implementations, workload and cognitive-burden measurements, and comparison on the coverage-risk-cost frontier.

O16. Open debt: Relevance-set soundness. Status: Rel\_t is a declared semantic relevance set, not a discovered ground truth; excluding a recorded parent can create false cleanliness while F12 remains formally true on the reduced set. Next evidence: seeded critical-parent exclusions, reviewer reconstruction of relevance decisions, receipts for exclusions that affect authority-sensitive use, and tests showing that an unjustified exclusion routes to review or block rather than silently lowering inherited debt.

O17. Open debt: Approximation-and-transfer ledger reliability. Status: the multi-axis ledger is a candidate coding instrument; its categories and workflow burden are not independently validated. Next evidence: preregistered blinded multi-rater coding of independently selected relations with source provenance cards, an adjudicator independent of label design, agreement reported by axis, confusion/error analysis, and comparison with a simpler rubric.



## Appendix E — Scientific Stress-Test Protocol

Workflow comparison. Primary endpoint: independently adjudicated unsupported authoritative use under a preregistered adequacy charter; secondary endpoints include reconstruction quality, reviewer burden, queue stability, latency, bypass, false-block, unsafe-pass, and total human minutes per governed artefact. Failure condition: no advantage over prompt-only, retrieval, provenance/assurance, or untyped evidence workflows, or any safety gain that depends on unstable review queues or disproportionate operational burden.

**Query fidelity. Primary endpoint:** fiber distortion; decoder success; transfer errors **Failure condition:** declared preserved queries fail under matched inputs

**Typed terminals. Primary endpoint:** selective risk; abstention precision/recall; contradiction recall **Failure condition:** forced answers or no diagnostic improvement

**Debt coding. Primary endpoint:** inter-rater reliability; outcome correlation **Failure condi-**

**tion:** low agreement or no relation to downstream failure

Approximation / transfer ledger. Primary endpoints: inter-rater agreement by source\_basis and compiler\_ops, confusion between UNRESOLVED and terminal UNKNOWN, detection of undeclared transfer/approximation, coding burden, and incremental discrimination versus a simpler free-form rubric. Design rule: coders receive source provenance cards; label-design authors do not serve as the sole ground truth; numeric sample size and decision thresholds are preregistered with the pilot rationale. Failure condition: unstable coding, circular adjudication, or no discriminating value at lower burden.

Mutation contract. Critical mutation families include: waiver clears debt; U is coerced to a clean numeric value; warranted green is rendered without a current valid gate receipt; Rel\_t suppresses a critical parent without a justified receipt; unknown dependency coverage is promoted to measured; or a typed non-answer is coerced to ANSWER without new evidence evaluation. Every mutation designated critical by the preregis-

tered threat model must be detected; the overall mutation score is reported only as a secondary diagnostic and cannot compensate for a surviving critical bypass.

**False green. Primary endpoint:** green after receipt/readback/policy/monitor mutation **Failure condition:** authoritative state survives a missing proof predicate

**Governance. Primary endpoint:** waiver rate; expiry compliance; contestation; bypass **Failure condition:** rubber-stamping, capture, or unrecorded bypass

Literature-compiled model. Primary endpoint: blind expert discrimination; trajectory error. Failure condition: systematic distinguishability beyond preregistered threshold



## **Appendix F — Layer-Specific Contribution Claims and Defeat Conditions**

This appendix compresses the scientific status of the trajectory. “Inherited” means that the core mechanism already exists in the cited literature. “Relabeling risk” names the selected comparison showing how the proposal could amount only to new terminology. “Claimed delta” is the additional typed relation or architecture. “Defeat condition” is the observation that would remove or materially weaken that delta.

### **Reticular Local Abstraction: Contribution Claim and Defeat Condition**

Inherited: abstraction/refinement, partitions, sufficient statistics, coarse-graining, information bottleneck, causal abstraction. Relabeling risk: a knowledge graph with level labels. Claimed delta:

query-indexed factorization, explicit fibers, side-information accounting, and loss receipts at every transmission. Defeat condition: no improved localization of lost distinctions over a matched abstraction formalism.

## **Compact Reticular Computability: Contribution Claim and Defeat Condition**

Inherited: computability, transition systems, type soundness, termination, model checking, abstract interpretation. Relabeling risk: finite workflow graph. Claimed delta: one compact acceptance contract coupling effectivity, scheduling, resources, terminals, receipts, and cross-layer obligations. Defeat condition: ordinary workflow specifications provide the same guarantees with no missing relation.

## **Optional      Bounded-Simulation Bridge: Contribution Claim and Defeat Condition**

Inherited: reductions, universality, simulation, bi-simulation, observational equivalence. Relabeling risk: analogy described as equivalence. Claimed delta: only when invoked, a bounded observable-specific simulation certificate with distortion and overhead. Defeat condition: no non-trivial certificate, success only after circular choice of observables, or complete removability from downstream admitted queries and governance states. This side bridge is not part of the mandatory architecture.

## **Epistemic Computational Units and Networks: Contribution Claim and Defeat Condition**

Inherited: partial transducers, labelled transition systems, workflow engines, selective prediction, conformal risk control, neural-symbolic systems, paraconsistent logic, provenance graphs, and human review. Relabeling risk: an agent pipeline or knowledge graph with status labels. Claimed del-

ta: a navigable multi-level epistemic reticulum with candidate/certified state separation, deterministic replay of the control plane, typed non-answer terminals, receipts, debt updates, and composition rules that forbid non-answer-to-answer or candidate-to-certified coercion. Defeat condition: a simpler baseline matches diagnosis, abstention quality, certification traceability, replay, and burden.

## **Reticular Observer Architectures: Contribution Claim and Defeat Condition**

Inherited: observer-relative modelling, provenance graphs, knowledge graphs, runtime assurance, safety cases, human-in-the-loop governance. Relabeling risk: a governed knowledge graph. Claimed delta: observer compilation as a versioned contract linking horizon, reticulum, local units, memory, monitors, threat model, governance, and accountable authorization. Defeat condition: the integrated contract adds no measurable reconstructability or error-localization advantage.



## **Epistemic Debt: Contribution Claim and Defeat Condition**

Inherited: evidence gaps, provenance, assurance deficits, technical debt, documentation, risk acceptance. Relabeling risk: a risk register renamed debt. Claimed delta: a use-independent componentwise liability attached to semantic objects before use, combined with a separate use-sensitive gate, typed discharge, debt-preserving waiver, and non-launders. Defeat condition: unreliable coding, no reduction in unsupported authoritative use, or unbounded reviewer burden relative to simpler controls.

### **Close Comparator Matrix**

*Comparator rule. None of these traditions is an inferior baseline by definition. The residual delta is accepted only if it improves reconstruction, failure localization, or protection against unsupported authoritative use at acceptable burden.*

| <b>Comparator</b>        | <b>Already supplies</b>                                                                         | <b>Residual ROA/ED question</b>                                                                                                                            |
|--------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Truth-maintenance / ATMS | Already maintain beliefs through reasons, dependencies, assumption contexts, and inconsistency. | ED keeps evidence status separate from named-use authorization; waiver retains residual warrant; semantic transformations can carry typed debt before use. |
| Justification logic      | Attaches explicit proof/evidence terms to propositions.                                         | ED adds the governance split among evidence state, residual debt, authority, named use, waiver, and discharge; practical value remains empirical.          |

| <b>Comparator</b>                    | <b>Already supplies</b>                                                             | <b>Residual ROA/ED question</b>                                                                                                                                |
|--------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nanopublications / micropublications | Provide claim-level assertions, provenance, evidence, and support/challenge graphs. | Residual delta is the use-sensitive gate plus debt-preserving waiver and explicit non-answer/governance terminals.                                             |
| Provenance semirings                 | Provide algebraic derivation provenance through transformations.                    | ED must show that policy-indexed debt propagation, relevance sets, discharge, and use-gate semantics add discriminating information beyond provenance algebra. |
| Information-flow / taint analysis    | Provides mature conservative propagation and exposes over-taint risk.               | Strict absorbing-unknown propagation is one declared policy profile, not a universal algebra.                                                                  |

| Comparator                                | Already supplies                                                              | Residual ROA/ED question                                                                                                                                                            |
|-------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Digital-twin<br>V&V / UQ<br>/ credibility | Supplies mature<br>fit-for-purpose<br>validation and<br>uncertainty practice. | A literature-<br>compiled epistemic<br>model is not a live-<br>coupled digital twin;<br>executability does<br>not establish fitness,<br>so independent validation remains required. |

## **Cross-Layer Necessity and Layer-Removal Criterion**

The mandatory architecture is cumulative only if each interface exports something the next layer cannot safely reconstruct after the fact. RLA exports preserved and collapsed distinctions; CRC exports executable closure and terminal semantics; ECU/ECNN exports typed local epistemic actions and receipts; ROA exports observer and governance context; ED exports residual warrant and use standing. Bounded simulation is a side obligation, not a required export: it is introduced only when a cross-formalism claim needs an explicit simulation certificate. F23 converts the cumulative intuition into a removal test: a layer may be replaced only when its complete interface export is effectively reconstructable from adjacent bundles and every admitted query, terminal, receipt predicate, debt update, threat assumption, and resource bound remains unchanged. Failure of any condition is evidence that the layer is load-bearing; success establishes only relative redundancy under the frozen horizon, not unique minimality.









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# **Methodological Note on Drafting Assisted by Artificial Intelligence**

Parts of drafting, restructuring, terminology harmonization, bibliography expansion, figure prototyping, source-concept mining, finite-model stress testing, and local consistency checking were supported by AI-based assistants used as tools under human direction. The author remains responsible for every conceptual claim, theorem statement, formal choice, citation, interpretation, omission, and error. Generated material was treated as candidate artefact: it entered the manuscript only after source comparison, domain checks, hostile review, and human release approval. No AI system is treated as an autonomous scientific authority, certifier, or truth oracle.





## **Visual Synthesis Atlas**

The plates below consolidate complementary visual explanations of constructs already argued in the monograph. They add no new scientific claim and do not replace the formal text; each plate is illustrative and remains governed by the evidential and falsification boundaries stated in the corresponding conceptual locus.



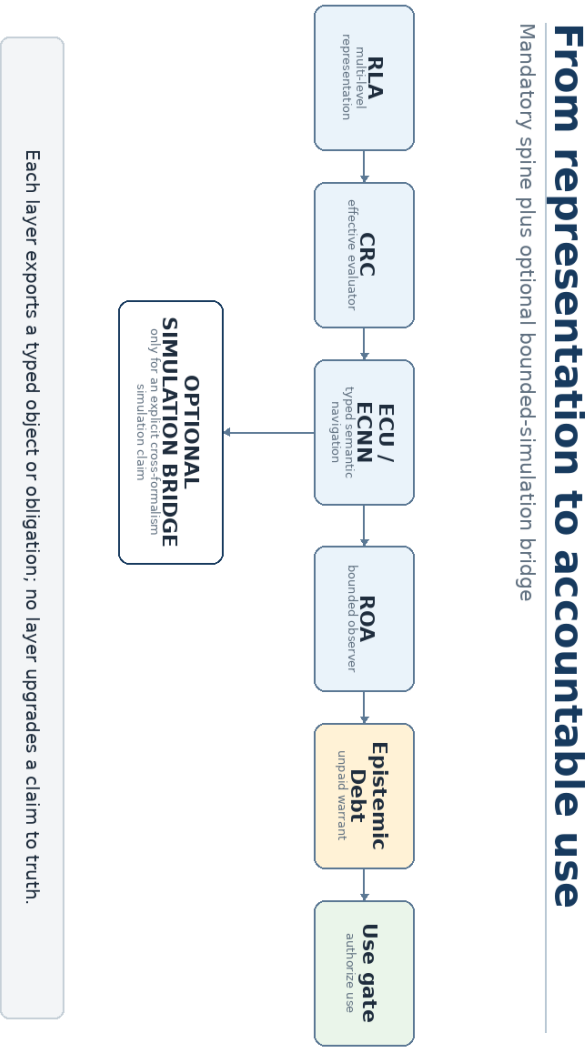


Figure 16. From representation to accountable use. The mandatory spine is RLA -> CRC -> ECU/ECNN -> ROA -> Epistemic Debt -> named-use gate; bounded simulation remains an optional side obligation.

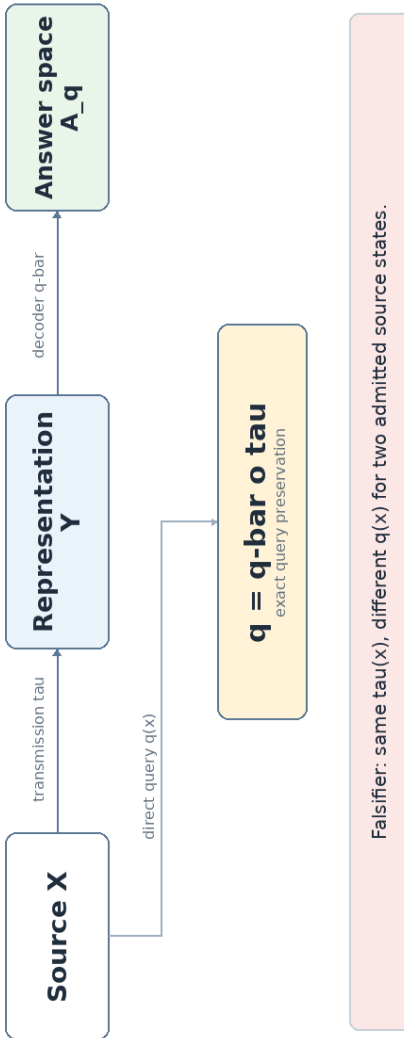


Figure 17. Query-relative fidelity by factorization. Exact preservation means the declared query can be recovered through the transmitted representation rather than requiring pointwise injectivity.

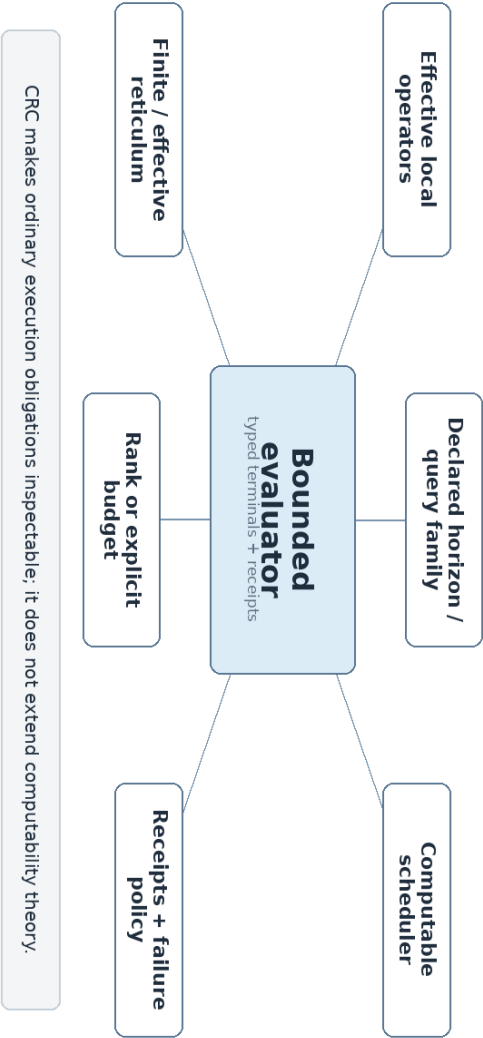


Figure 18. Compact Reticular Computability as an executable contract. Effective operators, a declared horizon, a computable scheduler, an effective reticulum, explicit resource bounds, and receipts surround a bounded evaluator.

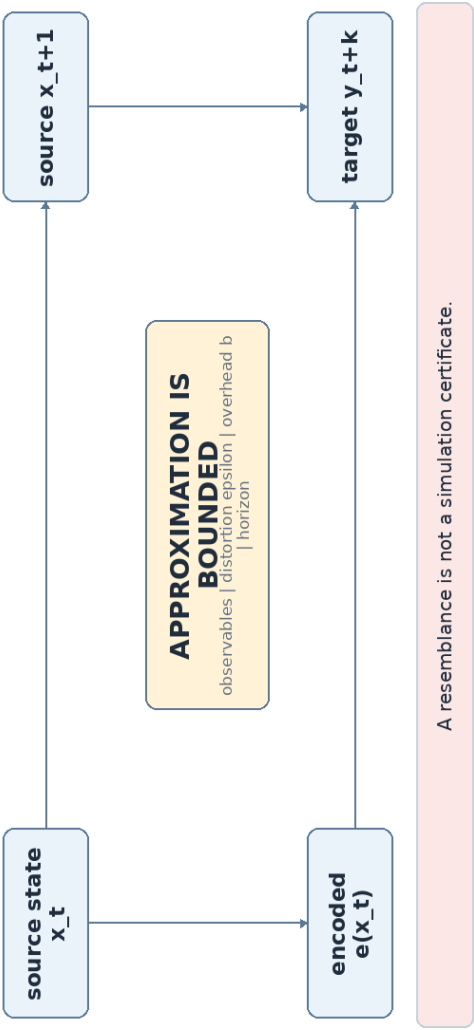
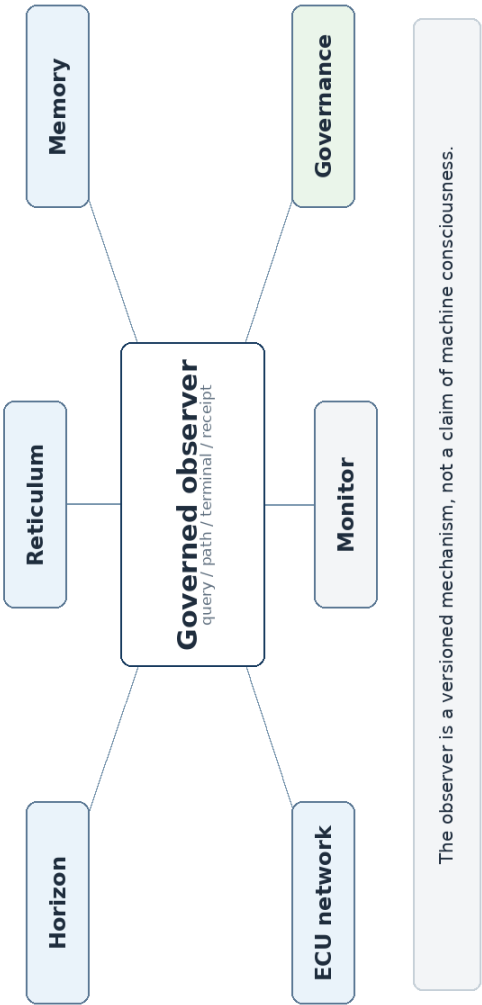


Figure 19. PCE bridge as a simulation obligation. A concrete encoding, declared observables, bounded distortion and overhead are required; one-way simulation is never silently promoted to equivalence.



Figure 20. Composition of Epistemic Computational Units and typed terminals. Answer, unknown, contradiction, out-of-horizon, review, timeout, and failure remain explicit under composition.



*Figure 21. Reticular Observer Architecture. A governed observer integrates horizon, reticulum, memory, ECU network, monitoring, and governance while remaining a versioned contract rather than a claim of consciousness.*



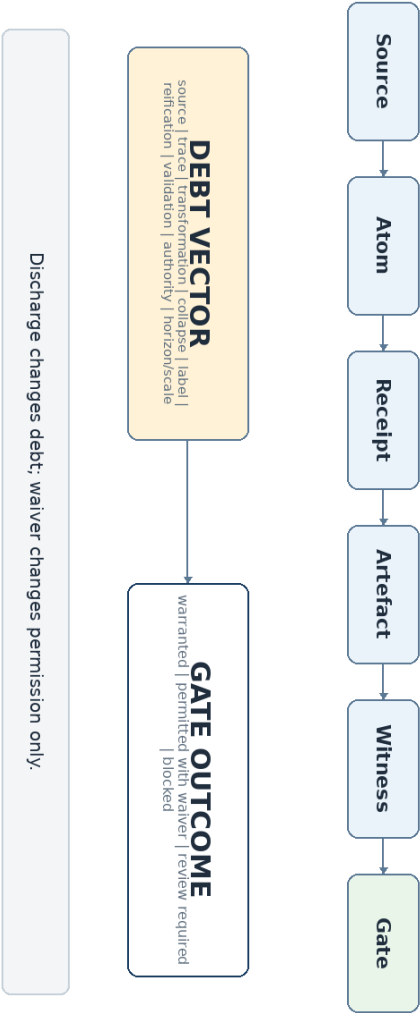
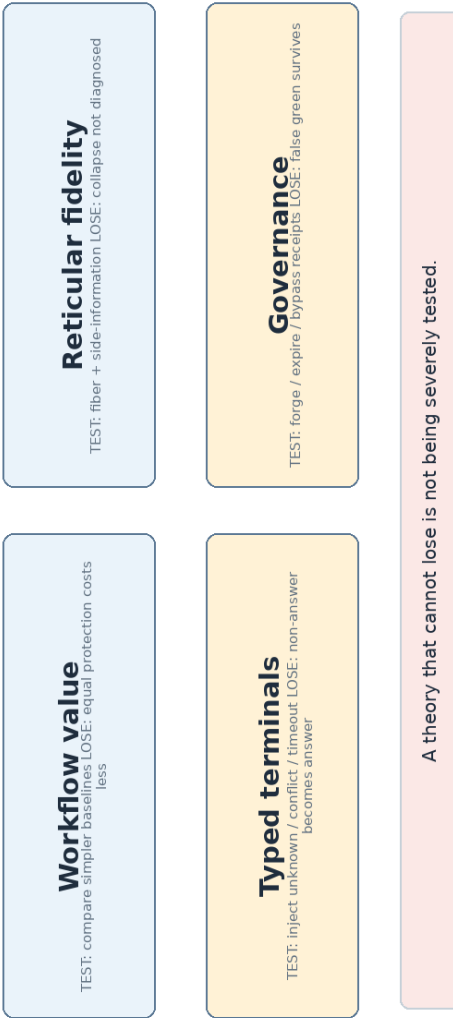


Figure 22. Epistemic Debt from artefact to authorized use. Debt follows the object through the proof chain until a named-use gate warrants, reviews, blocks, or explicitly permits with waiver.



*Figure 23. Severe-test programme, atlas view. Each lane names a hostile test and a concrete condition under which the architecture loses.*

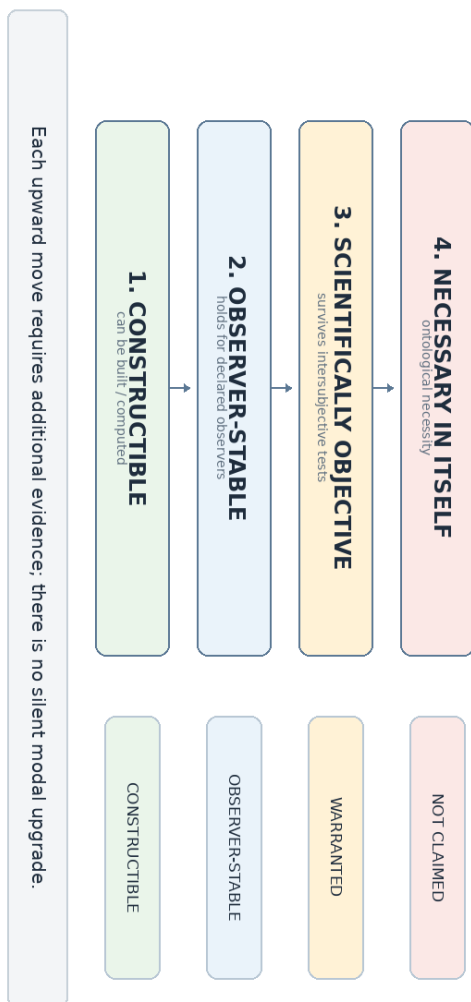
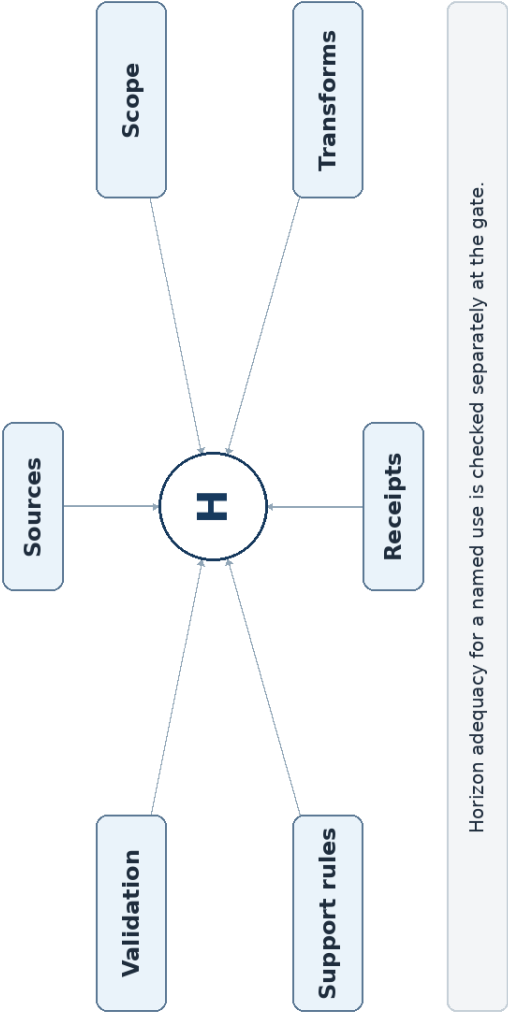


Figure 24. Four-step modal ladder. Constructibility, observer-stability, scientific objectivity, and necessity are distinct obligations; each upward move requires additional evidence.



*Figure 25. Declared epistemic horizon contract. The horizon bounds sources, scope, transformations, receipts, support rules, and validation; adequacy for a named use is checked separately.*

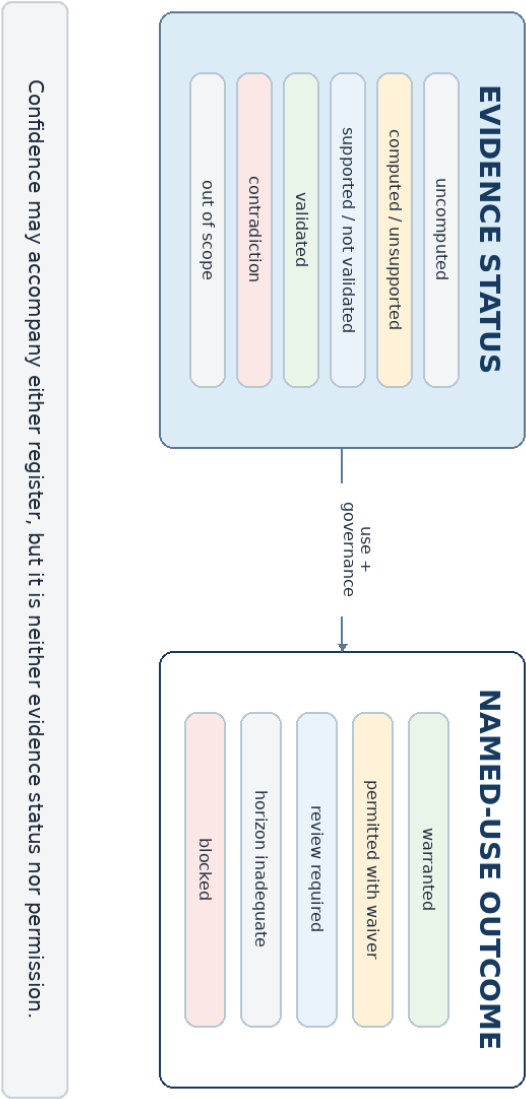


Figure 26. Evidence status versus use-gate outcome. What is earned by evidence and checks remains separate from what is authorized for a named use.

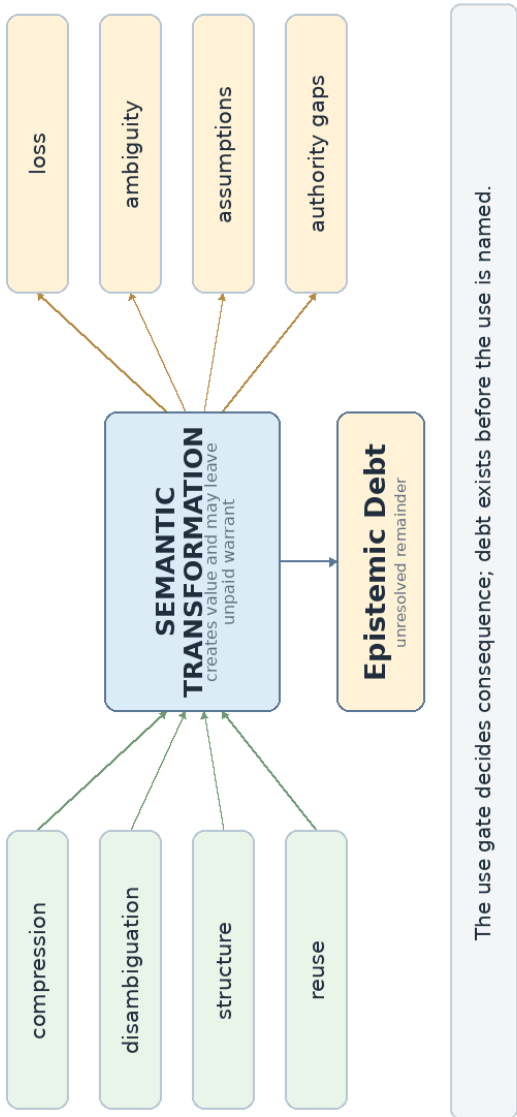


Figure 27. Asset and liability of semantic transformation. Compression, disambiguation, structure, and reuse can create value while loss, ambiguity, assumptions, and authority gaps create obligations.

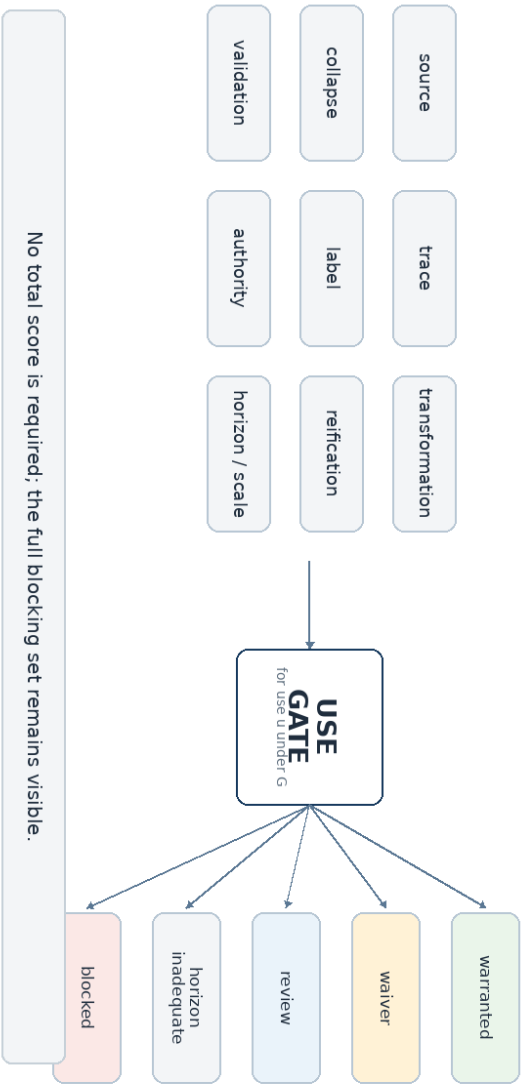


Figure 28. Debt vector and use gate. Componentwise debt feeds a use-specific gate that may warrant, permit with waiver, require review, find the horizon inadequate, or block.

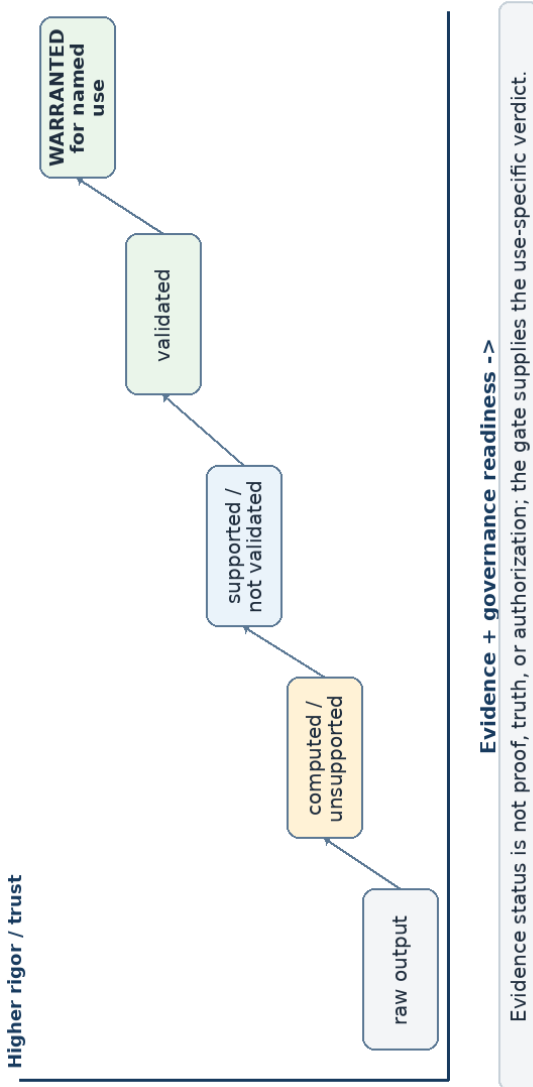


Figure 29. Measurement readiness and evidence status. Progress from raw output to supported, validated, and warranted use is staged and non-equivalent; evidence status is not proof, truth, or authorization.